二模参与网络中的中心性

Centrality in affiliation networks

Katherine Faust · 1997

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SOCIAL NETWORKS

中文译文 | G03-P001-001

社会网络

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Centrality in affiliation networks

中文译文 | G03-P001-003

二模参与网络的中心性

G03-P001-004 | 英文原文

Katherine Faust

中文译文 | G03-P001-004

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G03-P001-006 | 英文原文

Abstract

中文译文 | G03-P001-006

摘要

G03-P001-007 | 英文原文

This paper discusses the conceptualization, measurement, and interpretation of centrality in affiliation networks. Although centrality is a well-studied topic in social network analysis, and is one of the most widely used properties for studying affiliation networks, virtually all discussions of centrality and centralization have concerned themselves with one-mode networks. Bonacich's work on simultaneous group and individual centralities is a notable exception (*Social Networks*, 1991, 13, 155–168). I begin by outlining the distinctive features of affiliation networks and describe four motivations for centrality indices in affiliation networks. I then consider properties of some existing centrality indices for affiliation networks, including the relationship between centralities for actors and events in these networks, and present a new conceptualization of centrality that builds on the formal properties of affiliation networks and captures important theoretical insights about the positions of actors and events in these networks. These centralities are then illustrated on Galaskiewicz's data on club and board memberships of a sample of corporate executive officers (*Social Organization of an Urban Grants Economy*. New York: Academic Press, 1985). The conclusion to this paper discusses strengths and weaknesses of centrality indices when applied to affiliation networks. © 1997 Elsevier Science B.V.

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本文讨论二模参与网络中中心性的概念化、测量与解释。尽管中心性是社会网络分析中研究充分的主题，也是研究二模参与网络时使用最广泛的属性之一，但几乎所有关于中心性与中心势的讨论都针对单模网络。Bonacich关于群体与个体中心性同步测量的研究是一个显著例外（Social Networks，1991，13：155—168）。我首先概述二模参与网络的独特特征，并说明在此类网络中构建中心性指标的四项理由。接着，我考察现有若干中心性指标的性质，包括此类网络中行动者中心性与事件中心性的关系，并提出一种新的中心性概念。这一概念建立在二模参与网络的形式性质之上，能够体现有关行动者与事件在此类网络中所处位置的重要理论认识。随后，我利用Galaskiewicz关于一组企业高管的俱乐部成员资格及董事会任职情况的数据来说明这些中心性（Social Organization of an Urban Grants Economy，纽约：Academic Press，1985）。结论部分讨论了将中心性指标应用于二模参与网络时的优点与不足。© 1997 Elsevier Science B.V.。

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Keywords: Affiliation networks; Membership networks; Dual networks; Two-mode networks; Non-dyadic networks; Centrality

中文译文 | G03-P001-008

关键词："二模参与网络"、"成员资格网络"、"对偶网络；二模网络；非二元关系；中心性"

G03-P001-009 | 英文原文

1. Affiliation networks

中文译文 | G03-P001-009

1. 二模参与网络

G03-P001-010 | 英文原文

Many social network relations consist of the linkages among actors through their joint participation in social activities or membership in collectivities. This common activity creates a network of ties among actors. Similarly, collectivities, communities, or social occasions are linked to each other through the multiple memberships of actors. Such networks of actors tied to each other through their participation in collectivities, and collectivities linked through multiple memberships of actors, are referred to as *affiliation networks*, *membership networks*, *dual networks*, or *hypernetworks* (Breiger, 1974, 1990; Seidman, 1981, 1985; McPherson, 1982; Wasserman and Faust, 1994).

中文译文 | G03-P001-010

许多社会网络关系，是行动者因共同参加社会活动或同属某一集体而形成的联系。这种共同活动在行动者之间建立起关系网络。同样，集体、社群或社会活动场合，也会通过行动者同时拥有的多重成员资格而相互联系。行动者因参与集体而彼此联结、集体又通过行动者的多重成员资格而相互联结，这样的网络被称为参与网络、成员资格网络、对偶网络或超网络（Breiger，1974、1990；Seidman，1981、1985；McPherson，1982；Wasserman与Faust，1994）。

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Formally, an affiliation network consists of two key elements: a set of actors and a collection of subsets of actors (called *events*). Thus, an affiliation network is a two-mode, non-dyadic network. The two modes are the set of actors and the set of

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从形式上看，二模参与网络由两个关键要素构成：一组行动者，以及由这些行动者的若干子集组成的集合（这些子集称为事件）。因此，参与网络是一种二模、非二元关系网络。两个模分别是行动者集合与

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events. An affiliation network is non-dyadic because the affiliation relation relates each actor to a subset of events, and relates each event to a subset of actors. Affiliation networks have been referred to as dual networks because of the complementary perspectives through which actors are linked to each other as members of collectivities, and collectivities are linked to each other through shared members (Breiger, 1974,

中文译文 | G03-P002-002

事件集合。参与网络之所以属于非二元关系网络，是因为参与关系将每一名行动者同一组事件的子集相联系，也将每一个事件同一组行动者的子集相联系。参与网络还被称为对偶网络，因为它提供两种互为补充的视角：行动者作为集体成员彼此联系，而各个集体又通过共同成员相互联系（Breiger，1974，

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Substantive examples of affiliation networks are widespread, and include, for example, interlocking boards of directors (Levine, 1972; Mariolis, 1975; Sonquist and Koenig, 1975; Mintz and Schwartz, 1981a,b; Allen, 1982; Mizruchi, 1982; Bearden and Mintz, 1987); memberships in voluntary organizations (McPherson, 1982); club memberships (Bonacich, 1978); social gatherings (Davis et al., 1941; Homans, 1950; Breiger,

中文译文 | G03-P002-003

二模参与网络的具体实例十分常见，例如相互关联的董事会（Levine, 1972; Mariolis, 1975; Sonquist and Koenig, 1975; Mintz and Schwartz, 1981a,b; Allen, 1982; Mizruchi, 1982; Bearden and Mintz, 1987）；志愿组织的会员资格（McPherson, 1982）；俱乐部会员身份（Bonacich, 1978）；社交聚会（Davis et al., 1941; Homans, 1950; Breiger,

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1974); ceremonial events attended by community members (Foster and Seidman, 1984; Schweizer et al., 1993); observations of informal social interactions (Bernard et al.,

中文译文 | G03-P002-004

1974）；社区成员参与的仪式性事件（Foster and Seidman, 1984; Schweizer et al., 1993）；非正式社交互动的观察（Bernard et al.,

G03-P002-005 | 英文原文

1980, 1982; Freeman and Romney, 1987; Freeman et al., 1987; Freeman et al., 1989) and so on. Regardless of their substance, these applications share a formal similarity in that each consists of a set of actors and a collection of subsets of actors.

中文译文 | G03-P002-005

1980年、1982年；弗里曼与罗姆尼，1987年；弗里曼等，1987年；弗里曼等，1989年等。无论其性质如何，这些应用都具有形式上的相似性，即每个都由一组行动者以及该行动者的若干子集组成。

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In this paper I will use two examples of affiliation networks as illustrations. The first is a hypothetical affiliation network of six actors and three events from Wasserman and Faust (1994). The second example is a subset of the data collected by Galaskiewicz (1985) for his study of the urban grants economy in the Minneapolis-St. Paul area. This affiliation network consists of 26 corporate executive officers and their membership in

中文译文 | G03-P002-006

在本文中，我将以两个二模参与网络为例进行说明。第一个例子是 Wasserman 和 Faust（1994）提出的由六位行动者和三起事件构成的假设性二模参与网络。第二个例子则是 Galaskiewicz（1985）为研究明尼阿波利斯-圣保罗地区的城市补助经济而收集的数据的 子集，该二模参与网络包含 26 位企业高管及其成员资格

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15 clubs, boards of cultural organizations, and corporate boards. (See Galaskiewicz, 1985; Wasserman and Faust, 1994, for further discussion of these data.)

中文译文 | G03-P002-007

15个俱乐部、文化组织董事会和企业董事会中的成员资格组成。（有关这些数据的更多讨论，请参见 Galaskiewicz, 1985; Wasserman and Faust, 1994。）

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To illustrate an affiliation network, consider the hypothetical example of six actors and three events. The set of actors is denoted by $\mathcal{N} = \{n_1, n_2, \ldots, n_g\}$, and the set of events is denoted by $\mathcal{M} = \{m_1, m_2, \ldots, m_h\}$. There are g actors and h events. The affiliation network matrix for this example is presented in Table 1. This matrix, denoted by $\boldsymbol{A} = \{a_{ik}\}$, shows the affiliation of the actors with the events. A '1' in row i, column k of $\boldsymbol{A}$ indicates that actor n_i is affiliated with event m_k. Table 2 gives the one-mode matrix of actor co-memberships. This matrix, denoted by $\boldsymbol{X}^{\mathcal{N}}$, indicates the number of memberships shared by each pair of actors. Table 3 gives the one-mode matrix of event overlaps, denoted by $\boldsymbol{X}^{\mathcal{M}}$, which gives the number of actors shared by each pair of events.

中文译文 | G03-P002-008

为了说明二模参与网络，以六个行动者和三个事件这一假设示例为例。行动者集合表示为 ~'= {ni,n 2 ..... ng}，事件集合表示为 ~'= {mi,m 2 ..... mh}。共有 g 个行动者和 h 个事件。该示例的二模参与网络矩阵如表1所示。这个由 A = {aik} 表示的矩阵，显示了行动者与事件的关联关系。A中第i行第k列的'1'表示行动者 n i与事件m k相关。表2给出了行动者共同参与关系的单模矩阵。这个由 X ~ 表示的矩阵，显示了每对行动者之间的共同参与次数。表3给出了事件重叠的单模矩阵，由 X ~' 表示，它显示了每对事件中共同参与的行动者数量。

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Table 1
Affiliation network matrix

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表1 二模参与网络矩阵

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Table 2

中文译文 | G03-P003-002

表2

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Actor co-membership matrix

中文译文 | G03-P003-003

行动者成员资格矩阵

G03-P003-004 | 英文原文

n_1 n_2 n_3 n_4 n_5 n_6

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As Breiger (1974) has noted, the affiliation matrix is related to the actor co-membership matrix and to the event overlap matrix through the following equations:

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正如 Breiger（1974）所指出的，关联矩阵与行动者的成员资格矩阵以及事件重叠矩阵之间存在以下关系：

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$$\boldsymbol{X}^{\mathscr{N}} = \boldsymbol{A}\boldsymbol{A}' \qquad (1)$$

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G03-P003-007 | 英文原文

and

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并且

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The affiliation network matrix uniquely determines both $\boldsymbol{X}^{\mathscr{N}}$ and $\boldsymbol{X}^{\mathscr{M}}$, but the reverse is not true. A given actor co-membership matrix or event overlap matrix may be generated by a number of different affiliation network matrices (Breiger, 1990).

中文译文 | G03-P003-008

二模参与网络矩阵唯一决定了 X ~r 和 X ~，但反之则不成立。某个行动者的成员资格矩阵或事件重叠矩阵可能由多种不同的二模参与网络矩阵生成（Breiger, 1990）。

G03-P003-009 | 英文原文

An affiliation network can also be represented as a bipartite graph (Wilson, 1982). In a bipartite graph nodes can be partitioned into two subsets and all lines are between nodes from different subsets. In the bipartite graph for an affiliation network the two sets of nodes are the set of actors, $\mathscr{N}$, and the set of events, $\mathscr{M}$, so there are $g + h$ nodes. Since the lines indicate ties of affiliation they are always between actors and events. Fig.

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二模参与网络也可以被表示为一个二部图（Wilson，1982）。在二部图中，节点可以被划分为两个子集，且所有边都连接不同子集中的节点。对于二模参与网络而言，两个节点集合分别是行动者集合 ~' 和事件集合 ~'，因此共有 g + h 个节点。由于这些边表示二模参与关系，它们总是连接行动者与事件。图。

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1 shows the hypothetical affiliation network as a bipartite graph.

中文译文 | G03-P003-010

1展示了作为二部图的假设性二模参与网络。

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The bipartite graph can also be presented in a sociomatrix, denoted by $\boldsymbol{X}^{\mathcal{NM}}$. This matrix has $g + h$ rows and columns indexing actors and events, and has the form:

中文译文 | G03-P003-011

二部图也可以以社会矩阵的形式呈现，该矩阵用 X ~r 表示。此矩阵具有 g + h 行和列，用于标识行动者与事件，其形式如下：

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where actors occupy the first g rows and columns and events occupy the last h rows and columns.

中文译文 | G03-P003-012

其中，行动者占据前 g 行和列，事件占据后 h 行和列。

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In studying an affiliation network one can look at properties of actors in the one-mode relation of actor co-memberships, properties of events in one-mode relation of event overlaps, or properties of both actors and events in the two-mode affiliation relation.

中文译文 | G03-P003-013

在研究二模参与网络时，可以关注行动者在单模参与关系中的属性、事件在单模重叠关系中的属性，或者二模参与关系中行动者与事件两者的属性。

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Table 3

中文译文 | G03-P003-014

表 3

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Event overlap matrix

中文译文 | G03-P003-015

事件重叠矩阵

G03-P003-016 | 英文原文

m_1 m_2 m_3

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Fig. 1. Bipartite graph of affiliation network.

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图1. 二部二模参与网络图。

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Centrality is one network property that frequently has been used to study actors or events in affiliation networks (Bonacich, 1972, 1991; Mariolis, 1975; Stokman, 1977; Bonacich and Domhoff, 1981; Mintz and Schwartz, 1981a,b; Mizruchi and Bunting,

中文译文 | G03-P004-003

中心性是常用于研究二模参与网络中的行动者或事件的一种网络特性（Bonacich, 1972, 1991; Mariolis, 1975; Stokman, 1977; Bonacich和Domhoff, 1981; Mintz和Schwartz, 1981a,b; Mizruchi和Bunting,

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1981; Mariolis and Jones, 1982; Mizruchi, 1982, 1992; Rosenthal et al., 1985; Mizruchi et al., 1986; Fernandez and McAdam, 1988). However, virtually all analyses of centrality in affiliation networks neglect the inherent properties of these networks: the presence of two modes, the duality of actors and events, and the non-dyadic affiliation relation. In the following sections I address the issue of conceptualization and measurement of centrality in affiliation networks, with special attention to the interpretation of existing centrality measures as applied to affiliation networks and to possible future directions for developing centrality approaches that are especially appropriate for affiliation networks.

中文译文 | G03-P004-004

1981；Mariolis与Jones，1982；Mizruchi，1982、1992；Rosenthal等，1985；Mizruchi等，1986；Fernandez与McAdam，1988）。然而，几乎所有对二模参与网络中心性的分析，都忽视了此类网络的固有性质：存在两个模、行动者与事件之间的对偶性，以及非二元的参与关系。下文讨论二模参与网络中心性的概念化与测量，尤其关注现有中心性度量用于此类网络时应如何解释，并探讨未来如何发展更适合二模参与网络的中心性研究方法。

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2. Centrality

中文译文 | G03-P004-005

2. 中心性

G03-P004-006 | 英文原文

The general notion of centrality encompasses a number of different aspects of the 'importance' or 'visibility' of actors within a network. Discussions of centrality can be found in Freeman (1979), Knoke and Burt (1983), Friedkin (1991), Faust and Wasserman (1992) and Wasserman and Faust (1994). In general, there are four common motivations for centrality in one-mode dyadic networks:

中文译文 | G03-P004-006

中心性这一总体概念，涵盖行动者在网络中"重要性"或"可见度"的多个方面。有关讨论可参见Freeman（1979）、Knoke与Burt（1983）、Friedkin（1991）、Faust与Wasserman（1992），以及Wasserman与Faust（1994）。一般而言，在单模二元关系网络中，中心性有以下四种常见理论依据：

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- actors are central if they are active in the network (motivating degree centrality);
- actors are central if they can contact others through efficient (short) paths (motivating closeness centrality);
- actors are central if they have the potential to mediate flows of resources or

中文译文 | G03-P004-007

行动者若在网络中十分活跃，就具有中心性（对应度中心性）；行动者若能通过高效的短路径联系其他人，就具有中心性（对应接近中心性）；行动者若有可能在其他行动者之间的资源或

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information between other actors (motivating betweenness centrality); and
actors are central if they have ties to other actors that are themselves central (motivating eigenvector centrality).

中文译文 | G03-P004-008

信息流动中发挥中介作用，就具有中心性（对应中介中心性）；行动者若与本身也具有中心性的其他行动者相联，就具有中心性（对应特征向量中心性）。

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G03-P005-002 | 英文原文

Freeman (1979) presents three sets of indices that formalize the ideas of degree (activity), closeness (efficiency), and betweenness (control) actor centrality and graph centralization. From a slightly different perspective, Friedkin (1991) reviews the theoretical basis for centrality measures within the framework of models of interpersonal influence. Within this framework he distinguishes between 'total', 'immediate', and

中文译文 | G03-P005-002

Freeman（1979）提出了三组指数，用以形式化中心性、中心势以及行动者中心性与图中心势中的程度（活动）、接近性（效率）和中间性（控制）等概念。从略有不同的视角来看，Friedkin（1991）在人际影响模型框架内对中心性测度的理论基础进行了综述。在该框架中，他区分了"整体"和"直接"两类。

G03-P005-003 | 英文原文

'mediative' effects centrality, roughly analogous to the total influence of an actor, an actor's closeness to others, and the betweenness of the actor. At the conclusion of his paper Friedkin cites Coombs's position on the theoretical basis for measurement and scaling models in general, commenting:

中文译文 | G03-P005-003

"中介性"中心性，大致相当于行动者的总影响力、行动者与其他节点的距离，以及该行动者的中间性。在论文结尾处，弗里德金引用了库姆斯关于测量和缩放模型理论基础的观点，并评论道：

G03-P005-004 | 英文原文

Coombs reminds us that 'a measurement or scaling model is actually a theory about behavior, admittedly on a miniature level, but nevertheless theory' [Coombs,

中文译文 | G03-P005-004

Coombs 提醒我们："测量或缩放模型实际上是一种关于行为的理论，尽管是在微观层面，但依然属于理论" [Coombs,

G03-P005-005 | 英文原文

1964, p. 5]. By this criterion, every new proposal of a centrality measure presents new theoretical material. (Friedkin, 1991, p. 1498.)

中文译文 | G03-P005-005

1964, p. 5]。根据这一标准，每一种新的中心性度量提议都带来了新的理论内容。（Friedkin, 1991, p. 1498。）

G03-P005-006 | 英文原文

It is in this spirit that I intend to review centrality measures for affiliation networks. The theoretical foundations for affiliation networks as well as their formal properties differ in important ways from the more usual one-mode dyadic networks. Thus, centrality indices that have been developed for one-mode dyadic social networks may or may not be appropriate for studying affiliation networks. In addition, there may be theoretical insights gained from affiliation networks that will suggest new centrality approaches.

中文译文 | G03-P005-006

正是基于这种理念，我打算对二模参与网络的中心性度量进行探讨。二模参与网络的理论基础及其形式特性与常见的单模双点网络存在显著差异。因此，为单模双点社会网络所开发的中心性指标，可能适合也可能不适合用于研究二模参与网络。此外，从二模参与网络中获得的理论启示，也可能提出新的中心性研究方法。

G03-P005-007 | 英文原文

In the next section I discuss four motivations for centrality in affiliation networks, with special attention to those features of affiliation networks that are distinctive and lead to centrality motivations that are different from centrality motivations for one-mode networks.

中文译文 | G03-P005-007

在下一节中，我将讨论二模参与网络中心性的四种成因，并特别关注那些具有独特特性的二模参与网络特征，这些特征导致了与单模网络中心性成因不同的中心性成因。

G03-P005-008 | 英文原文

3. Centrality motivations for affiliation networks

中文译文 | G03-P005-008

3. 二模参与网络的中心性动机

G03-P005-009 | 英文原文

This section presents four ideas that can motivate centralities in affiliation networks. Two of these ideas arise from the formal properties of affiliation networks (two-mode, non-dyadic networks), and two arise from empirical and theoretical insights suggested by researchers who have studied affiliation networks. The four ideas that I present below are: centralities for both actors and events, centralities for subsets of actors and/or subsets of events, the importance of linkages created by actors and events, and the importance of subset–superset relationships in actors' affiliations and events' memberships.

中文译文 | G03-P005-009

本节提出四种可激发二模参与网络中中心性的理论观点。其中两种观点源自二模非二元关系网络的正式特性，另外两种则源于研究二模参与网络的研究者提出的经验与理论见解。我下面所提出的四种观点包括：针对行动者与事件的中心性；针对行动者子集与/或事件子集的中心性；行动者与事件所建立的关联的重要性；以及行动者关联与事件成员资格中的子集-超子集关系的重要性。

G03-P005-010 | 英文原文

3.1. Centralities for actors and events

中文译文 | G03-P005-010

3.1. 行动者与事件的中心性

G03-P005-011 | 英文原文

Since affiliation networks are two-mode networks, a complete analysis should give centrality indices for both actors and events. As we shall see below, calculating indices for both actors and events is straightforward. The important question is: what, if anything, is the relationship between the centralities of events and the centralities of the actors that belong to the events? Different authors have taken different perspectives on this question. In their discussion of the relationship between the centrality of corporate

中文译文 | G03-P005-011

由于二模参与网络是二模网络，完整的分析应给出所有行动者和事件的中心性指数。正如我们下面将看到的，计算所有行动者和事件的指数非常直接。关键问题是：事件的中心性与属于这些事件的行动者的中心性之间是否存在某种关系？不同作者对这个问题持有不同的观点。在他们对企业中心性关系的讨论中

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G03-P006-002 | 英文原文

boards and the centrality of individuals who sit on the boards, Perrucci and Lewis (1989, p. 216) argue that

中文译文 | G03-P006-002

Perrucci和Lewis（1989，第216页）认为，董事会成员及其成员资格与个体中心性之间存在关系……

G03-P006-003 | 英文原文

These two networks can reveal considerable overlap in that the organizations that occupy central positions in their network are also those organizations that are the primary affiliations of the leaders that hold central positions in their network. On the other hand, the two networks can be independent in that the central organizations and the central leaders are drawn from two different subgroups.

中文译文 | G03-P006-003

这两个网络存在显著的重叠，因为在其网络中处于核心位置的机构，也是其网络中处于核心位置的领导人所隶属的主要机构。另一方面，这两个网络也可以相互独立，因为核心机构与核心领导人来自两个不同的子群体。

G03-P006-004 | 英文原文

Perrucci and Lewis view the relationship between the centrality of an actor and the centralities of the events to which the actor belongs as an empirical question. In their view, a given network may or may not show any relationship between actor and event centralities. Bonacich (1991, p. 256) takes exactly the opposite position. He argues that

中文译文 | G03-P006-004

Perrucci和Lewis认为，行动者的中心性与其所参与的事件的中心性之间的关系是一个经验性问题。在他们看来，某个网络可能也未必能显示出行动者与事件中心性之间的关联。Bonacich（1991，第256页）则持完全相反的观点，他认为……

G03-P006-005 | 英文原文

Centrality involves ... the 'duality' of groups and individuals. A central firm gets its central position from the board membership patterns of its members.... Dually a central individual should be one who belongs to a variety of important firms. One kind of centrality cannot be defined without reference to the other.

中文译文 | G03-P006-005

中心性涉及群体与个体之间的"双重性"。一家核心企业之所以处于中心地位，源于其成员成员的成员资格模式……作为核心个体，应当属于多种重要企业中的一员。一种中心性无法脱离另一种中心性来定义。

G03-P006-006 | 英文原文

Bonacich's argument builds on the theoretically important duality in the relationship between actors and events. In keeping with this perspective, Bonacich (1991) presents a centrality measure for which the centrality of an event is proportional to the sum of the centralities of the actors that are members of the event, and the centrality of an actor is proportional to the sum of the centralities of the events to which it belongs. I discuss this measure, eigenvector centrality, in detail below.

中文译文 | G03-P006-006

Bonacich的论点基于行动者与事件之间关系中的理论重要对偶性。遵循这一视角，Bonacich（1991）提出了一种中心性度量方法：事件的中心性与其参与该事件的行动者中心性之和成正比；而行动者的中心性则与其所归属的事件中心性之和成正比。下面我将详细讨论这种度量方法——特征向量中心性。

G03-P006-007 | 英文原文

Thus, the first important feature of centralities for affiliation networks is that there should be centrality scores for both actors and events. Furthermore, there should be a clearly specifiable relationship between these quantities.

中文译文 | G03-P006-007

因此，二模参与网络中心性的第一个重要特征是，行动者和事件都应具有中心性得分。此外，这些数值之间应存在可明确指定的关系。

G03-P006-008 | 英文原文

3.2. Centralities for subsets

中文译文 | G03-P006-008

3.2. 子集的中心性

G03-P006-009 | 英文原文

Another important feature of an affiliation network is that the affiliation relation is *non-dyadic* (Wilson, 1982; Seidman, 1985). Unlike a dyadic relation that links pairs of actors (and thus is binary), a non-dyadic relation links members of subsets of arbitrary size (and is thus *n*-ary). The affiliation relation relates each event to a *subset* of actors and relates each actor to a *subset* of events.

中文译文 | G03-P006-009

二模参与网络的另一个重要特性是，其参与关系属于非二元关系（Wilson, 1982; Seidman, 1985）。与连接行动者对组的二元关系不同（即为二进制关系），非二元关系则连接任意大小的子集成员（即为n元关系）。参与关系将每个事件与一组行动者子集关联起来，同时将每个行动者与一组事件子集关联起来。

G03-P006-010 | 英文原文

What does this imply for centralities in these networks? Existing centrality measures apply to one of two levels: centrality indices for actors or centralization indices for whole networks (Freeman, 1979). But if we take seriously the idea that actors in affiliation networks are defined by the collection of the events to which they belong and events are defined by their collection of members (Simmel, 1955; Breiger, 1974), then the centrality of an actor should be a function of the collection of events to which it belongs and the centrality of an event should be a function of the centrality of its collection of members. In contrast to a centrality index for a node in a graph, a centrality index for a collection of nodes should quantify the importance of that collection within the entire graph. For example, a centrality index for an event in an affiliation network

中文译文 | G03-P006-010

这对这些网络中的中心性有何含义？现有的中心性度量方法适用于两种层面：行动者的中心性指数或整个网络的中心势指数（Freeman，1979）。但如果我们认真考虑这样一个观点，即二模参与网络中的行动者由其所归属的事件集合定义，而事件由其成员集合定义（Simmel，1955；Breiger，1974），那么行动者的中心性应该是其所属事件集合的函数，而事件的中心性则应该是其成员中心性的函数。与图中节点的中心性指数不同，节点集合的中心性指数应衡量该集合在整个图中的重要性。例如，二模参与网络中的事件中心性指数就反映了这一点。

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G03-P007-002 | 英文原文

should quantify the importance of the collection of actors belonging to that event. The idea is to extend centrality indices for individual nodes to centrality indices for collections of nodes in a graph (or actors or events in an affiliation network).

中文译文 | G03-P007-002

应量化参与该事件的所有行动者集合的重要性。其思路是将单个节点的中心性指标扩展为图中节点集合的中心性指标（或二模参与网络中的行动者或事件）。

G03-P007-003 | 英文原文

These proposals of centralities for two modes and for subsets of entities build on formal properties of affiliation networks. Next I discuss theoretical ideas and empirical insights that have been used to describe the 'importance' of actors and events in affiliation networks, and which thus suggest theoretically and substantively grounded motivations for centralities in affiliation networks.

中文译文 | G03-P007-003

这些关于二模参与网络与实体子集的中心性提议，基于二模参与网络的正式特性。接下来，我将讨论用于描述行动者与事件在二模参与网络中"重要性"的理论观点与实证见解，这些观点和见解从理论和实质层面为二模参与网络中的中心性提供了依据。

G03-P007-004 | 英文原文

3.3. Linkages between actors and events

中文译文 | G03-P007-004

3.3. 行动者与事件之间的关联

G03-P007-005 | 英文原文

A theoretically important property of an affiliation network is that actors create linkages between events and events create linkages between actors. Multiple memberships of actors provide conduits for the flow of information between events or for the coordination of activities between events. Events as collections of actors provide the opportunity for contacts between actors, for sharing of information or other resources among actors, and facilitate formation of pair-wise ties between actors (McPherson,

中文译文 | G03-P007-005

二模参与网络的一个理论上重要的特性是，行动者在事件之间建立联系，而事件又在行动者之间建立联系。行动者的多重成员身份为事件之间的信息流动或事件间活动的协调提供了通道。作为行动者集合的事件则为行动者之间的接触、信息或其他资源的共享提供了机会，并有助于形成行动者之间的配对关系（McPherson）

G03-P007-006 | 英文原文

1982; McPherson and Smith-Lovin, 1982). Bonacich and Domhoff (1981) observe that ''no groups are directly related to each other; they are only indirectly related to each other through individuals'' (p. 178), and similarly ''central individuals link groups which tend to be disjoint'' (p. 179).

中文译文 | G03-P007-006

1982；McPherson和Smith-Lovin，1982）。Bonacich和Domhoff（1981）指出："没有哪些群体之间存在直接关联；它们只是通过个体间接相互关联"（第178页），同样地，"核心个体将那些往往互不相关的群体连接在一起"（第179页）。

G03-P007-007 | 英文原文

The linking function of actors and events means that an actor is central in an affiliation network if it creates ties between events and an event is central if it creates ties between actors. An actor that belongs only to one event creates no ties between events. Thus, an actor's multiple memberships contribute to its centrality because that actor creates linkages between events. It is not simply the number of memberships that an actor has that is important. An actor that belongs to few events may nevertheless create a critical tie between two or more events that otherwise would not be linked. Similarly, an event is important because it creates linkages between its members. Although events with large membership lists create more ties between pairs of actors than do events with small membership lists, it is not simply the size of an event that is important. Even a small event may bring together actors that would not meet otherwise.

中文译文 | G03-P007-007

行动者与事件之间的关联功能意味着：如果某个行动者在二模参与网络中建立了事件与事件之间的联系，则该行动者具有中心性；如果某个事件在行动者与行动者之间建立了联系，则该事件具有中心性。仅属于一个事件的行动者无法在事件之间建立联系。因此，一个行动者的多重成员资格有助于其中心性，因为该行动者能在事件之间建立关联。重要的并非行动者拥有的成员资格数量。一个只属于少数事件的行动者仍可能在两个或更多事件之间建立关键联系，而这些事件原本不会相互关联。同样，一个事件之所以重要，是因为它能在其成员之间建立关联。尽管拥有大量成员名单的事件比拥有少量成员名单的事件能建立更多的行动者间联系，但重要的并非事件的大小。即使是小型事件也能将原本不会相遇的行动者聚集在一起。

G03-P007-008 | 英文原文

Thus, the linkages between events created by actors' multiple affiliations, and the linkages between actors created by events' collections of members are important considerations in determining the centrality of both actors and events in an affiliation network. Actors are always between events and events are always between actors. This suggests that some form of betweenness centrality will be appropriate for studying affiliation networks.

中文译文 | G03-P007-008

因此，由行动者多种二模参与网络所产生的事件间的联系，以及由事件成员集合所产生的行动者间的联系，是确定二模参与网络中行动者与事件中心性的重要因素。行动者始终位于事件之间，而事件也始终位于行动者之间。这表明某种中介中心性形式适用于研究二模参与网络。

G03-P007-009 | 英文原文

3.4. Subset–superset inclusions of actors and events

中文译文 | G03-P007-009

3.4. 行动者与事件的子集-超集包含关系

G03-P007-010 | 英文原文

A fourth distinctive motivation for centrality in affiliation networks arises from observations about the structure of memberships in these networks. Memberships in affiliation networks can be viewed as patterns of inclusions among events' membership

中文译文 | G03-P007-010

二模参与网络中中心性的第四个独特成因，源于对这些网络中成员资格结构的观察。二模参与网络中的成员资格可视为事件成员资格之间的包含关系模式

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G03-P008-002 | 英文原文

lists, as patterns of inclusions among actors' affiliations, or both (Bonacich, 1978). Here, we shall want to look not at the number of overlaps for events or co-memberships for actors, but at the patterns of inclusions among memberships. Specifically, a number of authors have observed that in terms of their participation in events, some actors' affiliations are included within the affiliations of other actors. This leads to the distinction between 'primary' and 'secondary' actors (Davis et al., 1941; Doreian,

中文译文 | G03-P008-002

列表，作为行动者归属中的包含模式，或两者兼而有之（Bonacich, 1978）。在此，我们关注的不是事件之间的重叠数量或行动者的共同归属情况，而是归属中的包含模式。具体而言，一些作者观察到，就参与事件的情况而言，某些行动者的归属被其他行动者的归属所包含。这便导致了"主要"与"次要"行动者之间的区分（Davis et al., 1941; Doreian）

G03-P008-003 | 英文原文

1979a,b; Freeman and White, 1993). Many of these insights have come from analyses of the data of Davis et al. (1941) on social activities of a community of Southern women. In their original analysis of these data, Davis et al. arrange the affiliation network matrix not only to highlight two 'cliques' of women, but also to indicate which women were primary and which were secondary within each 'clique'. In his reanalysis of these data using q-connectivity to locate cohesive subgroups, Doreian (1979a) finds essentially the same two subgroups that Davis et al. had found, and, moreover, he describes the internal structure of these subgroups. Doreian (1979a, pp. 224–225) comments that

中文译文 | G03-P008-003

1979a,b；Freeman和White，1993）。这些见解多<源自Davis等人在1941年对南方女性社区社会活动所进行的数据分析。在最初分析这些数据时，Davis等人不仅将二模参与网络矩阵进行整理以凸显两个"小团体"，还指出了每个"小团体"中哪些女性是核心成员，哪些属于次要成员。Doreian在利用q-连通性对这些数据进行重新分析以寻找凝聚性子群时，发现了与Davis等人完全相同的两个子群，此外，他还描述了这些子群的内部结构。Doreian（1979a，第224-225页）评论说：

G03-P008-004 | 英文原文

each of the subgroups can be viewed as having a core and a periphery, with higher connectivity within the core, lower connectivity with the periphery.

中文译文 | G03-P008-004

每个子群都可以被看作具有核心部分和边缘部分，核心内部具有更高的连通性，而与边缘部分的连通性则较低。

G03-P008-005 | 英文原文

Furthermore, Doreian identifies one of the women as a 'central core member' of one of the cliques based on her high level of connectivity to others. In their reanalysis of these data using Galois lattices (which I discuss below), Freeman and White (1993) specifically describe the nature of the internal structure of the subgroups. They observe that for secondary actors

中文译文 | G03-P008-005

此外，Doreian 根据某女性与他人的高度联系程度，将其认定为某个小团体中的“核心成员”。在利用伽罗瓦格子对这些数据进行重新分析时（我在下文中会讨论），Freeman 和 White（1993）具体描述了这些子群的内部结构特征。他们观察到，对于次要行动者而言……

G03-P008-006 | 英文原文

the events they attended are all subsets of the events attended by one or more primary actors. (p. 138.)

中文译文 | G03-P008-006

他们参与的事件都是一个或多个主要行动者参与的事件的子集。（第138页。）

G03-P008-007 | 英文原文

Freeman and White also note that events' memberships can be characterized by similar subset–superset relationships.

中文译文 | G03-P008-007

Freeman和White还指出，事件成员资格可以通过类似的子集-超集关系来描述。

G03-P008-008 | 英文原文

These observations suggest that centrality indices for affiliation networks should capture the subset–superset relationships in the affiliations of actors and events. The distinction between primary and secondary actors means that secondary actors are more likely to participate in events when primary actors are also present, and are unlikely to participate on their own. Primary actors, on the other hand, participate in events even when secondary actors are not present. The participation of secondary actors is conditional on the participation of primary actors, implying that the events to which secondary actors belong are a subset of the events to which primary actors belong (Freeman and White, 1993). In terms of centrality in affiliation networks, less central actors participate in events only in the presence of more central actors. Importantly, it is not simply the greater number of events to which more or less central actors belong, but the patterning of these events such that the events to which less central actors belong are a subset of the events to which more central actors belong (Bonacich, 1978). Similarly, one can consider the relative centrality of events using these ideas. Less central events have membership lists that are contained within the memberships of more central events.

中文译文 | G03-P008-008

这些观察结果表明，二模参与网络的中心性指标应能够反映行动者与事件成员资格中的子集-超子集关系。主要行动者与次要行动者之间的区别意味着：当主要行动者存在时，次要行动者更有可能参与事件，而单独存在时则不太可能参与。另一方面，即使次要行动者不存在，主要行动者仍会参与事件。次要行动者的参与依赖于主要行动者的参与，这意味着次要行动者所参与的事件属于主要行动者所参与的事件的子集（Freeman和White，1993）。就二模参与网络的中心性而言，中心性较低的行动者只有在中心性较高的行动者存在时才会参与事件。重要的是，并非简单地取决于中心性高低的行为者所参与的事件数量多少，而是这些事件的构成模式使得中心性较低的事件属于中心性较高的事件的子集（Bonacich，1978）。同样，可以利用这些概念来分析事件的相对中心性。中心性较低的事件其成员资格列表包含在中心性较高的事件的成员资格中。

G03-P008-009 | 英文原文（原段接续 / 分段）

Again, it is not simply the size of the events that determines their centrality, but rather it is the patterning of memberships in these events that is important.

中文译文 | G03-P008-009

再次强调，决定事件中心性并非其规模本身，而是这些事件中成员参与的模式才至关重要。

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G03-P009-002 | 英文原文

3.5. Summary

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G03-P009-003 | 英文原文

In summary, the properties of affiliation networks suggest that centralities for these networks should do four things:

中文译文 | G03-P009-003

总结来说，二模参与网络的特性表明，这些网络的中心性应具备四项特征：

G03-P009-004 | 英文原文

1. give centrality indices for both actors and events;
2. be extendable to subsets of actors and events;
3. focus on the linkages between actors and events through overlapping memberships;
4. capture subset–superset inclusion relations between actors and events.

中文译文 | G03-P009-004

1. 为行动者和事件提供中心性指数；2. 可扩展至行动者和事件的子集；3. 关注通过重叠成员身份形成的行动者与事件之间的关联；4. 捕捉行动者与事件之间子集与超集的包含关系。

G03-P009-005 | 英文原文

Analyses of affiliation networks have seldom used these ideas to study the centrality of actors and events in affiliation networks. Rather, more conventional centrality motivations for one-mode networks have been widely employed to study affiliation networks.

中文译文 | G03-P009-005

对二模参与网络的分析很少使用这些概念来研究行动者与事件在二模参与网络中的中心性。相反，用于研究单模网络的传统中心性理论被广泛应用于二模参与网络的研究中。

G03-P009-006 | 英文原文

In addition, virtually all analyses of affiliation networks study one-mode networks of actors or events derived from the original affiliation network. For example, many researchers have studied the affiliation of corporate executives with the corporate boards on which they sit. With the exception of Bonacich (1991) and Bonacich and Domhoff (1981), centrality analyses of these data always study either the board member ties of co-membership or the corporation ties of interlocking directorates, but do not study both simultaneously.

中文译文 | G03-P009-006

此外，几乎所有关于二模参与网络的分析，都研究源自原始二模参与网络的行动者或事件所构成的单模网络。例如，许多研究者研究了企业高管与其任职的董事会之间的关系。除了 Bonacich（1991）和 Bonacich与Domhoff（1981）的研究外，对这些数据的中心性分析通常只关注共同成员资格带来的董事会成员关系，或相互兼任董事制度带来的企业关联，而不会同时研究两者。

G03-P009-007 | 英文原文

Studying only a single mode from the two-mode network ignores the fundamental duality inherent in the affiliation relation. The relationship between the centrality of actors and the centrality of the events to which they belong, or the relationship between the centrality of events and the centrality of their members, cannot be studied directly by looking at the one-mode networks separately.

中文译文 | G03-P009-007

仅研究二模网络中的单模，会忽视参与关系中所固有的基本对偶性。行动者的中心性与其所涉及的事件的中心性之间的关系，或者事件的中心性与其成员的中心性之间的关系，无法通过单独研究单模网络来直接探究。

G03-P009-008 | 英文原文

In the next sections I consider five centrality indices: degree, eigenvector, closeness, betweenness, and flow betweenness. For each I describe the application of the index to affiliation network data and discuss the results with respect to the centrality motivations introduced above.

中文译文 | G03-P009-008

在后续章节中，我将探讨五个中心性指标：度、特征向量、接近性、流中介中心性以及流中介中心性。针对每个指标，我将说明该指标在二模参与网络数据中的应用，并结合上述中心性动机对结果进行讨论。

G03-P009-009 | 英文原文

4. Centralities for affiliation networks

中文译文 | G03-P009-009

4. 用于融合网络的中心性

G03-P009-010 | 英文原文

A review of analyses of centrality in affiliation networks reveals that virtually all applications use one of two approaches to centrality: (1) some form of degree centrality or (2) some form of eigenvector centrality. I begin by discussing degree and eigenvector centrality, since they have been widely used to study centrality in affiliation networks. I then discuss closeness, betweenness, and flow betweenness centralities. As we shall see, calculating basic centrality measures for both actors and events in an affiliation network is straightforward. To study both actors and events one can analyze the bipartite graph using standard centrality approaches. An important issue for each measure is to specify, if possible, the relationship between actor and event centralities.

中文译文 | G03-P009-010

对二模参与网络中中心性分析的综述表明，几乎所有研究都采用两种方法中的一种来计算中心性：(1) 度中心性或 (2) 特征向量中心性。首先，我将讨论度中心性和特征向量中心性，因为这两种方法已被广泛用于研究二模参与网络中的中心性。随后，我将探讨接近性中心性、流中介中心性以及流中介中心性。正如我们将看到的，计算二模参与网络中行动者与事件的基础中心性指标非常简单。为了研究行动者和事件，可以使用标准的中心性方法对二部图进行分析。对于每种指标而言，一个重要问题就是尽可能明确说明行动者与事件中心性之间的关系。

G03-P009-011 | 英文原文

Since centrality indices will be defined for several different kinds of graphs and networks (a one-mode dyadic network, a bipartite graph, a one-mode actor co-membership relation, or a one-mode event overlap relation) and indices will be presented for nodes in a graph, and for actors and events in affiliation networks, we shall need

中文译文 | G03-P009-011

由于不同类型的图与网络（单模双部网络、二部图、单模行动者成员资格关系或单模事件重叠关系）将定义不同的中心性指数，且这些指数将针对图中的节点以及二模参与网络中的行动者和事件进行呈现，因此我们需要

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G03-P010-001 | 英文原文

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G03-P010-002 | 英文原文

notation that reflects these distinctions. $C_A(\bullet)$ will refer to centrality definition 'A' for entity '●'. For example, $C_C(\bullet)$ will refer to closeness centrality. The entity for which the centrality is defined can be an actor, denoted by n_i, an event, denoted by m_k, or a node in a generic graph, denoted p_i. In addition, when the centrality definition applies to a specific kind of relation or matrix (for example the bipartite graph, or the actor co-membership relation) I will use the following superscripts: $\mathcal{NM}$ for the bipartite graph, $\mathcal{N}$ for the one-mode actor co-membership relation, and $\mathcal{M}$ for the one-mode event overlap relation. Thus, $C_C\mathcal{NM}(n_i)$ will refer to the closeness centrality of an actor calculated on the bipartite graph. In the absence of a superscript, the definition refers to a one-mode dyadic graph or network.

中文译文 | G03-P010-002

能够反映这些差异的标注方式。CA(Q) 指的是针对实体 O 的中心性定义 'A'。例如，Cc(O) 则指接近中心性。被定义中心性的实体可以是行动者，用 n i 表示；事件，用 m~ 表示；或者通用图中的节点，用 Pi 表示。此外，当中心性定义适用于特定类型的关系或矩阵时（例如二部图或行动者成员资格关系），我将使用以下上标：对于二部图，用 JT~" 表示；对于单模行动者成员资格关系，用 J/~ 表示；对于单模事件重叠关系，用 ~t" 表示。因此，CcH~{(ni)} 指的是在二部图上计算得到的行动者的接近中心性。如果没有上标，则指单模双边图或网络。

G03-P010-003 | 英文原文

4.1. Degree centralities

中文译文 | G03-P010-003

4.1. 度中心性

G03-P010-004 | 英文原文

Degree centrality, denoted by $C_D(p_i)$, is one of the most straightforward centrality indices. In an affiliation network motivations for degree centrality are that actors are important because of their level of activity or the number of contacts that they have, and events are important because of the size of their memberships (Stokman, 1977; Mizruchi and Bunting, 1981; Mariolis and Jones, 1982; Fernandez and McAdam, 1988).

中文译文 | G03-P010-004

度中心性，用 CD(Pi) 表示，是最直观的中心性指标之一。在二模参与网络中，采用度中心性的原因在于：行动者因其活动水平或联系数量而具有重要性，事件则因其成员规模而具有重要性（Stokman, 1977; Mizruchi and Bunting, 1981; Mariolis and Jones, 1982; Fernandez and McAdam, 1988）。

G03-P010-005 | 英文原文

In a graph for a single dichotomous relation, the degree of a node is the number of nodes adjacent to it. For a nondirectional relation the degree of a node is equal to the sum of the values in the row (or column) of the sociomatrix; $C_D(p_i) = \sum_{j=1}^{g} x_{ij}$. In an affiliation network there are several ways to calculate degree centrality depending on whether one focuses on the bipartite graph (including both actors and events) or on the one-mode networks of actor co-memberships or event overlaps. Interpretations of the results will differ.

中文译文 | G03-P010-005

在表示单重二分关系的图中，节点的度等于与其相邻的节点数量。对于非方向性关系，节点的度等于社会矩阵中对应行（或列）数值的总和：Co(Pi) = E g I Xij。在二模参与网络中，根据是关注二部图（包括行动者和事件）还是关注单模网络中的行动者联合成员关系或事件重叠情况，有几种不同的方法来计算度中心性。对结果的解释也会有所不同。

G03-P010-006 | 英文原文

First, focus on the bipartite graph in which both actors and events are represented as nodes and lines represent affiliation ties between actors and events. In the bipartite graph the degree centrality of an actor is the number of events with which it is affiliated and the degree centrality of an event is the number of actors affiliated with it. These quantities are also equal to the entries on the main diagonal of the actor co-membership matrix $\boldsymbol{X}^{\mathcal{N}}$ (for actors) or the event overlap matrix $\boldsymbol{X}^{\mathcal{M}}$ (for events), or to the row total (for actors) or the column total (for events) of the affiliation matrix $\boldsymbol{A}$. Thus, actor and event degree centralities in the bipartite graph are:

中文译文 | G03-P010-006

首先，关注那个二部图，其中行动者和事件都作为节点表示，连线则代表行动者与事件之间的关联关系。在二部图中，行动者的度中心性是指与其关联的事件数量，而事件的度中心性则是指与它相关联的行动者数量。这些数值也等于行动者共同成员矩阵 X yr（针对行动者）或事件重叠矩阵 X ~r（针对事件）主对角线上的值，或者等于关联矩阵 A 的行总和（针对行动者）或列总和（针对事件）。因此，二部图中的行动者与事件的度中心性分别为：

G03-P010-007 | 英文原文

$$C_D^{\mathcal{NM}}(n_i) = \sum_{k=1}^{\cdot} x_{ik}^{\mathcal{NM}} = x_{ii}^{\mathcal{N}} = a_{i+} \qquad (4)$$

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G03-P010-008 | 英文原文

$$C_D^{\mathcal{NM}}(m_k) = \sum^{g+h} x_{ik}^{\mathcal{NM}} = x_{kk}^{\mathcal{M}} = a_{+k}.$$

中文说明（原文编号 / 符号 / 数值保留） | G03-P010-008

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G03-P010-009 | 英文原文（原段接续 / 分段）

$$\sum_{i=1} x_{ik}^{\mathcal{NM}} = x_{kk}^{\mathcal{M}} = a_{+k}. \qquad (5)$$

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G03-P010-010 | 英文原文

Although degree centrality is usually defined for dichotomous relations, there are also a number of ways to define degree-like centralities for the one-mode valued relations of actor co-memberships and event overlaps. Both of these are valued, nondirectional relations where the values indicate the number of co-memberships for pairs of actors or the number of overlaps for pairs of events.

中文译文 | G03-P010-010

虽然度中心性通常用于二元关系，但也有多种方法可以定义单模值关系——即行动者成员关系与事件重叠关系中的类度中心性。这两种都属于有值的非方向性关系，其数值表示行动者对之间的成员关系数量或事件对之间的重叠数量。

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G03-P011-001 | 英文原文

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G03-P011-002 | 英文原文

Consider the entries in a given row (or column) of the actor co-membership matrix, $\boldsymbol{X}^{\mathcal{N}}$. The number of non-zero off-diagonal entries in the row gives the number of distinct actors with which the actor shares any event in common. However, we can also define other possible degree-like measures for this relation. Suppose we take the sum of the entries in a row. For actors in the co-membership relation this degree centrality is:

中文译文 | G03-P011-002

考虑行动者联合成员资格矩阵中某一行（或列）中的元素 X e"。该行中非零非对角线元素的个数，即表示与该行动者共同参与过某事件的不同行动者的数量。不过，我们也可以为这种关系定义其他类似度中心性的度量方式。假设我们取该行的元素之和。对于处于联合成员资格关系中的行动者而言，其度中心性为：

G03-P011-003 | 英文原文

$$C_D^{\mathcal{N}}(n_i) = \sum_{j=1}^{\tilde{}} x_{ij}^{\mathcal{N}} = x_{i+}^{\mathcal{N}}. \qquad (6)$$

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G03-P011-004 | 英文原文

Interestingly, this quantity is equal to the sum of the sizes of the events to which the actor belongs:

中文译文 | G03-P011-004

有趣的是，这个数值等于该行动者所参与事件的大小之和：

G03-P011-005 | 英文原文

$n_i \in m_k \qquad n_i \in m_k$

中文译文 | G03-P011-005

ni~m k ni~m k

G03-P011-006 | 英文原文

We can think of $C_D^{\mathcal{N}}(n_i)$ as the actor's total 'volume' of activity. It is the number of contacts that an actor has with other actors, counting other actors each time they are encountered.

中文译文 | G03-P011-006

我们可以将C~(n i)视为行动者活动的"总量"。它表示行动者与其他行动者之间的接触次数，每次遇到其他行动者时都计入一次。

G03-P011-007 | 英文原文

For events, the similar degree centrality is equal to

中文译文 | G03-P011-007

对于事件而言，类似的度中心性等于

G03-P011-008 | 英文原文

This quantity can be thought of as the total participation of the actors that belong to the event, and is equal to the sum of the number of memberships for all actors in the event:

中文译文 | G03-P011-008

这一数值可视为参与该事件的行动者总数，等于该事件中所有行动者会员数量之和。

G03-P011-009 | 英文原文

$$C_D^{\mathcal{M}}(m_k) = \sum x_{ii}^{\mathcal{N}} = \sum a_{i+}.$$

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G03-P011-010 | 英文原文（原段接续 / 分段）

(9)

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G03-P011-011 | 英文原文

$n_i \in m_k \qquad n_i \in m_k$

中文译文 | G03-P011-011

t~i~m k niEm k

G03-P011-012 | 英文原文

If these centralities are calculated including the diagonal elements of $X^{\mathcal{N}}$ or $X^{\mathcal{M}}$, then they are also equal to the number of two-step walks from the actor or event node in the bipartite graph.

中文译文 | G03-P011-012

如果包含 X :r 或 X ~e 的对角元素来计算这些中心性，那么它们也等于二部图中从行动者或事件节点出发的两步路径数量。

G03-P011-013 | 英文原文

Taken together, Eqs. (7) and (9) show that for the one-mode networks of actor co-memberships and event overlaps there is a specifiable relationship between an actor's centrality and the centrality of the events to which it belongs, and between an event's centrality and the centrality of its members:

中文译文 | G03-P011-013

综合来看，方程（7）和（9）表明，对于单模网络中的行动者成员关系与事件重叠情况，行动者的中心性与其所属于事件的中心性之间存在可确定的关系；同时，事件的中心性与其成员的中心性也存在这种关系。

G03-P011-014 | 英文原文

$$C_D^{\mathcal{N}}(n_i) = \sum C_D^{\mathcal{NM}}(m_k) \qquad (10)$$

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G03-P011-015 | 英文原文

$n_i \in m_k$

中文译文 | G03-P011-015

niC m k

G03-P011-016 | 英文原文

$$C_D^{\mathcal{M}}(m_k) = \sum C_D^{\mathcal{NM}}(n_i).$$

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G03-P011-017 | 英文原文（原段接续 / 分段）

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G03-P011-018 | 英文原文

$n_i \in m_k$

中文译文 | G03-P011-018

niEm k

G03-P011-019 | 英文原文

The degree centrality of an actor is equal to the sum of the sizes of its events (the degrees of these events in the bipartite graph) and the degree centrality of an event is equal to the sum of the number of memberships of its actors (the degrees of these actors in the bipartite graph).

中文译文 | G03-P011-019

行动者的度中心性等于其事件规模之和（这些事件在二部图中的度），而事件的度中心性则等于其行动者成员数之和（这些行动者在二部图中的度）。

G03-P011-020 | 英文原文

Degree centrality indices for actors and events in the bipartite graph for the hypothetical example are presented in Table 4. Table 5 presents the degree centralities

中文译文 | G03-P011-020

假设示例中二部图中各行动者与事件의度中心性指标见表4。表5则展示了度中心性数值。

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G03-P012-001 | 英文原文

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G03-P012-002 | 英文原文

Table 4
Centrality of

中文译文 | G03-P012-002

表4 中心性

G03-P012-003 | 英文原文

actors and events in the bipartite graph

中文译文 | G03-P012-003

二部图中行动者与事件

G03-P012-004 | 英文原文

Measure of centrality

中文译文 | G03-P012-004

中心性度量

G03-P012-005 | 英文原文

$C_D^{\mathcal{N}\mathcal{M}}$ $\sum d(i,k)$ $C_C^{\mathcal{N}\mathcal{M}}$ $C_B^{\mathcal{N}\mathcal{M}}$ $C_F^{\mathcal{N}\mathcal{M}}$ $C_E^{\mathcal{N}\mathcal{M}}$

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G03-P012-006 | 英文原文

Table 5
Centrality measures for actors and events in one-mode networks

中文译文 | G03-P012-006

表5 单模网络中行动者与事件的中心性度量

G03-P012-007 | 英文原文

Measure of centrality

中文译文 | G03-P012-007

中心性度量指标

G03-P012-008 | 英文原文

$C_D^{\mathcal{N}a}$	$C_D^{\mathcal{N}b}$	$C_B^{\mathcal{N}01}$	$C_F^{\mathcal{N}}$	$C_E^{\mathcal{N}}$	$C_D^{\mathcal{M}a}$	$C_D^{\mathcal{M}b}$	$C_B^{\mathcal{M}01}$	$C_F^{\mathcal{M}}$	$C_E^{\mathcal{M}}$

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G03-P012-009 | 英文原文

[a]Excludes diagonal. [b]Includes diagonal.

中文译文 | G03-P012-009

a排除对角线，包含对角线。

G03-P012-010 | 英文原文

for the one-mode networks of actor co-memberships and event overlaps, calculated both with diagonal entries included and with diagonal entries excluded. [1]

中文译文 | G03-P012-010

对于行动者成员资格与事件重叠的单模网络，既包含对角线项进行计算，也排除对角线项进行计算。1

G03-P012-011 | 英文原文

A number of authors have used versions of degree centrality to study centrality in affiliation networks. Romney and Weller (1984) use this index to measure the overall level of people's observed behavioral interaction in data from three groups studied by Bernard et al. (1980, Bernard et al., 1982). In their study of corporate interlock networks, Mariolis and Jones (1982) use this index as one measure of a firm's centrality, referring to it as 'the number of interlocks'. Similarly, in his study of delegations' co-sponsorship of bills within the United Nations, Stokman (1977) uses the number of co-sponsorship ties that a delegation has as an index of the delegation's centrality in the co-sponsorship network.

中文译文 | G03-P012-011

许多学者使用度中心性这一指标来研究二模参与网络中的中心性现象。Romney和Weller（1984）利用该指标来衡量Bernard等人在1980年和1982年所研究的三个群体中人们行为互动的整体水平。在对企业关联网络的研究中，Mariolis和Jones（1982）将这一指标作为衡量企业中心性的一种方式，称之为"关联数量"。同样，在对联合国内代表团共同提案行为的研究中，Stokman（1977）将代表团拥有的共同提案关系数量作为该代表团在共同提案网络中的中心性指标。

G03-P012-012 | 英文原文

Some authors have criticized degree centrality because it does not consider the centrality of the actors (or events) to which an actor (or event) is adjacent. According to

中文译文 | G03-P012-012

一些学者批评度中心性，因为该指标未考虑与行动者（或事件）相邻的行动者（或事件）的中心性。根据……

G03-P012-013 | 英文原文

[1] Centrality indices were calculated using *UCINET IV* (Borgatti et al., 1992).

中文译文 | G03-P012-013

中心性指标是通过UCINET IV软件计算得出的（Borgatti等，1992）。

原PDF第13页（原刊第169页）

G03-P013-002 | 英文原文

this argument, two actors may be adjacent to the same number of others, but an actor is more central if it has ties to actors that themselves are quite central. One way to deal with this issue is to incorporate the centrality of the actors to which a given actor is adjacent into the centrality index. This is what eigenvector centrality does.

中文译文 | G03-P013-002

根据这一论点，两个行动者可能与相同数量的其他行动者相邻，但如果一个行动者与本身具有较高中心性的行动者存在关联，则该行动者更具中心性。解决这一问题的一种方法是将与特定行动者相邻的行动者的中心性纳入中心性指数中。这正是特征向量中心性的作用。

G03-P013-003 | 英文原文

4.2. Eigenvector centrality

中文译文 | G03-P013-003

4.2. 特征向量中心性

G03-P013-004 | 英文原文

The rationale for eigenvector centrality in a one-mode network is that the centrality of an actor should be proportional to the strength of the actor's ties to other network members and the centrality of these other actors (Bonacich, 1972; Mizruchi, 1982; Mizruchi et al., 1986). Originally, Bonacich (1972) motivated this centrality index as a measure of popularity, related to the measures of relative standing or status proposed by Katz (1953) and Hubbell (1965). Recently, people have defended the use of this index as a measure of the extent to which actors are in a position to influence others in the network (Mizruchi and Bunting, 1981; Fernandez and McAdam, 1988; Friedkin, 1991). Friedkin (1991) describes this index as 'total effects centrality' in the context of network influence models.

中文译文 | G03-P013-004

单模网络中特征向量中心性的原理是：行动者的中心性应与该行动者与网络其他成员的联系强度以及这些其他行动者的中心性成正比（Bonacich，1972；Mizruchi，1982；Mizruchi等，1986）。最初，Bonacich（1972）提出这一中心性指标作为受欢迎程度的衡量标准，该指标与Katz（1953）和Hubbell（1965）提出的相对地位或身份度量相关。近期，人们支持将这一指标作为衡量行动者在多大程度上能够影响网络中的其他行动者的度量标准（Mizruchi和Bunting，1981；Fernandez和McAdam，1988；Friedkin，1991）。Friedkin（1991）在网络影响力模型的背景下将这一指标称为"总效应中心性"。

G03-P013-005 | 英文原文

Eigenvector centrality is widely used in studies of interlocking corporate boards of directors (Mariolis, 1975; Mintz and Schwartz, 1981a,b; Mizruchi and Bunting, 1981; Mariolis and Jones, 1982; Mizruchi, 1982; Roy, 1983; Rosenthal et al., 1985). For many of these researchers centrality *is* eigenvector centrality (Mariolis, 1975; Mintz and Schwartz, 1981a,b; Roy, 1983; Rosenthal et al., 1985).

中文译文 | G03-P013-005

特征向量中心性被广泛用于研究相互关联的董事会结构的研究中（Mariolis, 1975；Mintz和Schwartz, 1981 a,b；Mizruchi和Bunting, 1981；Mariolis和Jones, 1982；Mizruchi, 1982；Roy, 1983；Rosenthal等, 1985）。对于这些研究者中的许多来说，中心性即特征向量中心性（Mariolis, 1975；Mintz和Schwartz, 1981a,b；Roy, 1983；Rosenthal等, 1985）。

G03-P013-006 | 英文原文

In this section I review eigenvector centrality in general and then show how the centrality indices for actors are related to the centrality indices for events in an affiliation network.

中文译文 | G03-P013-006

在本节中，我将总体回顾特征向量中心性，然后说明二模参与网络中行动者的中心性指标与事件的中心性指标之间的关系。

G03-P013-007 | 英文原文

Denoting the eigenvector centrality of node p_i in a one-mode network by $C_E(p_i)$, eigenvector centrality is expressed as

中文译文 | G03-P013-007

将单模网络中节点Pi的特征向量中心性表示为CE(pi)，则特征向量中心性可表示为

G03-P013-008 | 英文原文

$$C_E(p_i) \approx C_E(p_j)\, x_{ij}.$$

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G03-P013-009 | 英文原文（原段接续 / 分段）

$$(12)$$

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G03-P013-010 | 英文原文

The centrality of a node is proportional to the centrality of the nodes to which it is adjacent, weighted by the value of the tie between the nodes.

中文译文 | G03-P013-010

节点的中心性与其相邻节点的中心性成正比，并以这些节点之间的关联值作为权重。

G03-P013-011 | 英文原文

Finding centrality values, $C_E(p_i)$, that satisfy this equation for all nodes in a graph involves solving a system of simultaneous linear equations. This standard eigenvector–eigenvalue problem is expressed by the equation

中文译文 | G03-P013-011

为图中的所有节点找到满足该方程的中心性值 CE(p)，需要解一组同时成立的线性方程组。这一标准的特征向量-特征值问题由以下方程表示

G03-P013-012 | 英文原文

$$\boldsymbol{X}\boldsymbol{c} = \lambda \boldsymbol{c} \qquad (13)$$

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G03-P013-013 | 英文原文

where $\boldsymbol{X}$ is a $g \times g$ sociomatrix, λ is its largest eigenvalue, and $\boldsymbol{c}$ is a vector of centrality scores (the eigenvector corresponding to the largest eigenvalue).

中文译文 | G03-P013-013

其中 X 是一个 g × g 社会矩阵，A 是其最大特征值，c 是中心性得分向量（对应最大特征值的特征向量）。

G03-P013-014 | 英文原文

I will denote the vector of eigenvector centrality scores for actors by $c^{\mathcal{N}}$ and the vector of eigenvector centrality scores for events by $c^{\mathcal{M}}$. One can find these scores by analyzing the one-mode actor co-membership matrix and the one-mode event overlap matrix:

中文译文 | G03-P013-014

我将行动者的特征向量中心性得分向量记为 c ~r，将事件的特征向量中心性得分向量记为 c ~'. 通过分析单模行动者成员资格矩阵和单模事件重叠矩阵，即可得到这些得分。

G03-P013-015 | 英文原文

$$\boldsymbol{X}^{\mathcal{N}}\boldsymbol{c}^{\mathcal{N}} = \boldsymbol{A}\boldsymbol{A}'\boldsymbol{c}^{\mathcal{N}} = \lambda^2\boldsymbol{c}^{\mathcal{N}} \qquad (14)$$

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G03-P013-016 | 英文原文

$$\boldsymbol{X}^{\mathcal{M}}\boldsymbol{c}^{\mathcal{M}} = \boldsymbol{A}'\boldsymbol{A}\boldsymbol{c}^{\mathcal{M}} = \lambda^2\boldsymbol{c}^{\mathcal{M}}.$$

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G03-P014-001 | 英文原文

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G03-P014-002 | 英文原文

Bonacich suggests that the scores be scaled so that $\sum C_E^{\mathcal{N}}(n_i)^2 = \sum C_E^{\mathcal{M}}(m_k)^2 = 1$, though other scalings are possible. The actor and event eigenvector centralities are related to each other through the set of equations:

中文译文 | G03-P014-002

博纳奇赫建议对分数进行缩放，使得 EC~(ni) 2 = ECE~(mk) 2 = 1，不过也可以采用其他缩放方式。行动者与事件特征向量中心性通过一组方程相互关联：

G03-P014-003 | 英文原文

$$\boldsymbol{A}\boldsymbol{c}^{\mathcal{N}} = \lambda \boldsymbol{c}^{\mathcal{M}} \qquad (16)$$

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G03-P014-004 | 英文原文

$$\boldsymbol{A}'\boldsymbol{c}^{\mathcal{M}} = \lambda \boldsymbol{c}^{\mathcal{N}}.$$

中文说明（原文编号 / 符号 / 数值保留） | G03-P014-004

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G03-P014-005 | 英文原文（原段接续 / 分段）

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G03-P014-006 | 英文原文

Eqs. (16) and (17) show the duality of actor and event centralities for the affiliation network. The centrality of an actor is proportional to the centralities of the events with which it is affiliated, and the centrality of an event is proportional to the centralities of its members.

中文译文 | G03-P014-006

方程（16）和（17）表明，在二模参与网络中，行动者与事件的中心性具有对偶性。一个行动者的中心性与其所关联的事件的中心性成正比，而一个事件的中心性则与其成员的中心性成正比。

G03-P014-007 | 英文原文

Bonacich (1991) shows that for an affiliation network one can find the actor and event centrality scores by analyzing a number of different matrices. Specifically, one can analyze the affiliation network matrix, $\boldsymbol{A}$, the pair of one-mode matrices, $\boldsymbol{X}^{\mathcal{N}}$ and $\boldsymbol{X}^{\mathcal{M}}$ (Eqs. (14) and (15)), or the matrix for the bipartite graph, $\boldsymbol{X}^{\mathcal{N}\mathcal{M}}$. Analysis of the bipartite graph is expressed in the equation:

中文译文 | G03-P014-007

Bonacich (1991) 表明，对于二模参与网络，可以通过分析多种不同的矩阵来得出行动者和事件中心性得分。具体来说，可以分析二模参与网络矩阵 A、双单模矩阵 X Jr 和 X '~r（公式 (14) 和 (15)），或者二部图的矩阵 X ~r~。对二部图的分析可用以下方程表示：

G03-P014-008 | 英文原文

(Bonacich, 1991, p. 158). The generality of this approach is apparent in the fact that the eigenvector centrality scores from any of these analyses are equivalent, once they are identically scaled.

中文译文 | G03-P014-008

（Bonacich，1991，第158页）。这种方法的普遍性在于，一旦进行相同的缩放处理，任何这种分析得出的特征向量中心性得分都是相等的。

G03-P014-009 | 英文原文

For eigenvector centralities there is a clear and specifiable relationship between the centralities for actors and the centralities for events. This relationship is more apparent if we rewrite Eqs. (16) and (17) in terms of individual actor and event centrality indices. An actor's centrality is a function of the centralities of the events to which it belongs:

中文译文 | G03-P014-009

对于特征向量中心性而言，行动者的中心性与事件的中心性之间存在明确且可确定的关系。如果我们用个体行动者与事件中心性指数来重新表述方程（16）和（17），这种关系就会更加明显。一个行动者的中心性取决于其所属事件的中心性：

G03-P014-010 | 英文原文

$$C_E^{\mathcal{N}}(n_i) = \frac{1}{\lambda} \sum^{h} C_E^{\mathcal{M}}(m_k)\, a_{ik}.$$

中文说明（原文编号 / 符号 / 数值保留） | G03-P014-010

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G03-P014-011 | 英文原文（原段接续 / 分段）

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G03-P014-012 | 英文原文

Similarly, the centrality of an event is a function of the centralities of its members:

中文译文 | G03-P014-012

同样，事件的中心性是其成员中心性的函数：

G03-P014-013 | 英文原文

$$C_E^{\mathcal{M}}(m_k) = \frac{1}{\lambda} \sum^{g} C_E^{\mathcal{N}}(n_i)\, a_{ik}.$$

中文说明（原文编号 / 符号 / 数值保留） | G03-P014-013

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G03-P014-014 | 英文原文（原段接续 / 分段）

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G03-P014-015 | 英文原文

Eigenvector centrality explicitly incorporates the duality between actor and event centralities (Bonacich, 1991). This duality is expressed in the pairs of Eqs. (16) and (17) or Eqs. (19) and (20) in which actors' centralities are proportional to the centralities of the events to which they belong, and events' centralities are proportional to the centralities of their members.

中文译文 | G03-P014-015

特征向量中心性明确体现了行动者与事件中心性之间的对偶关系（Bonacich，1991）。这种对偶关系体现在方程(16)和(17)或方程(19)和(20)中，其中行动者的中心性与其所属事件的中心性成正比，而事件的中心性与其成员的中心性成正比。

G03-P014-016 | 英文原文

A similar relationship between actor and event scores results from a correspondence analysis of the affiliation matrix, A (Bonacich, 1991; Wasserman and Faust, 1994). One of the goals of correspondence analysis is to assign scores to the rows and columns of a data array so that a row's score is proportional to the weighted average of the column scores and a column's score is proportional to the weighted average of the row scores; the weights are the relative frequencies in the columns or rows, respectively. For an affiliation network, an actor's score is proportional to the average of the scores for the events with which it is affiliated and an event's score is proportional to the average of

中文译文 | G03-P014-016

行动者与事件得分之间的类似关系源于二模参与网络的对应分析（Bonacich, 1991; Wasserman和Faust, 1994）。对应分析的一个目标是对数据数组的行和列分配分数，使得一行得分与列分数的加权平均值成正比，一列得分与行分数的加权平均值成正比；权重分别为该列或行的相对频率。对于二模参与网络，行动者的得分与其参与的事件得分的平均值成正比，而事件的得分则与它所参与的事件的平均得分成正比。

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G03-P015-002 | 英文原文

the scores for the actors that belong to the event. Correspondence analysis of an affiliation matrix, A, results in three sets of information:

中文译文 | G03-P015-002

属于该事件的行动者的得分。对关联矩阵A进行对应分析后，会得到三组信息：

G03-P015-003 | 英文原文

- a set of g actor scores on each of W dimensions $\{u_{iw}\}$ for $i = 1,2,\ldots,g$ and

中文译文 | G03-P015-003

• 对于第 i = 1,2 ..... g 的每一个维度 {Ug w}，有一组 g 个行动者的得分

G03-P015-004 | 英文原文

$w = 1,2,\ldots,W$;

- a set of h event scores on each of W dimensions $\{v_{kw}\}$ for $k = 1,2,\ldots,h$ and $w = 1,2,\ldots,W$, and
- a set of W principal inertias $\{\eta_w^2\}$ for $w = 1,2,\ldots,W$.

Correspondence analysis scores for rows (actors) and columns (events) are related to each other through the following equations:

中文译文 | G03-P015-004

w = 1,2,...,W；对于k= 1,2 ..... h，每个维度{v~w}上的h个事件得分集合；且w = 1,2 ..... W时，有W个主惯性量{r/~2~}的集合。行（行动者）与列（事件）的对应分析得分通过以下方程相互关联：

G03-P015-005 | 英文原文

$$\eta_w u_{iw} = \sum_{k=1}^{h} \frac{a_{ik}}{a_{i+}} v_{kw} \qquad (21)$$

$g \quad a_{\cdot\cdot}$

中文译文 | G03-P015-005

h TIwUiw E aik -- Pkw k= 1 ai+ (21) g

G03-P015-006 | 英文原文

$$\eta_w v_{kw} = \sum^{g} \frac{a_{ik}}{a} u_{iw}.$$

中文译文 | G03-P015-006

aik ~w Pkw E Uiw.

G03-P015-007 | 英文原文（原段接续 / 分段）

$\overline{i=1}\ a_{+k}$

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G03-P015-008 | 英文原文

These row and column scores are found through a singular value decomposition of an appropriately scaled affiliation matrix: $\tilde{A} = \{a_{ik}/\sqrt{a_{i+}a_{+k}}\}$. Although the relationship

中文译文 | G03-P015-008

这些行与列得分是通过对适当缩放的关联网络矩阵进行奇异值分解得到的：P~= {ai~ / ~ } 。

G03-P015-009 | 英文原文（原段接续 / 分段）

appropriately scaled affiliation matrix: $\tilde{A} = \{a_{ik}/\sqrt{a_{i+}a_{+k}}\}$. Although the relationship between actor and event correspondence analysis scores, Eq. (21), appears similar to the relationship between actor and event eigenvector centrality scores, Eqs. (19) and (20), the resulting scores can be quite different. This difference is due to the fact that eigenvector centralities result from a decomposition of A whereas correspondence analysis scores result from a decomposition of $\tilde{A}$.

中文译文 | G03-P015-009

尽管方程（21）中行动者与事件对应分析得分与方程（19）和（20）中行动者与特征向量中心性得分之间的关系相似，但最终的得分却可能大相径庭。这种差异源于特征向量中心性是由矩阵A的分解得出的结果，而对应分析得分则是由矩阵A的分解得出的结果。

G03-P015-010 | 英文原文

The eigenvector centralities for actors and events in the hypothetical example are presented in Table 4 (for the bipartite graph) and Table 5 (for the one-mode relations of actor co-memberships and event overlaps). To illustrate the duality of actor and event eigenvector centralities, consider the eigenvector centrality of n_1. For the one-mode actor co-membership relation, $\lambda = 2.775$ and $C_E^{\mathcal{N}}(n_1) = 0.404$. Actor n_1 is affiliated with events m_1 and m_3, with centralities in the event overlap relation of $C_E^{\mathcal{M}}(m_1) = 0.515$ and $C_E^{\mathcal{M}}(m_3) = 0.606$. Thus, n_1's centrality is equal to

中文译文 | G03-P015-010

假设示例中行动者与事件的特征向量中心性分别见表4（针对二部图）和表5（针对单模行动者成员资格与事件重叠关系）。为了说明行动者与事件特征向量中心性的对偶性，以 n~ 的特征向量中心性为例。对于单模行动者成员资格关系，h = 2.775，C~r(nl)= 0.404。行动者 nl 与事件 m1 和 m3 相关，其在事件重叠关系中的中心性分别为 C~(m~) = 0.515 和 CEae(m3) = 0.606。因此，nl 的中心性等于

G03-P015-011 | 英文原文

A number of people have argued that the magnitudes of eigenvector centralities for events in the event overlap relation are undesirably affected by differences in the sizes of events (Mariolis, 1975; Mizruchi and Bunting, 1981; Mariolis and Jones, 1982; Mizruchi, 1982; Bonacich, 1991). Thus, they argue, steps should be taken to remove this effect. Two approaches have been proposed: (1) standardize the event overlap measure $x_{kl}^{\mathcal{M}}$ prior to analysis, or (2) remove from the centrality index that component which is

中文译文 | G03-P015-011

许多人认为，事件重叠关系中事件的特征向量中心性数值不希望受到事件规模差异的影响（Mariolis, 1975; Mizruchi and Bunting, 1981; Mariolis and Jones, 1982; Mizruchi, 1982; Bonacich, 1991）。因此，他们认为应采取措施消除这种影响。已有两种方法被提出：(1) 在分析之前对事件重叠度量 x~ 进行标准化；或 (2) 从中心性指数中剔除该成分。

G03-P015-012 | 英文原文

due to the degree (i.e. the size) of the event.

中文译文 | G03-P015-012

由于事件的度（即规模）的影响。

G03-P015-013 | 英文原文

The first way to accommodate different event sizes is to standardize the entries in the event overlap matrix, $\boldsymbol{X}^{\mathcal{M}}$, prior to finding eigenvector centralities. Mariolis (1975) suggests that standardized values $\boldsymbol{X}^{*\mathcal{M}} = \{x_{kl}^{*\mathcal{M}}\}$ be defined as:

中文译文 | G03-P015-013

容纳不同事件规模的第一种方法是在寻找特征向量中心性之前，对事件重叠矩阵中的条目进行标准化处理，即 X ~e。Mariolis（1975）建议，标准化值 X * ~ = {x~(} 可定义为：

G03-P015-014 | 英文原文

$$\overset{*\mathcal{M}}{kl} = \frac{x^{\mathcal{M}}_{kl}}{\sqrt{\quad\quad}} . \qquad (23)$$

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G03-P016-002 | 英文原文

Since $x^{\mathcal{M}}_{kk}$ and $x^{\mathcal{M}}_{ll}$ are the sizes of events m_k and m_l respectively, this standardization takes event sizes into account by dividing the measure of overlap by the square root of the product of the event sizes. However, Bonacich (1991) demonstrates that this standardization fails to remove size differences. Moreover, the event eigenvector centralities from this standardized overlap measure will not be identical to event eigenvector centralities for the bipartite graph, unless entries in $\boldsymbol{X}^{\mathcal{NM}}$ are also appropriately standardized. The appropriate standardization defines a new affiliation network matrix $\boldsymbol{A}^* = \{a^*_{ik}\}$ where $a^*_{ik} = a_{ik}/\sqrt{a_{+k}}$. The new sociomatrix for the bipartite graph, $\boldsymbol{X}^{*\mathcal{NM}}$, then has the form

中文译文 | G03-P016-002

由于 x~ 和 xff 分别是事件 m k 和 m l 的大小，这种标准化方法通过将重叠度量除以事件大小乘积的平方根来考虑事件大小。然而，Bonacich（1991）指出，这种标准化无法消除大小差异。此外，除非 X~¢~ 中的条目也进行适当标准化，否则这种标准化的重叠度量所得到的事件特征向量中心性将不会与二部图中的事件特征向量中心性相同。适当的标准化定义了一个新的二模参与网络矩阵 A * = {a/k}，其中 ai~ = aik / v"a+k。那么，二部图的新的社会矩阵 X * ~'g 就具有如下形式

G03-P016-003 | 英文原文

The event eigenvector centralities for $\boldsymbol{X}^{*\mathcal{NM}}$ will be identical to the event eigenvector centralities for $\boldsymbol{X}^{*\mathcal{M}}$, once they are scaled as described above. However, the actor eigenvector centralities from $\boldsymbol{X}^{*\mathcal{NM}}$ will *not* be identical to actor eigenvector centralities from $\boldsymbol{X}^{\mathcal{N}}$. Thus, by standardizing $\boldsymbol{X}^{\mathcal{M}}$ to 'remove' differences in event sizes, one loses the important duality between actor and event centralities.

中文译文 | G03-P016-003

按照上述方式进行缩放后，X * ~~"的事件特征向量中心性将与X * ~'的事件特征向量中心性相同。然而，X * ~A中的行动者特征向量中心性将不会与X x中的行动者特征向量中心性相同。因此，通过将X ~"标准化以“消除”事件规模差异，就会失去行动者与事件中心性之间的关键对偶性。

G03-P016-004 | 英文原文

A second way to deal with differences in event sizes exploits information about the components of the eigenvector centrality index. Bonacich (1991) shows that this index is the sum of a weighted series of components, each of which quantifies the contribution to the overall centrality of the node due to paths of length 1,2,...,∞ from the node. One can remove the effect of event sizes by 'removing' the component of centrality due to paths of length 1 (the size of the event). The advantage of this approach is that one can analyze the bipartite graph and obtain both actor and event centrality scores, thus maintaining the duality between actor and event centralities.

中文译文 | G03-P016-004

处理事件规模差异的第二种方法是利用关于特征向量中心性指数各组成部分的信息。Bonacich（1991）指出，该指数是若干加权成分的总和，每一成分都反映了从节点出发长度为1、2……~的路径对节点整体中心性的贡献。通过"移除"由长度为1的路径产生的中心性成分，就可以消除事件规模的影响。这种方法的优势在于，可以分析二部图并获取行动者与事件的中心性得分，从而保持行动者与事件中心性之间的对偶性。

G03-P016-005 | 英文原文

One potentially undesirable feature of eigenvector centralities for studying affiliation networks is that actors that belong to a single event (and thus create no ties between events) and events that have only a single member (and thus create no ties between actors) can have non-zero eigenvector centrality. For example, in the hypothetical affiliation network, n_2, which belongs to only one event, has eigenvector centrality of 0.154. This is due to the centrality that actors and events derive from their ties to others that are central. Non-zero centrality is inconsistent with the theoretical argument that actors in affiliation networks are central to the extent that they link events, and events are central to the extent that they link actors. However, it is consistent with the 'status' or 'popularity' motivation for this index that central actors derive their centrality through their ties to central actors.

中文译文 | G03-P016-005

研究二模参与网络时，特征向量中心性的一个潜在不理想特性是：属于单一事件的行动者（因此无法在事件之间建立联系），以及仅有一个成员的事件（因此也无法在行动者之间建立联系），其特征向量中心性可能不为零。例如，在假设性的二模参与网络中，仅属于一个事件的 n 2 具有 0.154 的特征向量中心性。这源于行动者与事件因与核心他人的联系而获得的中心性。非零中心性与理论论点相矛盾，该论点认为行动者在二模参与网络中的中心性在于它们连接了事件，而事件的中心性则在于它们连接了行动者。然而，这种特性与该指数的"地位"或"受欢迎程度"动机相一致，即核心行动者通过其与核心行动者的联系而获得中心性。

G03-P016-006 | 英文原文

Both degree centrality and eigenvector centrality have been widely used to study affiliation networks. However, only Bonacich and Domhoff (1981) and Bonacich (1991) consider both actor and event centralities and the relationship between them. In the next sections I discuss centralities that were not designed to study affiliation networks and have not been used to analyze these networks.

中文译文 | G03-P016-006

度中心性和特征向量中心性都被广泛用于研究二模参与网络。然而，只有 Bonacich 和 Domhoff (1981) 以及 Bonacich (1991) 考虑了行动者与事件中心性及其之间的关系。在后续章节中，我将讨论那些并非为研究二模参与网络而设计、也未被用于分析这些网络的中心性。

G03-P016-007 | 英文原文

4.3. Closeness centrality

中文译文 | G03-P016-007

4.3. 接近中心性

G03-P016-008 | 英文原文

Closeness centrality is based on the geodesic distances between nodes in a graph, and is not applicable to valued relations. Thus, I will restrict my attention to closeness

中文译文 | G03-P016-008

接近中心性基于图中节点之间的测地距离，不适用于具有价值的关系。因此，我将把注意力集中在接近中心性上

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G03-P017-002 | 英文原文

centralities for the bipartite graph. The basis for the closeness centrality index is the average geodesic distance that a node is from all other nodes in the graph. Thus it begins with the 'farness' of a node from other nodes in the graph. Distances between nodes are summarized in the matrix $\boldsymbol{D} = \{d(i,j)\}$, where the entry in cell (i,j) of this matrix is the geodesic distance from node i to node j. In other words, it is the length of any shortest path between the nodes. Since the bipartite graph is nondirectional, the distance from i to j is the same as the distance from j to i, and $\boldsymbol{D}$ is symmetric.

中文译文 | G03-P017-002

二部图的中心性。接近中心性指数的基础是节点与图中所有其他节点之间的平均测地距离。因此，它始于节点与图中其他节点的"距离"。节点间的距离被汇总在矩阵 D = {d(i,j)} 中，该矩阵中单元格 (i,j) 的条目即为从节点 i 到节点 j 的测地距离。换句话说，它就是节点间任意最短路径的长度。由于二部图是非定向的，从 i 到 j 的距离与从 j 到 i 的距离相同，且 D 是对称的。

G03-P017-003 | 英文原文

In general, closeness centrality, $C_C(p_i)$, is the inverse of the average geodesic distance between the node and all other nodes in the graph, and is calculated as

中文译文 | G03-P017-003

通常而言，中心性中心度 Cc(Pi) 是节点与图中所有其他节点之间的平均测地距离的倒数，其计算方式为

G03-P017-004 | 英文原文

If the graph is disconnected, then closeness centrality is undefined, since some pairs of nodes are unreachable and the distance between them is infinite.

中文译文 | G03-P017-004

如果图是不连通的，则接近中心性无法定义，因为某些节点对无法到达且它们之间的距离为无穷大。

G03-P017-005 | 英文原文

The numerator of Eq. (25) is the sum of the distances from node i to all other nodes in the graph. For an actor in an affiliation network this is the distance from the actor to other actors plus the distance from the actor to all events. However, since the affiliation network is a bipartite graph, actors are only adjacent to events, and all paths emanating from an actor must first pass through the events to which the actor belongs. Similarly, since events are only adjacent to actors, all paths from events must pass through the actors that are their members.

中文译文 | G03-P017-005

方程（25）的分子是节点i到图中所有其他节点的距离之和。在二模参与网络中，对于一个行动者而言，这等于该行动者与其他行动者的距离加上该行动者与所有事件的距离。然而，由于二模参与网络是一个二部图，行动者与事件仅存在邻接关系，且从某个行动者出发的所有路径都必须先经过该行动者所属于的事件。同样，由于事件与行动者仅存在邻接关系，从事件出发的所有路径也必须经过作为其成员的行动者。

G03-P017-006 | 英文原文

Consider the relationship between the closeness centrality of an actor and the closeness centralities of the events to which the actor belongs, and the relationship between the closeness centrality of an event and the closeness centralities of its actors. First focus on the distances from an actor in the bipartite graph as a function of the distances from the events to which it belongs. The distance from node i representing an actor to any node j (either actor or event) is $d(i,j) = 1 + \min_k d(k,j)$, for event nodes k adjacent to i. Given this property, the numerator of Eq. (25) for the closeness centrality of an actor in the bipartite graph can be expressed as a function of the distances from the actor's events, k:

中文译文 | G03-P017-006

考虑一个行动者的接近中心性与该行动者所归属的事件之间的接近中心性之间的关系，以及一个事件与其行动者之间的接近中心性与该行动者中心性之间的关系。首先关注二部图中与行动者相关的距离，作为该行动者所归属的事件距离的函数。代表行动者的节点 i 到任意节点 j（无论是行动者还是事件）的距离，为 d(i,j) = 1 + min~ d(k,j)，其中 k 是与 i 相邻的事件节点。基于这一性质，方程 (25) 中二部图中行动者中心性的分子可以表示为与行动者所归属的事件距离的函数：k:

G03-P017-007 | 英文原文

$$\sum_{j=1}^{g+h} d(i,j) = \sum_{j=1}^{g+h} \left[1 + \min_k d(k,j)\right] = g + h - 1 + \sum_{j=1}^{g+h} \min_k d(k,j) \qquad (26)$$

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G03-P017-008 | 英文原文

for k adjacent to i, and j not equal to i. Combining Eqs. (25) and (26) gives the following expression for the closeness centrality of an actor as a function of the geodesic distances from its events to other actors and events:

中文译文 | G03-P017-008

对于与 i 相邻的 k，且 j 不等于 i。结合方程（25）和（26），可以得到行动者的接近中心性的表达式，该表达式取决于其事件与其他行动者及事件之间的测地距离：

G03-P017-009 | 英文原文

$$C_C^{\mathcal{N}\mathcal{M}}(n_i) = \left[1 + \frac{\sum_{j=1}^{g+h} \min_k d(k,j)}{g+h-1}\right]^{-1} \qquad (27)$$

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G03-P017-010 | 英文原文

for events k adjacent to actor i. Thus, the closeness centrality of an actor is a function of the minimum distances from any of its events to other actors and events in the bipartite graph.

中文译文 | G03-P017-010

对于与行动者 i 相邻的事件 k。因此，行动者的接近中心性取决于其任何事件与二部图中的其他行动者及事件之间的最小距离。

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G03-P018-002 | 英文原文

Similarly, the closeness centrality of an event is a function of the minimum distances from its actors to other actors and to other events. Eq. (27) can be written for events as:

中文译文 | G03-P018-002

同样，事件的接近中心性是其行动者与其他行动者及其他事件之间最小距离的函数。方程（27）可针对事件表示为：

G03-P018-003 | 英文原文

$$= \left[1 + \frac{\sum_{i=1}^{g+h} \min_j d(i,j)}{g+h-1}\right]^{-1} \qquad (28)$$

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G03-P018-004 | 英文原文

for actors i adjacent to event k.

中文译文 | G03-P018-004

对于与事件 k 相邻的行动者 i。

G03-P018-005 | 英文原文

The combination of Eqs. (27) and (28) shows that there is a clear relationship between the closeness centralities of actors and the closeness centralities of events in an affiliation network. The closeness centrality of an actor is a function of the minimum distances to its events, and the closeness centrality of an event is a function of the minimum distances to its actors.

中文译文 | G03-P018-005

方程（27）和（28）的结合表明，在二模参与网络中，行动者的接近中心性与事件의接近中心性之间存在明显的关联。行动者的接近中心性取决于其到相关事件的最小距离，而事件의接近中心性则取决于其到相关行动者的最小距离。

G03-P018-006 | 英文原文

Table 4 gives the closeness centralities for actors and events in the bipartite graph for the hypothetical example.

中文译文 | G03-P018-006

表 4 给出了假设示例中二部二模参与网络中行动者与事件的接近中心性。

G03-P018-007 | 英文原文

4.4. Betweenness centrality

中文译文 | G03-P018-007

4.4. 中介中心性

G03-P018-008 | 英文原文

Betweenness centrality for a one-mode dyadic network focuses on the extent to which actors sit on geodesic paths between other pairs of actors (Freeman, 1979). Since calculating betweenness centrality requires considering *all* geodesics in a graph, it seems unlikely that it will be possible to express the betweenness of a node in a graph (an actor or event in an affiliation network) as a function simply of the betweenness of the nodes to which it is adjacent. Nevertheless, this section presents some informal observations about the relationship between actor and event betweenness centralities in affiliation networks. Betweenness centrality is defined for dichotomous, nondirectional relations, so I will concentrate betweenness of actors and events in the bipartite graph for the affiliation network. However, at the end of this section I consider what would happen if one were to analyze the one-mode relations of actor co-memberships or event overlaps.

中文译文 | G03-P018-008

单模双部网络的中介中心性，指的是行动者在其他行动者对之间测地路径上的位置程度（Freeman，1979）。由于计算中介中心性需要考虑图中的所有测地路径，因此似乎不太可能仅通过与其相邻节点的中介中心性来表达图中一个节点（即二模参与网络中的行动者或事件）的中介中心性。尽管如此，本节仍对二模参与网络中行动者与事件的中介中心性之间的关系提供一些非正式观察。中介中心性是针对二分、非定向关系定义的，因此我将重点研究二模参与网络中的双部图中行动者与事件的中介中心性。不过，在本节末尾，我将探讨如果分析单模关系下的行动者共属或事件重叠情况会产生什么结果。

G03-P018-009 | 英文原文

An intermediate step in calculating betweenness centrality is to find the 'partial betweenness' of nodes in the network (Freeman, 1979). Node p_i's partial betweenness counts the number of pairs of other nodes whose geodesic(s) contain node p_i. If there is more than one geodesic between a given pair of nodes, then p_i receives fractional credit, where the fraction is reciprocal of the number of geodesics between the pair. Let g_{jk} be the number of geodesics between p_j and p_k, and let $g_{jk}(p_i)$ be the number of geodesics between p_j and p_k that contain p_i. If all geodesics are equally likely, then the probability that a geodesic between p_j and p_k contains node p_i is equal to $g_{jk}(p_i)/g_{jk}$. The betweenness centrality of node p_i, denoted by $C_B(p_i)$, is defined as the sum of these quantities across all pairs of nodes:

中文译文 | G03-P018-009

计算中介中心性的一个中间步骤是找出网络中各节点的"部分中介中心性"（Freeman，1979）。节点pi的部分中介中心性用于统计包含节点Pi的其他节点对的数量。如果给定节点对之间存在多个测地线，则pi会获得分数，该分数等于该对节点之间测地线数量的反比。设gjk为pj与Pk之间的测地线数量，gj~(p~)为包含Pi的pj与Pk之间的测地线数量。如果所有测地线的出现概率相同，那么pj与p~之间的测地线包含节点pi的概率等于gak(P)/gik"。节点p~的中介中心性，记为CB(p~)，定义为所有节点对中这些数值的总和。

G03-P018-010 | 英文原文

$$C_B(p_i) = \sum_{j<k} \frac{g_{jk}(p_i)}{g_{jk}}. \qquad (29)$$

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G03-P018-011 | 英文原文

For a graph with g nodes, $C_B(p_i)$ reaches its maximum value of $(g-1)(g-2)/2$ when node p_i is on geodesics between all other pairs of nodes.

中文译文 | G03-P018-011

对于具有 g 个节点的图，当节点 pg 位于所有其他节点对之间的最短路径上时，C~(pi) 达到其最大值 (g-1)(g- 2)/2。

G03-P018-012 | 英文原文

Returning to an affiliation network, recall that linkages between pairs of actors are always through actors' participation in events, thus events are always on geodesics

中文译文 | G03-P018-012

回到二模参与网络的话题上，需记住：行动者之间的关联始终是通过其参与事件实现的，因此事件总是沿着最短路径进行

原PDF第19页（原刊第175页）

G03-P019-002 | 英文原文

between actors. Similarly, linkages between pairs of events are always through the joint memberships of actors, thus actors are always on geodesics between events. Consider the relationship between the betweenness centrality of an actor and the betweenness centrality of the events with which it is affiliated, and the relationship between the betweenness centrality of an event and the betweenness of its members.

中文译文 | G03-P019-002

在行动者之间。同样，事件对之间的联系总是通过行动者的共同参与实现的，因此行动者始终位于事件之间的最短路径上。考虑一个行动者的中介中心性与其关联事件的中介中心性之间的关系，以及一个事件的中介中心性与其成员的中介中心性之间的关系。

G03-P019-003 | 英文原文

In calculating the betweenness centrality of an event, m_k, in an affiliation network, focus on the collection of actors that belong to that event. Event m_k is on a geodesic between all pairs of actors that are members of it. Since event m_k has a_{+k} members,

中文译文 | G03-P019-003

在计算二模参与网络中事件 m k 的中介中心性时，需关注属于该事件的行动者集合。事件 m~ 位于所有属于该事件的行动者之间的测地线上。由于事件 m k 有 a+k 个成员，

G03-P019-004 | 英文原文

there are a_{+k}^2 pairs of actors that have a geodesic that contains m_k. If a given pair of

中文译文 | G03-P019-004

有2对行动者之间存在包含m k的测地线。如果给定的一对行动者之间有a+

G03-P019-005 | 英文原文

actors, (n_i, n_j), only shares event m_k in common (thus $x_{ij}^{\mathcal{N}} = 1$) then m_k is on the only geodesic between them, and m_k's betweenness is incremented by 1. Actually, n_i and n_j

中文译文 | G03-P019-005

jr= 1) 那么 m~ 是唯一的行动者，(ni,nj)，仅共享事件 rn k (因此它们之间存在 xij 测地线，mk 的介数增加 1。实际上，n i 和 nj

G03-P019-006 | 英文原文

share $x_{ij}^{\mathcal{N}}$ memberships, thus m_k's betweenness is incremented by $1/x_{ij}^{\mathcal{N}}$ for each pair of actors (n_i, n_j) in m_k. Thus, a portion of the betweenness centrality of event m_k can be expressed in terms of the number of co-memberships of pairs of its members as:

中文译文 | G03-P019-006

/r 成员关系，因此 mk 的中介中心性会随着 m k 中每对行动者 (ni, nj) 的共享比例 xij 增加 1/xi。因此，事件 m k 的中介中心性的一部分可以用其成员之间共同成员关系的数量来表示：

G03-P019-007 | 英文原文

$$\frac{1}{2}\sum_{n_i, n_j \in m_k} \frac{1}{x_{ij}^{\mathcal{N}}}. \qquad (30)$$

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G03-P019-008 | 英文原文

From this we see that an event gains betweenness centrality to the extent that pairs of its members meet only in that event.

中文译文 | G03-P019-008

由此可以看出，一个事件获得中介中心性程度取决于其成员仅在该事件中相遇。

G03-P019-009 | 英文原文

In addition, an event gains betweenness centrality if an individual actor belongs only to that event. In that case, all geodesics from such an actor must contain the event. Since there are $g + h$ nodes (actors and events) in the affiliation network, an event gains $g + h - 2$ betweenness 'points' for each of its members that belongs to no other event. This quantity is not independent of the count in Eq. (30).

中文译文 | G03-P019-009

此外，如果某个行动者仅属于该事件，则该事件会获得中介中心性。在这种情况下，该行动者所经过的所有路径都必须包含该事件。由于二模参与网络中存在 g + h 个节点（行动者和事件），因此该事件对于每个不属于其他事件的成员都会获得 g + h - 2 个中介中心性“点”。这一数值与方程（30）中的计数无关。

G03-P019-010 | 英文原文

These observations suggest that the betweenness centrality of an event increases to the extent that its members belong to no other events. Since such single-membership actors are not between any pairs of events, an event gains betweenness centrality if it contains 'non-central' actors. Also, the betweenness centrality of an event increases to the extent that pairs of actors share only that event in common.

中文译文 | G03-P019-010

这些观察结果表明，某事件的中介中心性会增强，其程度取决于该事件的所有成员都不属于其他任何事件。由于这种单一成员资格的行动者不会出现在任何事件对之间，因此如果某个事件包含“非中心”行动者，则该事件的中介中心性就会增强。此外，某事件的中介中心性还会增强，其程度取决于存在多少行动者仅共享该事件这一共同点。

G03-P019-011 | 英文原文

Similar properties of betweenness centrality hold for actors. An actor gains $g + h - 2$ betweenness centrality 'points' if it is the only member of an event. An actor gains

中文译文 | G03-P019-011

中介中心性具有类似的特性适用于行动者。如果某行动者是该事件的唯一成员，则该行动者获得 g + h - 2 中介中心性“点”。

G03-P019-012 | 英文原文

$1/x_{kl}^{\prime}$ betweenness 'points' for all pairs of events to which it belongs.

中文译文 | G03-P019-012

所有所属事件对之间的 l/xft 中介中心性“点”。

G03-P019-013 | 英文原文（原段接续 / 分段）

Betweenness centralities for the actors and events in the bipartite graph for the hypothetical example are presented in Table 4. Notice that $C_B(n_2) = C_B(n_4) = 0.00$ since actors n_2 and n_4 each belong to only one event, and thus create no linkages between either actors or events.

中文译文 | G03-P019-013

对于该假设示例中的二部图中的行动者和事件，其中间性中心性在表4中给出。注意，由于行动者n 2和n 4各自仅属于一个事件，因此Cs(n 2) = CB(n 4) = 0.00，从而行动者与事件之间不存在关联。

G03-P019-014 | 英文原文

As mentioned above, betweenness centrality is defined for networks in which relations are dichotomous and nondirectional. However, sometimes it is tempting to analyze valued nondirectional relations by dichotomizing the original values and analyzing the new dichotomous relation. For example, one could define a dichotomous actor co-membership relation, $x_{ij}^{\mathscr{N}01}$, where

中文译文 | G03-P019-014

如上所述，中介中心性是针对关系为二分且非定向的网络定义的。然而，有时人们倾向于通过将原始值进行二分处理并分析新的二分关系来分析具有价值的非定向关系。例如，可以定义一种二分行动者共同成员资格关系，-d(0，其中

G03-P019-015 | 英文原文

$$x^{\mathscr{N}01} = \begin{cases} 0 & \text{if} \quad x_{ij}^{\mathscr{N}} = 0 \end{cases} \tag{31}$$

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G03-P019-016 | 英文原文

$$\begin{cases} 1 & \text{if} \quad x_{ij}^{\mathscr{N}} > 0. \end{cases}$$

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G03-P020-002 | 英文原文

$\bar{n}_4$

a. Dichotomous actor co-membership relation

中文译文 | G03-P020-002

T~ 4 a. 二分型行动者成员资格关系

G03-P020-003 | 英文原文

$\bar{m}_2$

b. Dichotomous event overlap relation

中文译文 | G03-P020-003

712 2 b. 二分型事件重叠关系

G03-P020-004 | 英文原文

Fig. 2. Graphs for dichotomous one-mode relations.

中文译文 | G03-P020-004

图 2. 用于二分型单模关系的图。

G03-P020-005 | 英文原文

In this dichotomous relation actors are adjacent if they ever belong to the same event. This relation has been defined by Seidman (1985, p. 368) as the 'q-overlap graph', where actors are adjacent if they share at least q events, and $q = 1$. Doreian (1969) presents a similar idea for valued graphs. One could also define a dichotomous event overlap relation $x_{kl}^{\prime\prime 01}$ where

中文译文 | G03-P020-005

在这种二分关系中，如果行动者属于同一事件，则它们被视为相邻。Seidman（1985，第368页）将这种关系定义为"q-重叠图"，即如果行动者至少共享q个事件，则视为相邻，其中q = 1。Doreian（1969）为价值图提出了类似的概念。人们也可以定义一种二分事件重叠关系x~ 0'。

G03-P020-006 | 英文原文

if $x_{kl}^{\prime\prime} = 0$

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G03-P020-007 | 英文原文

$$\tilde{}_{kl} \quad \begin{cases} 1 & \text{if } x_{kl}^{\prime\prime} > 0. \end{cases}$$

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G03-P020-008 | 英文原文

In this dichotomous relation events are adjacent if they share any actor in common. Stokman (1977) describes either of these relations as creating a new graph by 'induction' where nodes in the new graph are adjacent if they are adjacent to the same node in the original graph.

中文译文 | G03-P020-008

在这种二分关系中，如果事件存在共同的行动者，则它们属于相邻关系。Stokman（1977）将这两种关系均描述为通过"归纳"生成新图的过程：新图中的节点与原始图中同一节点的相邻节点相连时，即视为相邻关系。

G03-P020-009 | 英文原文

Graphs for these two dichotomous relations for the hypothetical example are presented in Fig. 2 Notice that the graph for the dichotomous event overlap relation is complete (all events are adjacent) since each pair of events shares at least one actor. Since the graph is complete there is no differentiation among events; all are equally central in this graph. In contrast, the actor co-membership graph is not complete (some pairs of actors never belong to the same event).

中文译文 | G03-P020-009

该假设示例中这两种二分关系的相关图见图2。请注意，关于事件重叠关系的图是完整的（所有事件均相邻），因为每对事件至少包含一个行动者。由于该图是完整的，因此各事件之间不存在差异；所有事件在该图中都具有同等重要性。相比之下，行动者成员资格图则并不完整（某些行动者对从未属于同一事件）。

G03-P020-010 | 英文原文

What happens if we now find actor and event betweenness centralities, $C_B^{\mathcal{N}_{01}}(n_i)$ and $C_B^{\mathcal{N}_{01}}(m_i)$, for these relations? These centralities are presented in Table 5. Several points are important to note about these results. First, as anticipated above, all events in this

中文译文 | G03-P020-010

如果我们现在发现行动者与事件之间的中间中心性值 C~ o~ (n i) 和 C~°'(mi)，这些关系会如何变化呢？这些中心性值在表5中呈现。关于这些结果有几个重要点需要注意。首先，正如上述所预料的那样，这里的所有事件都……

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G03-P021-002 | 英文原文

example have equal betweenness centrality for the dichotomous one-mode network of event overlaps. In contrast, in the bipartite graph, events m_2 and m_3 are clearly more central than is event m_1. The dichotomous event overlap relation lacks information about the number of actors shared by pairs of events, and thus lacks information about the number of geodesics between events. Second, notice that for the dichotomous actor co-membership relation, actors n_2 and n_4 have betweenness centrality equal to 0. These actors are adjacent to other actors on this relation but they are not on geodesics between them. Actors n_2 and n_4 are each affiliated with only a single event. Thus, if one of these

中文译文 | G03-P021-002

示例表明，在二部单模事件重叠网络中，成员资格具有相同的中介中心性。相比之下，在二部图中，事件 m 2 和 m 3 的中心性明显高于事件 m 1。这种二部事件重叠关系缺乏关于事件对之间共享的行动者数量的信息，也缺乏关于事件之间的测地线数量的信息。其次，注意到对于二部行动者共同成员资格关系而言，行动者 n 2 和 174 的中介中心性为 0。这些行动者在该关系中与其他行动者相邻，但彼此之间并非处于测地线上。行动者 n 2 和 rt 4 各自仅与一个事件相关。因此，如果其中某个……

G03-P021-003 | 英文原文

'single-membership' actors is adjacent to two other actors (for example, n_2 is adjacent to n_5 and n_6) those two actors must be adjacent to each other since all three must belong to the single event to which the 'single-membership' actor belongs. Actors that belong to only one event cannot be on geodesics between other actors.

中文译文 | G03-P021-003

"单一成员资格"行动者与另外两个行动者相邻（例如，n 2 与 n 5 和 n 6 相邻），那两个行动者必须彼此相邻，因为三者都必须属于"单一成员资格"行动者所归属的同一事件。只属于一个事件的行动者无法位于其他行动者之间的测地线上。

G03-P021-004 | 英文原文

These observations about betweenness centrality in affiliation networks are informal. Clearly, it remains for further research to express the exact relationship between actor and event betweenness centrality.

中文译文 | G03-P021-004

关于二模参与网络中中介中心性的这些观察结果只是初步的。显然，进一步研究仍需揭示行动者与事件之间的确切中介中心性关系。

G03-P021-005 | 英文原文

4.5. Flow betweenness centrality

中文译文 | G03-P021-005

4.5. 流中介中心性

G03-P021-006 | 英文原文

Recently, Freeman et al. (1991) have proposed an extension of betweenness centrality that is applicable to valued relations. For a pair of actors, the value of the relation might be ''their amount of interaction, the time they spend in one another's company, the range of different social settings in which they interact...'' (Freeman et al., 1991, p.

中文译文 | G03-P021-006

最近，Freeman等人（1991）提出了一种适用于有价值关系的中介中心性扩展方法。对于一对行动者而言，这种关系的值可能包括"他们的互动程度、在对方陪伴下的时间、以及他们互动时所处的不同社交环境范围……"（Freeman等人，1991，第某页）

G03-P021-007 | 英文原文

145). Thus, this approach is clearly applicable to one-mode relations of actor co-memberships or event overlaps.

中文译文 | G03-P021-007

145). 因此，这种方法显然适用于行动者成员资格或事件重叠的单模关系。

G03-P021-008 | 英文原文

Flow betweenness centrality extends betweenness centrality in two ways. First it considers all paths between nodes, rather than just geodesics. Second, it is appropriate for both graphs and valued graphs in which larger values indicate stronger ties between actors. Values in the graph are considered to represent the potential for 'flow' of information or resources between nodes, with the following constraints. First, the flow between a pair of adjacent nodes, f_{ij}, is constrained to be less than or equal to the value of the tie between the nodes: $f_{ij} \leq x_{ij}$. Second the flow 'into' a node is equal to the flow

中文译文 | G03-P021-008

流中介中心性以两种方式扩展中介中心性。首先，它考虑节点之间的所有路径，而不仅仅是直线路径。其次，它适用于图以及具有较大数值表示行动者之间联系更紧密的权值图。图中的数值被视作节点间信息或资源流动的潜力，并受到以下约束：第一，相邻节点对 .f~j 之间的流动量必须小于或等于该节点间联系的数值：f,.j _< x~. 第二，"进入"节点的流动量等于流入该节点的流动量。

G03-P021-009 | 英文原文

'out of' the node. The fact that this approach focuses on the betweenness of nodes in a graph and can be used to study valued relations suggests that it should be quite appropriate for studying centrality in affiliation networks.

中文译文 | G03-P021-009

从节点"脱离"出来。这种方法侧重于图中节点的中间性，且可用于研究价值关系，这表明它非常适合用于研究二模参与网络中的中心性。

G03-P021-010 | 英文原文

To define the flow betweenness centrality of node p_i, one considers the extent to which the maximum flow between other pairs of nodes in the graph depends on paths that include node p_i. Roughly, the flow centrality index for a node in a valued graph quantifies the extent to which flow between other pairs of nodes in the graph would be reduced if that node were removed from the graph. Defining $F_{kl}(p_i)$ as the amount of flow between nodes p_k and p_l that passes through p_i, the flow betweenness centrality of p_i is defined as the total flow between all pairs of nodes that depends on node p_i. The flow betweenness for node p_i, denoted by $C_F(p_i)$, is calculated as:

中文译文 | G03-P021-010

为了定义节点Pi的流中介中心性，需要考虑图中其他节点对之间的最大流量在多大程度上依赖于包含节点Pl的路径。大致而言，有值图中的某节点的流中心性指数，用于衡量如果该节点从图中移除，图中其他节点对之间的流量会减少多少。将F~(p~)定义为通过节点Pi的节点Pk与p~之间的流量总量，则p~的流中介中心性被定义为所有依赖于节点Pi的节点对之间的总流量。节点p~的流中介中心性，记为CF(P~)，可按以下方式计算：

G03-P021-011 | 英文原文

$$C_F(p_i) = \sum F_{kl}(p_i).$$

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G03-P021-012 | 英文原文（原段接续 / 分段）

$$\sum_{k<l} F_{kl}(p_i). \qquad (33)$$

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G03-P022-002 | 英文原文

This index is described in detail in Freeman et al. (1991). Flow betweenness may be calculated for the one-mode networks of actor co-memberships and event overlaps, or for the bipartite graph.

中文译文 | G03-P022-002

该指数在 Freeman等人（1991）的研究中得到了详细描述。流中介中心性可以针对单模网络中的行动者成员关系与事件重叠情况来计算，也可以针对二部图来计算。

G03-P022-003 | 英文原文

Table 4 presents flow betweenness for actors and events in the bipartite graph. Table 5 presents results for the one-mode networks of actor co-memberships and event overlaps. Two results are worth noting in these tables. First, notice that actors n_2 and n_4 which each belong to only a single event have flow betweenness centralities equal to zero in the bipartite graph, but have flow centralities equal to four in the actor co-membership relation. In the one-mode network these actors cannot reside on geodesics between other actors, but they can reside on paths between other actors. Since flow betweenness considers 'flows' along all paths (not just geodesics), actors that belong to a single event can have non-zero flow betweenness in the one-mode actor co-membership relation.

中文译文 | G03-P022-003

表4展示了二部图中行动者与事件之间的流中介中心性。表5则给出了单模网络中行动者成员资格与事件重叠的相关结果。这些表格中有两个结果值得注意。首先，注意到行动者n 2和n 4各自仅属于一个事件，在二部图中它们的流中介中心性为零，但在行动者成员关系中其流中心性为四。在单模网络中，这些行动者无法位于其他行动者之间的测地线上，但可以在其他行动者之间的路径上存在。由于流中介中心性考虑了所有路径上的"流"（而不仅仅是测地线），属于单个事件的行动者在单模行动者成员关系中可以具有非零的流中介中心性。

G03-P022-004 | 英文原文

A second puzzling result is seen in Table 4. Notice that for the bipartite graph actor n_5 (which is affiliated with all three events) has flow betweenness centrality equal to three, but actors n_1 and n_3 (which are each affiliated with two events) have centralities equal to five. However, looking at either the bipartite graph (Fig. 1) or the affiliation network matrix (Table 1), we see that the events to which actors n_1 and n_3 belong are subsets of the events to which n_5 belongs; n_1 and n_3 are never in an event unless n_5 is also a member. This anomalous result deserves further investigation.

中文译文 | G03-P022-004

表4中出现了第二个令人困惑的结果。注意，对于二部图中行动者n 5（与三个事件均相关），其流中介中心性为3，而行动者n~和n 3（各与两个事件相关）的中心性为5。然而，无论是从二部图（图1）还是从二模参与网络矩阵（表1）来看，都可以发现：行动者n~和n 3所涉及的事件，都是行动者n 5所涉及的事件的子集；除非n 5也是该事件的成员，否则n~和n 3永远不会出现在某个事件中。这一异常结果值得进一步研究。

G03-P022-005 | 英文原文

5. Extensions

中文译文 | G03-P022-005

5. 扩展

G03-P022-006 | 英文原文

The centrality approaches discussed so far (degree, eigenvector, closeness, betweenness, and flow betweenness) were all designed for one-mode networks. As we have seen, all can be used to study affiliation networks by analyzing the bipartite graph. In this section I present two possible extensions of centrality using Galois lattices and graph covers. Both of these approaches are designed to study two-mode data, both focus on subsets and relations between subsets, and both allow insights into the difference between 'primary' and 'secondary' actors and events. Thus, these extensions incorporate the distinctive features of affiliation networks and might lead to centrality approaches that more naturally capture important features of these networks.

中文译文 | G03-P022-006

目前所讨论的中心性指标（如度、特征向量、接近度、流中介中心性以及流中介中心性）都是为单模网络设计的。正如我们所见，这些指标都可以用于通过分析二部图来研究二模参与网络。在本节中，我将介绍两种利用Galois格子与图覆盖技术来扩展中心性指标的方法。这两种方法均旨在研究二模数据，都关注子集及其之间的关系，并且都能帮助理解"主要"行动者与事件与"次要"行动者与事件之间的差异。因此，这些扩展方法能够体现二模参与网络的独特特征，或许能带来更自然地捕捉这些网络重要特性的中心性分析方法。

G03-P022-007 | 英文原文

5.1. Galois lattices

中文译文 | G03-P022-007

5.1. Galois格子

G03-P022-008 | 英文原文

Patterns of inclusion and overlap in events' memberships and in actors' affiliations are theoretically and empirically important components of actor and event centrality. As an early attempt to capture these relationships, Bonacich (1978) proposed using homomorphisms of Boolean algebras. The resulting homomorphisms were represented as lattices showing unions and intersections of events' members (with respect to a subset of actors), or of actors' affiliations (with respect to a subset of events). As noted by Bonacich, the height of an event or an actor in the lattice is related to its relative centrality. Bonacich's representation nicely shows the structure of inclusions among actors or among events, but it is limited in that it requires two separate representations

中文译文 | G03-P022-008

事件成员资格与行动者隶属关系中的包含与重叠模式，在理论及实证层面都是行动者与事件中心性的重要构成要素。作为捕捉这些关系的早期尝试，Bonacich（1978）提出使用布尔代数的同态映射。所得到的同态映射以格子形式呈现，显示出事件成员（针对某一子集行动者）的合并与交集，或行动者隶属关系（针对某一子集事件）的合并与交集。正如Bonacich所指出的，格子中某事件或某行动者的高度与其相对中心性相关。Bonacich的表示方式很好地揭示了行动者之间或事件之间的包含结构，但其局限性在于需要两种独立的表示方式。

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G03-P023-001 | 英文原文

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G03-P023-002 | 英文原文

(one for actors and one for events). In addition, it captures only a portion of the inclusions among actors and events because the lattice for actors is constructed with respect to only a subset of events, and the lattice for events is constructed with respect to only a subset of actors. A Galois lattice provides a more appropriate approach since a single representation presents patterns of inclusions for all actors and events simultaneously.

中文译文 | G03-P023-002

（一个用于行动者，另一个用于事件）。此外，它仅能反映行动者与事件之间部分包含关系，因为行动者的格结构是基于部分事件构建的，而事件的格结构则基于部分行动者构建。伽罗瓦格更为合适，因为单一表示法能够同时呈现所有行动者与事件之间的包含模式。

G03-P023-003 | 英文原文

Galois lattices have only recently been used to study affiliation networks (Schweizer, 1991; Freeman and White, 1993; Wasserman and Faust, 1994), though applications to other kinds of data are more widespread. General discussions of Galois lattices can be found in Birkhoff (1940) and Wille (1984).

中文译文 | G03-P023-003

伽罗瓦格子直到最近才被用于研究二模参与网络（Schweizer，1991；Freeman和White，1993；Wasserman和Faust，1994），尽管其在其他类型数据中的应用更为广泛。关于伽罗瓦格子的综合讨论可参见Birkhoff（1940）和Wille（1984）的著作。

G03-P023-004 | 英文原文

A Galois lattice represents the relationship between two sets of objects in terms of inclusion mappings between subsets of objects from each set. For an affiliation network we have the two sets: $\mathcal{N} = \{n_1, n_2, \ldots, n_g\}$ and $\mathcal{M} = \{m_1, m_2, \ldots, m_h\}$. We also have the affiliation relation, denoted by α, from $\mathcal{N}$ to $\mathcal{M}$, where $n_i \alpha m_k$ if n_i 'is affiliated with' m_k. Thus $n_i \alpha m_k$ if actor n_i belongs to event m_k. The inverse relation, α^{-1}, is a mapping from $\mathcal{M}$ to $\mathcal{N}$, where $m_k \alpha^{-1} n_i$ if event m_k includes actor n_i.

中文译文 | G03-P023-004

伽罗瓦格架描述了两组对象之间的关系，通过每组对象子集之间的包含映射来实现。对于二模参与网络而言，我们有两组集合：A/= {n l,n2 ..... ng} 和 ~¢.= {m l,m 2 ..... mh}。我们还存在一种参与关系，用 a 表示，表示从 JV 到 A' 的参与关系，当 niam k 与 m k 存在参与关系时即为如此。因此，当行动者 n i 属于事件 m 时，niam k 即成立。逆关系 a -l 则是从 ~" 到 A r 的映射，当事件 m, 包含行动者 ni 时即为如此。

G03-P023-005 | 英文原文（原段接续 / 分段）

Extending the relations α and α^{-1} from individual actors and events to subsets of actors and events, we get two complementary mappings: $\uparrow$ from subsets of actors to subsets of events and $\downarrow$ from subsets of events to subsets of actors. The $\uparrow$ mapping maps a subset of entities in $\mathcal{N}$ to a subset of entities in $\mathcal{M}$. For subsets $\mathcal{N}_S \subset \mathcal{N}$ and $\mathcal{M}_S \subset \mathcal{M}$, we define $\mathcal{N}_S \uparrow \mathcal{M}_S$ if $n_i \alpha m_k$ for all $n_i \in \mathcal{N}_S$, and $m_k \in \mathcal{M}_S$. For an affiliation

中文译文 | G03-P023-005

将个体行动者和事件与行动者及事件的子集之间的关系从 a 和 a-r 映射中推广开来，我们得到两个互补的映射：1' 是从行动者的子集到事件的子集的映射，而 $ 是从事件的子集到行动者的子集的映射。~' 映射则将 JV" 中的实体子集映射到 ~' 中的实体子集。对于 ~/'s c Ar 和 ~'s ~Z/ 这样的子集，我们定义 ~/V s 1" -Z/s，如果对所有 n i EJUS 都有 nio~m k，且 m k ~。

G03-P023-006 | 英文原文（原段接续 / 分段）

$\mathscr{M}_S \subset \mathscr{M}$, we define $\mathscr{N}_S \uparrow \mathscr{M}_S$ if $n_i \alpha m_k$ for all $n_i \in \mathscr{N}_S$, and $m_k \in \mathscr{M}_S$. For an affiliation network, the $\uparrow$ mapping maps a subset of actors to the subset of events attended by all of the actors in that subset.

中文译文 | G03-P023-006

对于二模参与网络而言，*$映射*将部分行动者映射到该部分行动者所参与的所有事件构成的子集。

G03-P023-007 | 英文原文

The dual mapping, $\downarrow$, goes from subsets of $\mathscr{M}$ to subsets of $\mathscr{N}$. For subsets $\mathscr{N}_S \subset \mathscr{N}$ and $\mathscr{M}_S \subset \mathscr{M}$, we define $\mathscr{M}_S \downarrow \mathscr{N}_S$ if $m_k \alpha^{-1} n_i$ for all $n_i \in \mathscr{N}_S$ and $m_k \in \mathscr{M}_S$. In an affiliation network the $\downarrow$ mapping maps a subset of events to the subset of actors all of which attend all events in the subset.

中文译文 | G03-P023-007

双映射 $ 将 ~" 的子集映射到 A r 的子集。对于 ~A/~. c A/ 和 l~'s c~' 这样的子集，我们定义 ~fsSA/s，当所有 ni~J/s 和 mk~/Z/s 满足 mka-lni 时。在二模参与网络中，该映射将事件的子集映射到行动者子集，且这些行动者都会参加该子集中的所有事件。

G03-P023-008 | 英文原文

A Galois lattice represents the relation between the sets $\mathscr{N}$ and $\mathscr{M}$ in terms of the $\uparrow$ and $\downarrow$ mappings between subsets of entities from each set. A Galois lattice is presented in a diagram in which points represent entities or subsets of entities from the two sets, and lines represent subset–superset relations in the following way. There is a line or sequence of lines in the diagram ascending from object $n_i \in \mathscr{N}$ to object $m_k \in \mathscr{M}$ if

中文译文 | G03-P023-008

伽罗瓦格子描述了集合 JV 与 ~¢" 之间的关系，通过每个集合中实体子集之间的 $ 和 $ 映射来体现。伽罗瓦格子以图表形式呈现，其中点代表两个集合中的实体或实体子集，线条则按以下方式表示子集与超子集的关系：如果图表中存在一条或一系列线条，从对象 n~ e S" 到对象 m k evg/，则表明...

G03-P023-009 | 英文原文

$n_i \alpha m_k$, and there is a line or sequence of lines descending from object $m_k \in \mathscr{M}$ to object

中文译文 | G03-P023-009

nio~m k，并且存在一条或一系列从对象 m k ~g" 到对象的下落线或序列

G03-P023-010 | 英文原文

$n_i \in \mathscr{N}$ if $m_k \alpha^{-1} n_i$.

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G03-P023-011 | 英文原文（原段接续 / 分段）

Fig. 3 presents the Galois lattice for the hypothetical example of six actors and three events (this figure is adapted from Wasserman and Faust, 1994, p. 333). In this figure an ascending line or sequence of lines connects each actor to the subset of events with which it is affiliated and a descending line or sequence of lines connects each event to the actors that are affiliated with it. One can also look at lines or sequences of lines between actors or between events. A line or sequence of lines ascending from actor n_i to actor n_j means that the events to which actor n_j belongs are a subset of the events to which actor n_i belongs. Similarly, a line or sequence of lines descending from event m_k to event m_l means that the actors that are affiliated with event m_l are a subset of the actors that are affiliated with event m_k.

中文译文 | G03-P023-011

图3展示了六个行动者和三个事件这一假设示例的Galois格子（该图改编自Wasserman和Faust，1994年，第333页）。在图中，一条上升线或一系列线条将每个行动者与与其相关的子集事件连接起来，而一条下降线或一系列线条则将每个事件与与其相关的行动者连接起来。人们也可以观察行动者之间或事件之间的线条或系列线条。从行动者n i到行动者nj的上升线条或系列线条意味着行动者t/j所涉及的事件是行动者ni所涉及的事件的子集。同样，从事件m k到事件m~的下降线条或系列线条意味着与事件m t相关的行动者是与事件m k相关的行动者的子集。

G03-P023-012 | 英文原文

Viewing centrality in terms of subset–superset relations, actors that are relatively low

中文译文 | G03-P023-012

从子集-超集关系的角度来看中心性，相对较低的行动者

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m_2

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G03-P024-003 | 英文原文

Fig. 3. Galois lattice for hypothetical example.

中文译文 | G03-P024-003

图3. 假设性示例的伽罗瓦格子。

G03-P024-004 | 英文原文

in the diagram are relatively more central than are actors that are relatively high in the diagram. Similarly, events that are relatively high in the diagram are relatively more central than are events that are relatively low in the diagram (Freeman and White, 1993). For the hypothetical example in Fig. 3, actor n_5 is the most central actor, followed by n_1, n_3, and n_6; n_4 and n_2 are least central. In this example the events are indistinguishable in terms of their centrality.

中文译文 | G03-P024-004

在图中，那些中心性较高的行动者，其地位相对比那些中心性较低的行动者更为核心。同样，在图中中心性较高的事件，其中心性也相对高于那些中心性较低的事件（Freeman和White，1993）。以图3中的假设示例为例，行动者n 5是中心性最高的行动者，其次是n~、n 3和n6；n 4和n 2的中心性最低。在这个例子中，这些事件的中心性并无差异。

G03-P024-005 | 英文原文

A Galois lattice is especially appropriate for studying affiliation networks because it explicitly represents the non-dyadic relation of inclusion between subsets of entities from the two modes. Furthermore, it represents an aspect of centrality in these networks, the subset–superset inclusions between actors and events, that is not represented by other centrality approaches. Freeman and White (1993) use Galois lattices in their reanalysis of the data of Davis et al. (1941) on Southern club women. Their results clearly showed which actors and events were primary and which were secondary in terms of the upward containment structure for actors and the downward containment structure for events.

中文译文 | G03-P024-005

伽罗瓦格子非常适合用于研究二模参与网络，因为它明确体现了来自两种模式的实体子集之间的非二元关系。此外，它还体现了这些网络中的中心性特征——行动者与事件之间的子集-超子集包含关系，这种特征其他中心性分析方法无法体现。Freeman和White（1993）在重新分析Davis等人（1941）关于南方俱乐部女性的数据时使用了伽罗瓦格子。他们的结果清楚地揭示了在行动者的向上包含结构以及事件的向下包含结构中，哪些行动者与事件属于主要角色，哪些属于次要角色。

G03-P024-006 | 英文原文

5.2. Graph covers

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G03-P024-007 | 英文原文

The idea of *graph covers* was introduced by Seidman (1985) to formalize the extent to which actors potentially have access to information about different kinds of structural features of an affiliation network. An actor that is located in such a way that it can 'see' important features of other actors' membership patterns (such as pair-wise attendances or cohesive subgroups) might be well located to initiate or coordinate actions within the network. Seidman uses the idea of a graph cover to formalize exactly what features of an affiliation network an actor or a set of actors 'sees'.

中文译文 | G03-P024-007

图覆盖的概念由Seidman在1985年提出，旨在明确行动者能够获取二模参与网络中不同结构特征信息的程度。一个处于能够“看到”其他行动者成员资格模式的重要特征的位置的行动者，可能适合发起或协调网络内的行动。Seidman利用图覆盖的概念，来明确一个行动者或一组行动者能够“看到”二模参与网络中的哪些特征。

G03-P024-008 | 英文原文

We begin with a hypergraph $\mathscr{H}(\mathscr{N},\mathscr{M})$ consisting of a set of points $\mathscr{N}$ and a set of edges $\mathscr{M}$, where each edge is a subset of points from $\mathscr{N}$. For an affiliation network it seems natural to start with the set of points being the set of actors, $\mathscr{N}$, and the set of edges being the set of events, $\mathscr{M}$. (We could also reverse the roles of $\mathscr{N}$ and $\mathscr{M}$ and have the dual hypergraph $\mathscr{H}^*(\mathscr{M},\mathscr{N})$.) We next define a duality function, $\delta(n_i)$ for points. This duality function is a mapping from point n_i to the set of all edges with which it is incident. In an affiliation network, this function for actors maps the actor to

中文译文 | G03-P024-008

我们首先考虑一个超图 ~(J,~')，它由一组点JK和一组边~'组成，其中每条边都是~'中点的子集。对于二模参与网络而言，将点集视为行动者集合X，将边集视为事件集合~'似乎是很自然的。（我们也可以颠倒X和~'的角色，得到对应的双超图~* (~/,~¢')。）接下来，我们定义一个针对点的对偶函数6(n i)。这个对偶函数是将点n i映射到与之相关的所有边的集合。在二模参与网络中，对于行动者而言，该函数将行动者映射为

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G03-P025-002 | 英文原文

the set of all events with which it is affiliated. Formally we can express this duality function as

中文译文 | G03-P025-002

它所关联的所有事件的集合。正式地来说，我们可以把这种对偶函数表达为

G03-P025-003 | 英文原文

$$\delta(n_i) = \{m_k \in \mathscr{M} | n_i \in m_k\}.$$

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G03-P025-005 | 英文原文

Using the notation from Galois lattices (above), $\delta(n_i)$ is the subset of events $\mathscr{M}_s$ for which $n_i \alpha m_k$.

中文译文 | G03-P025-005

使用上述Galois格子中的符号，6(n,)是那些满足nicem k的事件~¢'s的子集。

G03-P025-006 | 英文原文

Seidman (1985) then uses this duality function to define two different kinds of graph covers: a *strict cover* and a *wide cover*. A set is a wide (or strict) cover (as defined below) of some set of objects $\mathscr{A}$, where the objects are a specified set of entities. For example, the set $\mathscr{A}$ might be a subset of actors $\{n_1, n_2, n_3\}$ or it might be a subset of pairs of actors $\{(n_1, n_2), (n_1, n_3), (n_2, n_3)\}$.

中文译文 | G03-P025-006

随后，Seidman（1985）利用这一对偶函数定义了两种图覆盖：严格覆盖（strict cover）和广覆盖（wide cover）。一个集合可以是某个对象集A的广覆盖或严格覆盖（定义见下文）；这里的对象是某一指定集合中的实体。例如，A可以是行动者的一个子集{n1, n2, n3}，也可以是行动者对的一个子集{(n1, n2), (n1, n3), (n2, n3)}。

G03-P025-007 | 英文原文

In general, for some set of objects $\mathscr{A}$, a set $\mathscr{N}_s$ is a *wide cover* of $\mathscr{A}$ if the union of the duals of elements of $\mathscr{A}$ is included in the union of the duals of elements of $\mathscr{N}_s$. Formally, for object set $\mathscr{A}$ (a set of actors), $\mathscr{N}_s$ is a wide cover of $\mathscr{A}$ if

中文译文 | G03-P025-007

一般而言，对于某个对象集A，如果A中各元素的对偶之并集包含在N_s中各元素的对偶之并集中，则集合N_s是A的广覆盖。形式化地说，当对象集A是一个行动者集合时，N_s是A的广覆盖，当且仅当：

G03-P025-008 | 英文原文

$\bigcup \delta(a) \subset \bigcup \delta(n_i).$

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G03-P025-009 | 英文原文（原段接续 / 分段）

$$\bigcup_{a \in A} \delta(a) \subset \bigcup_{n_i \in \mathscr{N}_s} \delta(n_i). \qquad (35)$$

中文说明（原文编号 / 符号 / 数值保留） | G03-P025-009

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G03-P025-010 | 英文原文

Since the duality function $\delta(a)$ maps an actor to the set of events with which it is affiliated, for a subset of actors the union of these mappings is the collection of all events with which any actor in the subset is affiliated. Consider the case in which $\mathscr{A} = \mathscr{N}_t$ (a subset of actors). Another subset of actors, $\mathscr{N}_s$, is a wide cover of $\mathscr{N}_t$ if some actor in $\mathscr{N}_s$ was present at each and every event at which any actor from $\mathscr{N}_t$ was present. The events to which members of $\mathscr{N}_t$ belong are a subset of the events to which members of $\mathscr{N}_s$ belong. If $\mathscr{N}_s$ consists of a single actor, $\mathscr{N}_s = \{n_i\}$, then n_i is a wide cover of the set of all actors that were never present unless n_i was present.

中文译文 | G03-P025-010

由于对偶函数δ(a)把一个行动者映射到与其关联的事件集合，对于行动者的某个子集，这些映射的并集就是该子集中任一行动者所参与的全部事件。考虑A=N_t的情形，其中N_t是行动者的一个子集。如果N_t中任一行动者出现的每一个事件中，N_s中都至少有一名行动者出现，那么另一行动者子集N_s就是N_t的广覆盖。N_t成员所参与的事件集合，是N_s成员所参与的事件集合的子集。如果N_s只含一个行动者，即N_s={n_i}，那么n_i就是这样一组行动者的广覆盖：他们从不在n_i缺席的情况下出现。

G03-P025-011 | 英文原文

To illustrate, in the hypothetical affiliation network of six actors and three events, n_5 is a wide cover of the set consisting of the remaining five actors, $\{n_1, n_2, n_3, n_4, n_6\}$; n_3 is a wide cover of the set $\{n_2, n_4\}$; n_6 is a wide cover of $\{n_2\}$; and n_1 is a wide cover of $\{n_4\}$. The set $\{n_1, n_6\}$ is a wide cover of the set $\{n_2, n_4\}$.

中文译文 | G03-P025-011

例如，在由六名行动者和三个事件构成的假设性二模参与网络中，n5是由其余五名行动者组成的集合{n1, n2, n3, n4, n6}的广覆盖；n3是集合{n2, n4}的广覆盖；n6是{n2}的广覆盖；n1是{n4}的广覆盖。集合{n1, n6}是集合{n2, n4}的广覆盖。

G03-P025-012 | 英文原文

The second kind of graph cover is a *strict cover*. A strict cover is more general than a wide cover. For object set $\mathscr{A}$ (a set of actors), $\mathscr{N}_s$ is a strict cover of $\mathscr{A}$ if there exists some $n_i \in \mathscr{N}_s$ such that

中文译文 | G03-P025-012

第二种图覆盖是严格覆盖。严格覆盖比广覆盖更为一般。对于对象集A（一个行动者集合），如果存在某个属于N_s的n_i，使得下式成立，那么N_s就是A的严格覆盖：

G03-P025-013 | 英文原文

$$\delta(n_i) \bigcap \delta(a) \neq \varnothing.$$

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G03-P025-014 | 英文原文（原段接续 / 分段）

$$\bigcap_{a \in \mathscr{A}} \delta(a) \neq \varnothing. \qquad (36)$$

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G03-P025-015 | 英文原文

This means that for each object $a \in \mathscr{A}$ there exists some $n_i \in \mathscr{N}_s$ such that the dual of a intersects the dual of n_i. In an affiliation network, $\mathscr{N}_s$ is a strict cover of $\mathscr{A}$ if, for every actor in $\mathscr{A}$, some actor in $\mathscr{N}_s$ belongs to some event with that actor. To illustrate, for the hypothetical example n_6 is a strict cover of $\{n_1, n_2, n_3, n_5\}$.

中文译文 | G03-P025-015

这意味着：对于每一个属于A的对象a，都存在某个属于N_s的n_i，使a的对偶与n_i的对偶相交。在二模参与网络中，如果A中的每一名行动者都至少与N_s中的某一名行动者共同参与一个事件，那么N_s就是A的严格覆盖。例如，在前述假设性示例中，n6是集合{n1, n2, n3, n5}的严格覆盖。

G03-P025-016 | 英文原文

Every wide cover is a strict cover, but the reverse is not true. In an affiliation network, if $\mathscr{N}_s$ is a wide cover of $\mathscr{A}$, some actor from $\mathscr{N}_s$ is present each and every time any member of $\mathscr{A}$ is present. In other words, a member of $\mathscr{A}$ is never in an event unless a member of $\mathscr{N}_s$ is also present. In contrast, if $\mathscr{N}_s$ is a strict cover of $\mathscr{A}$, then each member of $\mathscr{A}$ must belong to at least one event with some member of $\mathscr{N}_s$, though members of $\mathscr{A}$ might belong to events where no member of $\mathscr{N}_s$ is present.

中文译文 | G03-P025-016

每一个广覆盖都是严格覆盖，但反之不成立。在二模参与网络中，如果集合N_s是集合A的广覆盖，那么A的任何成员每次参与某个事件时，N_s中都至少有一名行动者也参与该事件。换言之，A的成员不会参加一个没有N_s成员参加的事件。相比之下，若N_s是A的严格覆盖，则A的每个成员都必须至少与N_s中的某个成员共同参与一个事件，但A的成员仍可能参与没有N_s成员参加的其他事件。（集合符号的原字形见上方英文。）

G03-P025-017 | 英文原文

Returning to the Galois lattice diagram for an affiliation network, we see that a single

中文译文 | G03-P025-017

回到二模参与网络的伽罗瓦格图，我们看到一个单一的

原PDF第26页（原刊第182页）

G03-P026-002 | 英文原文

actor, n_i, is a wide cover of the set of all actors that reside on lines ascending from it. The events to which any of these other actors belong are a subset of the events to which n_i belongs. A set of actors is a wide cover of that set of actors that reside on lines ascending from any of the actors in the set. A single actor is a strict cover of the set of all actors with which it is ever a co-member. In the Galois lattice this is the set of all actors from which lines intersect either ascending or descending lines from n_i at any point before the topmost point in the diagram. Any subset of actors is a strict cover of the set of all actors that ever belonged to any event with any actor in the set. In the Galois lattice this is the set of all actors whose lines (ascending or descending) intersect a line from any member of the cover subset of actors, before the topmost point in the diagram.

中文译文 | G03-P026-002

行动者，n i，是所有位于从它开始上升的线上的行动者的广泛覆盖集。这些其他行动者所属于的事件，是n i所属于的事件的子集。一组行动者是那些位于该组中任何行动者上升线上上的行动者的广泛覆盖集。单个行动者是所有行动者的严格覆盖集，因为它始终是该集中的成员之一。在伽罗瓦格子中，这是所有行动者的集合，其线在图表中最顶点之前的任何时刻，都会与n i的上升或下降线相交。任何行动者的子集都是所有属于该组中任何行动者的事件的严格覆盖集。在伽罗瓦格子中，这是所有行动者的集合，其线（上升或下降）在图表中最顶点之前，会与覆盖子集中的任何成员所代表的线相交。

G03-P026-003 | 英文原文

Seidman (1985) elaborates in detail how graph covers can be used to study structural features of hypergraphs by varying the definition of the elements of $\mathscr{A}$. For example, if $\mathscr{A}$ consists of a set of pairs of actors, then a wide cover of $\mathscr{A}$ is a set of actors that are

中文译文 | G03-P026-003

Seidman (1985) 详细阐述了如何通过改变5g元素的定义，利用图覆盖来研究超图的结构特征。例如，如果5a¢由一组行动者对组成，那么5~'的广覆盖集则是一组具有广泛覆盖性的行动者。

G03-P026-004 | 英文原文

always present at any event at which any pair of actors in $\mathscr{A}$ is present.

中文译文 | G03-P026-004

在任何存在 5g 中任意一对行动者的事件中始终存在的行动者。

G03-P026-005 | 英文原文

5.3. Summary

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G03-P026-006 | 英文原文

Both the Galois lattice and the concept of graph covers nicely present subset–superset relationships among actors and events. This subset–superset relation is exactly the centrality idea of primary and secondary actors, as suggested by Davis et al. (1941), Doreian (1979a) and Freeman and White (1993). The Galois lattice incorporates the duality in the relation between actors and events in a single representation and makes the relative centrality of actors and events quite apparent in their vertical positions in the diagram. The distinction between primary events and actors and secondary events and actors is easily seen in the relative vertical positions of actors and events and in the patterns of inclusions among them.

中文译文 | G03-P026-006

Galois格子与图覆盖概念都能很好地体现行动者与事件之间的子集-超子集关系。这种子集-超子集关系正是Davis等人（1941年）、Doreian（1979a）以及Freeman和White（1993年）所提出的初级行动者与次级行动者的中心性概念。Galois格子将行动者与事件之间的关系中的对偶性整合在单一表示中，使得行动者与事件的相对中心性在图表中的垂直位置中十分明显。初级事件与行动者、次级事件与行动者之间的区别，可以通过行动者与事件的相对垂直位置以及它们之间的包含关系轻松看出。

G03-P026-007 | 英文原文

The concept of graph cover formalizes the idea of the 'importance' of an actor (or a subset of actors) in terms of the 'view' that the actor has of the network. One might think of a graph cover in terms of the kind and extent of information about an affiliation network that is available to an actor or to a subset of actors. An actor that 'covers' a large number of other actors, or 'covers' a large portion of the co-memberships between pairs of actors potentially has access to information about those features of the network.

中文译文 | G03-P026-007

图覆盖概念将行动者（或行动者的子集）的"重要性"这一理念形式化，该重要性取决于行动者对网络的"看法"。人们可以将图覆盖理解为关于二模参与网络的信息类型和范围，这些信息是行动者或行动者的子集所拥有的。一个能够"覆盖"大量其他行动者，或"覆盖"大量行动者之间关联成员关系的行动者，可能能够获取有关网络这些特征的信息。

G03-P026-008 | 英文原文

Both Galois lattices and graph covers provide a set of concepts for thinking about centrality in affiliation networks that is not provided by any of the other centrality indices. Both are naturally appropriate for two-mode non-dyadic networks and focus on subset–superset relations between actors and/or events in a way that captures theoretical and empirical insights about the relative importance of actors in these networks.

中文译文 | G03-P026-008

伽罗瓦格子与图覆盖都提供了一些用于思考二模参与网络中心性问题的概念，而这些概念是其他任何中心性指标所不具备的。这两种方法天然适用于二模非二元关系网络，它们关注行动者与/或事件之间的子集-超子集关系，从而能够捕捉到关于这些网络中行动者相对重要性的理论与实证见解。

G03-P026-009 | 英文原文

Neither the Galois lattice nor the formalism of a graph cover provides an index or quantification of actors' or events' centrality. One possibility would be to define a new relation between actors based on subset–superset relationships between their memberships. For example, one could define $x_{ij} = 1$ if $\delta(n_j) \subset \delta(n_i)$, and $x_{ij} = 0$ otherwise. The quantity $\sum_j x_{ij}$ would give the number of actors for which n_i is a wide cover, while $\sum_j x_{ji}$ would give the number of actors that are wide covers of n_i. Exploring other possible indices is a fruitful arena for further research.

中文译文 | G03-P026-009

无论是伽罗瓦格子还是图覆盖形式化，都无法提供衡量行动者或事件中心性程度的指标或量化方式。一种可能的办法是依据行动者成员资格之间的子集-超集关系，定义一种新的关联。例如，可以规定若 6(nj) c t~(ni)，则 Xij = 1；否则 xij = 0。量值 Ejxq 将表示满足 n~ 为广泛覆盖的行动者数量，而 S, jxj~ 则表示是 n~ 的广泛覆盖的行动者数量。探索其他可能的指标是进一步研究的重要方向。

G03-P026-010 | 英文原文

Now let us turn to another illustration.

中文译文 | G03-P026-010

现在让我们来看另一个例子。

原PDF第27页（原刊第183页）

G03-P027-002 | 英文原文

6. Another exampl

中文译文 | G03-P027-002

6. 另一个例子

G03-P027-003 | 英文原文

As another example, consider the affiliation network of club and board memberships

中文译文 | G03-P027-003

作为另一个例子，以Galaskiewicz（1985）研究的俱乐部及董事会二模参与网络为例

G03-P027-004 | 英文原文

of corporate executive officers studied by Galaskiewicz (1985) as part of his extensive

中文译文 | G03-P027-004

Galaskiewicz (1985) 研究的企业高管人员群体，作为其广泛研究的一部分

G03-P027-005 | 英文原文

research on the urban grants economy in Minneapolis-St.Paul. I will focus on a subset of

中文译文 | G03-P027-005

对明尼阿波利斯-圣保罗市资助经济的研究。我将重点研究其中一个子集

G03-P027-006 | 英文原文

26 CEOs and 15 clubs/boards from Galaskiewicz's data. The affiliation network matrix

中文译文 | G03-P027-006

26位CEO和15个俱乐部/董事会成员，来自Galaskiewicz的数据集。二模参与网络矩阵

G03-P027-007 | 英文原文

for this example is presented in Table 6. Centralities for CEOs and for the boards/clubs

中文译文 | G03-P027-007

该示例数据见于表6。CEO们以及董事会/俱乐部的中心性值

G03-P027-008 | 英文原文

are in Table 7 (for the bipartite graph) and Table 8 (for the one-mode relations).

中文译文 | G03-P027-008

这些结果分别见于表7（针对二部图）和表8（针对单模关系）。

G03-P027-009 | 英文原文

Overall there is considerable agreement among the centrality measures in their

中文译文 | G03-P027-009

总体而言，各中心性度量指标之间存在相当大的一致性

G03-P027-010 | 英文原文

identification of which clubs and CEOs are relatively central. Tables 9–11 present

中文译文 | G03-P027-010

识别哪些俱乐部和CEO相对处于中心地位。表9-11给出了相关结果。

G03-P027-011 | 英文原文

correlation matrices for the five centrality indices, calculated on the bipartite graph

中文译文 | G03-P027-011

针对五种中心性指数的相关矩阵，在二部图上计算得出

G03-P027-012 | 英文原文

(including all nodes, and for clubs and CEOs separately) and on the one-mode relations

中文译文 | G03-P027-012

（包括所有节点，并分别针对俱乐部和CEO）以及基于单模关系

G03-P027-013 | 英文原文

of club overlaps and actor co-memberships. The correlations between the measures of

中文译文 | G03-P027-013

与俱乐部重叠及行动者成员资格相关的指标。这些指标之间的相关性

G03-P027-014 | 英文原文

centrality for the clubs in the bipartite graph range from a low of 0.890 to a high of

中文译文 | G03-P027-014

二部图中俱乐部的中心性范围从低值0.890到高值

G03-P027-015 | 英文原文

0.992 with a median of 0.968; for CEOs in the bipartite graph the range is from 0.041 to

中文译文 | G03-P027-015

0.992，中位数为0.968；在二部图中，CEO的中心性范围为0.041到

G03-P027-016 | 英文原文

0.867 with a median of 0.643. For the one-mode relations, the correlation between the

中文译文 | G03-P027-016

0.867，中位数为0.643。对于单模关系，两者之间的相关性……

G03-P027-017 | 英文原文

centrality measures for clubs ranges from 0.918 to 0.993 with a median of 0.954, and for

中文译文 | G03-P027-017

俱乐部的中心性度量值介于 0.918 至 0.993 之间，中位数为 0.954。

G03-P027-018 | 英文原文

CEOs the range is from 0.348 to 0.989 with a median of 0.711. It appears that there is

中文译文 | G03-P027-018

CEO们的数值范围为 0.348 至 0.989，中位数为 0.711。看起来存在……

G03-P027-019 | 英文原文

Table 6
Affiliation network matrix for CEOs and clubs

中文译文 | G03-P027-019

表6 二模参与网络矩阵：CEO与俱乐部

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G03-P028-001 | 英文原文

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G03-P028-002 | 英文原文

Table 7
Centrality of CEOs and clubs in the bipartite graph

中文译文 | G03-P028-002

表7 二部图中CEO及俱乐部的中心性

G03-P028-003 | 英文原文

Measure of centrality

中文译文 | G03-P028-003

中心性度量

G03-P028-004 | 英文原文

C_D^{NM} $\sum d(i,k)$ C_C^{NM} C_B^{NM} C_E^{NM} C_F^{NM}

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G03-P028-005 | 英文原文

m_1	3	112	35.71	5.53	0.057	49
m_2	11	90	44.44	52.57	0.276	235
m_3	22	62	64.52	374.10	0.476	527
m_4	12	84	47.62	139.41	0.236	271
m_5	3	112	35.71	3.91	0.064	64

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G03-P028-006 | 英文原文（原段接续 / 分段）

m_5	3	112	35.71	3.91	0.064	64
m_6	4	110	36.36	5.76	0.102	60
m_7	4	114	35.09	12.61	0.049	125
m_8	4	110	36.36	13.16	0.068	106
m_9	6	104	38.46	17.13	0.114	156
m_{10}	3	116	34.48	5.32	0.037	79

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G03-P028-007 | 英文原文（原段接续 / 分段）

m_{10}	3	116	34.48	5.32	0.037	79
m_{11}	4	110	36.36	5.52	0.118	45
m_{12}	5	108	37.04	11.01	0.120	98
m_{13}	5	106	37.74	12.56	0.119	100
m_{14}	3	122	32.79	2.63	0.057	85

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G03-P028-008 | 英文原文（原段接续 / 分段）

m_{14}	3	122	32.79	2.63	0.057	85
m_{15}	9	94	42.55	37.77	0.237	146
n_1	3	89	44.94	9.36	0.126	7
n_2	3	93	43.01	20.88	0.090	58
n_3	2	97	41.24	3.46	0.091	14
n_4	3	95	42.11	3.24	0.151	5

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G03-P028-009 | 英文原文（原段接续 / 分段）

n_4	3	95	42.11	3.24	0.151	5
n_5	3	95	42.11	14.37	0.099	50
n_6	3	97	41.24	11.75	0.123	38
n_7	4	87	45.98	36.34	0.132	58
n_8	3	117	34.19	9.09	0.049	59

中文说明（原文编号 / 符号 / 数值保留）| G03-P028-009

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G03-P028-010 | 英文原文（原段接续 / 分段）

n_8	3	117	34.19	9.09	0.049	59
n_9	4	111	36.04	17.55	0.061	93
n_{10}	2	97	41.24	3.12	0.090	6
n_{11}	3	95	42.11	7.21	0.132	8
n_{12}	2	119	33.61	3.02	0.043	23
n_{13}	4	87	45.98	23.26	0.136	43

中文说明（原文编号 / 符号 / 数值保留）| G03-P028-010

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G03-P028-011 | 英文原文（原段接续 / 分段）

n_{13}	4	87	45.98	23.26	0.136	43
n_{14}	7	81	49.38	49.62	0.241	52
n_{15}	5	91	43.96	18.19	0.184	39
n_{16}	5	91	43.96	35.86	0.149	79
n_{17}	6	89	44.94	34.20	0.197	75

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G03-P028-012 | 英文原文（原段接续 / 分段）

n_{18}	5	91	43.96	24.93	0.126	35
n_{19}	5	85	47.06	44.54	0.142	64
n_{20}	5	85	47.06	23.79	0.205	27
n_{21}	3	89	44.94	14.25	0.119	23
n_{22}	3	93	43.01	11.12	0.119	25

中文说明（原文编号 / 符号 / 数值保留）| G03-P028-012

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G03-P028-013 | 英文原文（原段接续 / 分段）

n_{22}	3	93	43.01	11.12	0.119	25
n_{23}	4	93	43.01	10.11	0.166	26
n_{24}	5	85	47.06	37.87	0.169	61
n_{25}	3	95	42.11	5.75	0.133	16
n_{26}	3	95	42.11	6.08	0.133	16

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G03-P028-014 | 英文原文

greater consistency across centrality measures in the identification of central clubs than

中文译文 | G03-P028-014

在识别中心俱乐部方面，各中心性指标之间的一致性更高

G03-P028-015 | 英文原文

in the identification of central CEOs.

中文译文 | G03-P028-015

在识别核心CEO方面则并非如此

G03-P028-016 | 英文原文

We can also look at the relationship between centrality measures for the bipartite

中文译文 | G03-P028-016

我们还可以观察二部图与单模网络中各中心性指标之间的关系

G03-P028-017 | 英文原文

graph and for the one-mode networks. Table 12 presents the correlation between each

中文译文 | G03-P028-017

表12展示了各指标之间的相关性

原PDF第29页（原刊第185页）

G03-P029-001 | 英文原文

中文说明（原文编号 / 符号 / 数值保留） | G03-P029-001

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G03-P029-002 | 英文原文

Table 8
Centrality measures for CEOs and clubs in one-mode networks

中文译文 | G03-P029-002

表8 单模网络中CEO和俱乐部的中心性度量

G03-P029-003 | 英文原文

Measure of centrality

中文译文 | G03-P029-003

中心性度量方法

G03-P029-004 | 英文原文

m_1	11	14	1.85	0.080	55
m_2	36	47	4.07	0.391	122
m_3	62	84	10.70	0.673	280
m_4	38	50	6.97	0.333	200
m_5	10	13	1.09	0.091	35
m_6	15	19	1.27	0.144	52

中文说明（原文编号 / 符号 / 数值保留） | G03-P029-004

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G03-P029-005 | 英文原文（原段接续 / 分段）

m_6	15	19	1.27	0.144	52
m_7	9	13	0.70	0.070	45
m_8	11	15	2.10	0.096	49
m_9	16	22	1.99	0.162	51
m_{10}	8	11	0.95	0.052	41
m_{11}	18	22	1.51	0.167	55

中文说明（原文编号 / 符号 / 数值保留） | G03-P029-005

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G03-P029-006 | 英文原文（原段接续 / 分段）

m_{11}	18	22	1.51	0.167	55
m_{12}	18	22	0.29	0.170	51
m_{13}	18	23	2.11	0.169	57
m_{14}	8	11	0.17	0.080	28
m_{15}	34	43	3.24	0.336	106
n_1	37	40	3.85	0.178	407

中文说明（原文编号 / 符号 / 数值保留） | G03-P029-006

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G03-P029-007 | 英文原文（原段接续 / 分段）

n_2	26	29	2.60	0.127	278
n_3	25	27	0.15	0.128	248
n_4	39	42	0.15	0.213	337
n_5	27	30	0.15	0.140	274
n_6	33	36	0.00	0.174	312
n_7	37	41	3.85	0.187	392

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G03-P029-008 | 英文原文（原段接续 / 分段）

n_7	37	41	3.85	0.187	392
n_8	16	19	0.13	0.069	207
n_9	18	22	0.43	0.086	206
n_{10}	26	28	0.15	0.127	269
n_{11}	36	39	0.15	0.187	345
n_{12}	14	16	0.13	0.061	172

中文说明（原文编号 / 符号 / 数值保留） | G03-P029-008

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G03-P029-009 | 英文原文（原段接续 / 分段）

n_{12}	14	16	0.13	0.061	172
n_{13}	39	43	3.85	0.192	429
n_{14}	61	68	3.85	0.341	586
n_{15}	46	51	0.15	0.261	406
n_{16}	39	44	1.09	0.211	387
n_{17}	48	54	0.15	0.278	431

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G03-P029-010 | 英文原文（原段接续 / 分段）

n_{18}	32	37	3.09	0.178	294
n_{19}	42	47	3.85	0.201	504
n_{20}	53	58	3.85	0.289	483
n_{21}	35	38	3.85	0.168	388
n_{22}	32	35	1.24	0.168	301
n_{23}	42	46	0.15	0.235	354

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G03-P029-011 | 英文原文（原段接续 / 分段）

n_{23}	42	46	0.15	0.235	354
n_{24}	45	50	3.85	0.239	437
n_{25}	35	38	0.15	0.188	323
n_{26}	35	38	0.15	0.188	324

中文说明（原文编号 / 符号 / 数值保留） | G03-P029-011

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G03-P029-012 | 英文原文

[a]Excludes diagonal. [b]Includes diagonal.

中文译文 | G03-P029-012

排除对角线，包含对角线。

G03-P029-013 | 英文原文

kind of centrality index calculated on the bipartite graph and on the one-mode relation, for CEOs and clubs separately. For example, the first line in the body of the table gives the correlation between the degree centralities for CEOs and for clubs in the bipartite

中文译文 | G03-P029-013

一种基于二部图与单模关系计算得出的中心性指数，分别用于CEO和俱乐部。例如，表格正文中的第一行给出了CEO与俱乐部的度中心性之间的相关性。

原PDF第30页（原刊第186页）

G03-P030-002 | 英文原文

Table 9
Correlations between centrality

中文译文 | G03-P030-002

表9 中心性之间的相关性

G03-P030-003 | 英文原文

measures: clubs and CEOs in the bipartite graph

中文译文 | G03-P030-003

测量指标：二部图中的俱乐部和CEO

G03-P030-004 | 英文原文

	C_D^{NM}	C_C^{NM}	C_B^{NM}	C_E^{NM}
C_C^{NM}				
All nodes	0.694			
Clubs	0.992			
CEOs	0.627			
C_B^{NM}				
All nodes	0.928	0.752		
Clubs	0.948	0.968		
CEOs	0.867	0.659		
C_E^{NM}				
All nodes	0.889	0.872	0.816	
Clubs	0.980	0.967	0.890	
CEOs	0.780	0.823	0.576	

中文译文 | G03-P030-004

CJ C# ~" CJ C[~' c~ 所有节点 0.694 俱乐部 0.992 首席执行官 0.627 c~ ~ 所有节点 0.928 0.752 俱乐部 0.948 0.968 首席执行官 0.867 0.659 cF' 所有节点 0.889 0.872 俱乐部 0.980 0.967 首席执行官 0.780 0.823

G03-P030-005 | 英文原文

All nodes	0.938	0.485
Clubs	0.977	0.969
CEOs	0.572	0.041

中文译文 | G03-P030-005

所有节点 0.938 0.485 俱乐部 0.977 0.969 CEOs 0.572 0.041

G03-P030-006 | 英文原文

Table 10
Correlations between centrality measures: one-mode matrix of club overlaps

中文译文 | G03-P030-006

表10 中心性测量指标之间的相关性：俱乐部的单模矩阵

G03-P030-007 | 英文原文

C_D^M	C_C^M	C_B^M	C_E^M

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G03-P030-008 | 英文原文

C_C^M	0.958			
C_B^M	0.937	0.954		
C_E^M	0.993	0.930	0.918	
C_F^M	0.959	0.954	0.985	0.936

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G03-P030-009 | 英文原文

Table 11
Correlations between centrality measures: one-mode matrix of CEO co-memberships

中文译文 | G03-P030-009

表11 中心性测量指标之间的相关性：CEO共成员资格的单模矩阵

G03-P030-010 | 英文原文

Table 12
Correlations between centralities for the bipartite graph and the one-mode relations, for CEOs and clubs

中文译文 | G03-P030-010

表12 二部图的中心性与CEO和俱乐部之间的单模关系之间的相关性

G03-P030-011 | 英文原文

Centrality measure	CEOs	Clubs

中文译文 | G03-P030-011

CEO与俱乐部中心性指标

G03-P030-012 | 英文原文

Degree	0.767	0.979
Betweenness[a]	0.634	0.946
Eigenvector	1.000	1.000
Flow	0.137	0.944

中文译文 | G03-P030-012

度值：0.767、0.979；介于间性值：0.634、0.946；特征向量1.000、1.000；流量值：0.137、0.944

G03-P030-013 | 英文原文

[a]Betweenness centralities for dichotomized one-mode relation.

中文译文 | G03-P030-013

二分化单模关系下的介于间中心性

原PDF第31页（原刊第187页）

G03-P031-001 | 英文原文

中文说明（原文编号 / 符号 / 数值保留） | G03-P031-001

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G03-P031-002 | 英文原文

Fig. 4. Galois lattice for CEOs and clubs.

中文译文 | G03-P031-002

图 4. CEOs和俱乐部的高斯图结构。

G03-P031-003 | 英文原文

graph and in the one-mode actor co-membership relation (the $C_D^{\mathcal{N}\mathcal{A}}(n_i)$ and the $C_D^{\mathcal{N}}(n_i)$ and the $C_D^{\mathcal{N}\mathcal{A}}(m_k)$ and the $C_D^{\mathcal{A}}(m_k)$). For these data, there is greater discrepancy for actors (the CEOs) than for events (the clubs) between the centralities from the bipartite graph and the centralities from the one-mode relation.

中文译文 | G03-P031-003

在单模行动者二部成员资格关系中（即 CoX~(ni)、C~(n i)、Co~(m~) 和 C~(mk)）。对于这些数据而言，行动者（首席执行官）与事件（俱乐部）之间的中心性差异大于二部图中的中心性与单模关系中的中心性之间的差异。

G03-P031-004 | 英文原文

Fig. 4 presents the Galois lattice for the CEOs and clubs. [2] First, consider the relative centrality of the clubs. Clubs 3 and 4 are the only clubs whose memberships contain the memberships of other clubs (club 4 contains clubs 1 and 10; club 3 contains clubs 2, 5, 6, 11 and 14). Clubs 3 and 4 are located toward the top of the diagram in Fig. 4 and the containments are indicated by the lines descending from these clubs. More complicated patterns of containment are present among the CEOs. CEO 14 contains 25, 26 and 20 (who in turn contains 4); 15 contains 23 and 25; 17 contains 26 (who in turn contains 3); 8 contains 12; 11 contains 10; 13 contains 1 (who in turn contains 10); 19 contains both

中文译文 | G03-P031-004

图4展示了CEO们和俱乐部的Galois格子。首先，考虑俱乐部的中心性。俱乐部3和4是唯一包含其他俱乐部的成员资格的俱乐部（俱乐部4包含俱乐部1和10；俱乐部3包含俱乐部2、5、6、11和14）。俱乐部3和4位于图4图表的顶部，其包含关系由从这些俱乐部向下延伸的线条表示。CEO们中则存在更复杂的包含模式。CEO 14包含25、26和20（而20又包含4）；15包含23和25；17包含26（而26又包含3）；8包含12；11包含10；13包含l（而l又包含10）；19同时包含两者。

G03-P031-005 | 英文原文

1 and 12; and 16 contains 6. This suggests at least three levels of centrality for CEOs. The most central CEOs are 8, 11, 13, 14, 15, 16, 17 and 19; second are 1, 6, 12, 20, 23, 25 and 26; and at the third level are 3, 4 and 23. The memberships of the remaining CEOs are not included within the memberships of others, nor do they include others, thus, their centrality relative to others cannot be determined using this approach. Further analyses show that other indices of centrality for the CEOs are consistent with these three levels in the Galois lattice. CEOs in the first level have higher centrality on other indices than do CEOs in the second level. CEOs in the second level have higher centrality values than do the CEOs in the third level. CEOs whose centrality is

中文译文 | G03-P031-005

1和12；16包含6。这表明CEO至少有三个中心性层级。最中心的CEO为8、11、13、14、15、16、17和19；第二中心的是1、6、12、20、23、25和26；第三中心的是3、4和23。其余CEO的成员身份既不属于其他人的成员身份，也不包含其他人，因此无法用此方法确定他们相对于其他人的中心性。进一步分析表明，CEO的其他中心性指标与Galois格中的这三个层级一致。第一层级的CEO在其他指标上的中心性高于第二层级的CEO。第二层级的CEO的中心性值高于第三层级的CEO。中心性较高的CEO为

G03-P031-006 | 英文原文

[2] This figure was produced using the program *DIAGRAM* (Vogt and Bliegener, 1990).

中文译文 | G03-P031-006

2 该图使用DIAGRAM程序生成（Vogt和Bliegener，1990）。

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G03-P032-002 | 英文原文

indeterminate in the Galois lattice have moderate to high levels of centrality on other centrality indices.

中文译文 | G03-P032-002

在Galois格中不确定的参与关系在其他中心性指标上具有中等到高的中心性水平。

G03-P032-003 | 英文原文

In conclusion, let us consider some of the strengths and weaknesses of centrality indices for studying affiliation networks.

中文译文 | G03-P032-003

最后，让我们考察使用中心性指标研究二模参与网络时的一些优点和不足。

G03-P032-004 | 英文原文

7. Discussion

中文译文 | G03-P032-004

7. 讨论

G03-P032-005 | 英文原文

The introduction to this paper presented four ideas that might motivate centrality approaches that would be especially appropriate for affiliation networks. Let us now consider existing centrality indices and possible extensions in light of these ideas.

中文译文 | G03-P032-005

本文引言提出了四种思路，可能据此发展特别适合二模参与网络的中心性方法。现在，我们依据这些思路，考察现有中心性指标及其可能的扩展。

G03-P032-006 | 英文原文

First, it is important to have centrality scores for both actors and events in an affiliation network. This is not a problem for any of the approaches, since all can be used to analyze the bipartite graph. However, because of the duality in the relationship between actors and events, we might hope that the centralities of actors should be related in specifiable ways to the centralities of the events with which they are affiliated, and similarly the centralities of events should be related in specifiable ways to the centralities of the actors they include. These relationships hold for degree centralities calculated on the one-mode networks of actor co-memberships and event overlaps (Eqs. (10) and (11)), for closeness centrality in the bipartite graph (Eqs. (27) and (28)), and for eigenvector centralities in the bipartite graph or in the pair of one-mode matrices (Eqs. (16) and (17) or Eqs. (19) and (20)). Relationships between actor and event centralities are not as straightforward for betweenness centrality and flow betweenness centrality.

中文译文 | G03-P032-006

首先，在二模参与网络中，同时获得行动者与事件的中心性得分十分重要。对本文讨论的任何一种方法而言，这都不是问题，因为它们都可以用于分析二部图。不过，由于行动者与事件关系中的对偶性，我们可能希望：行动者的中心性与其所参与事件的中心性之间，应具有可以明确表述的关系；同样，事件的中心性与其中行动者的中心性之间，也应具有可以明确表述的关系。以下情况确实存在这些关系：在行动者共同成员关系与事件重叠关系的单模网络中计算的度中心性〔式（10）、（11）〕，二部图中的接近中心性〔式（27）、（28）〕，以及在二部图或一对单模矩阵中计算的特征向量中心性〔式（16）、（17），或式（19）、（20）〕。但在中介中心性与流中介中心性中，行动者与事件中心性的关系并没有这么直接。

G03-P032-007 | 英文原文

A second feature of affiliation networks is that the affiliation relation is non-dyadic, and thus focuses on subsets. To what extent do centralities in affiliation networks allow us to quantify the centrality of a subset of actors or a subset of events? In one sense, any index for which an actor's centrality is a function of the centralities of its events, or an event's centrality is a function of the centralities of its members, does capture the centralities of subsets. An actor's centrality is a function of the centrality of the subset of events to which it belongs. An event's centrality is a function of the subset of actors belonging to it. As noted above, these properties hold for degree, closeness, and eigenvector centralities, but not for betweenness and flow betweenness.

中文译文 | G03-P032-007

二模参与网络的第二个特征是：参与关系并非二元关系，因而着眼于子集。这类网络中的中心性，在多大程度上能够量化行动者子集或事件子集的中心性？从某种意义上说，如果某项指标将行动者的中心性定义为其所参与事件中心性的函数，或将事件的中心性定义为其成员中心性的函数，它就体现了子集的中心性。行动者的中心性，是其所属事件子集的中心性的函数；事件的中心性，是其中行动者子集的函数。如上所述，度中心性、接近中心性和特征向量中心性具有这些性质，但中介中心性与流中介中心性没有。

G03-P032-008 | 英文原文

A third important theoretical property of an affiliation network is that actors create linkages between events and events create linkages between actors. Central actors should link events and central events should link actors. Theoretically, actors that do not link events, and events that do not link actors, should not be considered central. The property of linkages among actors and/or events seems most naturally embodied in betweenness, flow betweenness, and eigenvector centrality, but is also most problematic for these approaches. Part of the problem is that misleading results arise when one analyzes the one-mode relations of actor co-memberships or event overlaps. These relations lack important information about the pattern of affiliation ties. When used to analyze the bipartite graph, betweenness centrality and flow betweenness centrality give centralities equal to zero for actors that belong only to one event and to events with only a single member, as desired. However, eigenvector centrality gives non-zero values to such actors and events, even in the bipartite graph. This property is consistent with the

中文译文 | G03-P032-008

二模参与网络的第三个重要理论性质是：行动者在事件之间建立联系，事件又在行动者之间建立联系。具有中心性的行动者应联结不同事件，具有中心性的事件应联结不同行动者。从理论上说，不联结不同事件的行动者，以及不联结不同行动者的事件，都不应被视为具有中心性。行动者和 / 或事件之间的联结性质，似乎最自然地体现于中介中心性、流中介中心性与特征向量中心性，但也正是这些方法中问题最多的部分。部分问题在于：分析行动者共同成员关系或事件重叠关系的单模网络时，会得到可能误导人的结果。这些关系缺少关于参与关系模式的重要信息。在二部图中，中介中心性与流中介中心性会如理论所预期的那样，将仅参与一个事件的行动者、以及仅有一名成员的事件的中心性设为0。然而，即使在二部图中，特征向量中心性也会为这样的行动者和事件赋予非零值。这一性质符合

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G03-P033-002 | 英文原文

intuitive motivation for eigenvector centrality that a node's centrality is a function of the centralities of the nodes to which it is adjacent, but is inconsistent with the idea that centrality in affiliation networks is a result of the linkages created by actors and events.

中文译文 | G03-P033-002

特征向量中心性的直观依据，即某节点的中心性是其邻接节点中心性的函数；但它却不符合另一种思路：二模参与网络中的中心性，应来自行动者与事件所建立的联结。

G03-P033-003 | 英文原文

The fourth empirically and theoretically important aspect of centrality in an affiliation network is the subset–superset (i.e. inclusion) relationships between actors' affiliations and events' memberships. These relationships capture the distinction between 'primary' and 'secondary' actors and events observed by numerous researchers. However, this distinction is not quantified by any of the existing centrality indices reviewed here. Subset–superset relationships are nicely displayed in the Galois lattice and are formalized by the idea of graph covers. Developing a centrality index based on these inclusion relations seems like a fruitful area for future research.

中文译文 | G03-P033-003

二模参与网络中心性的第四个具有经验及理论意义的方面，是行动者参与事件与事件成员构成之间的子集—超集关系，即包含关系。这些关系体现了许多研究者观察到的"主要"与"次要"行动者和事件之间的区别。然而，本文考察的现有中心性指标，没有任何一种能够量化这种区别。子集—超集关系可以在伽罗瓦格中清晰展示，并通过图覆盖这一概念加以形式化。基于这些包含关系构建中心性指标，看来是未来值得开展的研究方向。

G03-P033-004 | 英文原文

A final practical note is in order. Seidman (1981) discusses the misleading conclusions that can result when one studies one-mode networks of actor co-memberships or event overlaps for an affiliation network. Specifically, he demonstrates that cliques of actors in the co-membership relation (or events in the overlap relation) are not necessarily complete subgraphs in the affiliation relation. Thus, one is not necessarily justified in drawing conclusions about collections larger than pairs from the one-mode

中文译文 | G03-P033-004

最后有必要提出一项实际提醒。Seidman（1981）讨论过：研究二模参与网络时，如果考察行动者共同成员关系或事件重叠关系的单模网络，可能得出误导性结论。具体而言，他证明：共同成员关系中的行动者团（或重叠关系中的事件团），在参与关系中不一定构成完全子图。因此，仅凭单模

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relations. As we have seen, similar cautions are in order for centrality analyses of the one-mode networks derived from the affiliation network. In going from the affiliation relation to either the actor co-membership relation or the event overlap relation, one loses information about the patterns of affiliation between actors and events. Thus, one needs to be cautious when interpreting centralities for these one-mode relations.

中文译文 | G03-P033-005

关系，并不一定有理由对超过两个成员的集合下结论。如前文所见，对从二模参与网络导出的单模网络进行中心性分析时，也需要采取类似的谨慎态度。从参与关系转化为行动者共同成员关系或事件重叠关系的过程中，会丢失有关行动者与事件之间参与模式的信息。因此，解释这些单模关系的中心性时必须谨慎。

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Acknowledgements

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G03-P033-007 | 英文原文

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中文译文 | G03-P033-007

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SOCIAL NETWORKS

Centrality in affiliation networks

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Abstract

This paper discusses the conceptualization, measurement, and interpretation of centrality in affiliation networks. Although centrality is a well-studied topic in social network analysis, and is one of the most widely used properties for studying affiliation networks, virtually all discussions of centrality and centralization have concerned themselves with one-mode networks. Bonacich's work on simultaneous group and individual centralities is a notable exception (*Social Networks*, 1991, 13, 155–168). I begin by outlining the distinctive features of affiliation networks and describe four motivations for centrality indices in affiliation networks. I then consider properties of some existing centrality indices for affiliation networks, including the relationship between centralities for actors and events in these networks, and present a new conceptualization of centrality that builds on the formal properties of affiliation networks and captures important theoretical insights about the positions of actors and events in these networks. These centralities are then illustrated on Galaskiewicz's data on club and board memberships of a sample of corporate executive officers (*Social Organization of an Urban Grants Economy*. New York: Academic Press, 1985). The conclusion to this paper discusses strengths and weaknesses of centrality indices when applied to affiliation networks. © 1997 Elsevier Science B.V.

Keywords: Affiliation networks; Membership networks; Dual networks; Two-mode networks; Non-dyadic networks; Centrality

1. Affiliation networks

Many social network relations consist of the linkages among actors through their joint participation in social activities or membership in collectivities. This common activity creates a network of ties among actors. Similarly, collectivities, communities, or social occasions are linked to each other through the multiple memberships of actors. Such networks of actors tied to each other through their participation in collectivities, and collectivities linked through multiple memberships of actors, are referred to as *affiliation networks*, *membership networks*, *dual networks*, or *hypernetworks* (Breiger, 1974, 1990; Seidman, 1981, 1985; McPherson, 1982; Wasserman and Faust, 1994).

Formally, an affiliation network consists of two key elements: a set of actors and a collection of subsets of actors (called *events*). Thus, an affiliation network is a two-mode, non-dyadic network. The two modes are the set of actors and the set of

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events. An affiliation network is non-dyadic because the affiliation relation relates each actor to a subset of events, and relates each event to a subset of actors. Affiliation networks have been referred to as dual networks because of the complementary perspectives through which actors are linked to each other as members of collectivities, and collectivities are linked to each other through shared members (Breiger, 1974, 1990).

Substantive examples of affiliation networks are widespread, and include, for example, interlocking boards of directors (Levine, 1972; Mariolis, 1975; Sonquist and Koenig, 1975; Mintz and Schwartz, 1981a,b; Allen, 1982; Mizruchi, 1982; Bearden and Mintz, 1987); memberships in voluntary organizations (McPherson, 1982); club memberships (Bonacich, 1978); social gatherings (Davis et al., 1941; Homans, 1950; Breiger, 1974); ceremonial events attended by community members (Foster and Seidman, 1984; Schweizer et al., 1993); observations of informal social interactions (Bernard et al., 1980, 1982; Freeman and Romney, 1987; Freeman et al., 1987; Freeman et al., 1989) and so on. Regardless of their substance, these applications share a formal similarity in that each consists of a set of actors and a collection of subsets of actors.

In this paper I will use two examples of affiliation networks as illustrations. The first is a hypothetical affiliation network of six actors and three events from Wasserman and Faust (1994). The second example is a subset of the data collected by Galaskiewicz (1985) for his study of the urban grants economy in the Minneapolis-St. Paul area. This affiliation network consists of 26 corporate executive officers and their membership in 15 clubs, boards of cultural organizations, and corporate boards. (See Galaskiewicz, 1985; Wasserman and Faust, 1994, for further discussion of these data.)

To illustrate an affiliation network, consider the hypothetical example of six actors and three events. The set of actors is denoted by $\mathcal{N} = \{n_1, n_2, \ldots, n_g\}$, and the set of events is denoted by $\mathcal{M} = \{m_1, m_2, \ldots, m_h\}$. There are g actors and h events. The affiliation network matrix for this example is presented in Table 1. This matrix, denoted by $\boldsymbol{A} = \{a_{ik}\}$, shows the affiliation of the actors with the events. A '1' in row i, column k of $\boldsymbol{A}$ indicates that actor n_i is affiliated with event m_k. Table 2 gives the one-mode matrix of actor co-memberships. This matrix, denoted by $\boldsymbol{X}^{\mathcal{N}}$, indicates the number of memberships shared by each pair of actors. Table 3 gives the one-mode matrix of event overlaps, denoted by $\boldsymbol{X}^{\mathcal{M}}$, which gives the number of actors shared by each pair of events.

Table 1
Affiliation network matrix

	m_1	m_2	m_3
n_1	1	0	1
n_2	0	1	0
n_3	0	1	1
n_4	0	0	1
n_5	1	1	1
n_6	1	1	0

Table 2
Actor co-membership matrix

	n_1	n_2	n_3	n_4	n_5	n_6
n_1	2	0	1	1	2	1
n_2	0	1	1	0	1	1
n_3	1	1	2	1	2	1
n_4	1	0	1	1	1	0
n_5	2	1	2	1	3	2
n_6	1	1	1	0	2	2

As Breiger (1974) has noted, the affiliation matrix is related to the actor co-membership matrix and to the event overlap matrix through the following equations:

$$\boldsymbol{X}^{\mathcal{N}} = \boldsymbol{A}\boldsymbol{A}' \tag{1}$$

and

$$\boldsymbol{X}^{\mathcal{M}} = \boldsymbol{A}'\boldsymbol{A}. \tag{2}$$

The affiliation network matrix uniquely determines both $\boldsymbol{X}^{\mathcal{N}}$ and $\boldsymbol{X}^{\mathcal{M}}$, but the reverse is not true. A given actor co-membership matrix or event overlap matrix may be generated by a number of different affiliation network matrices (Breiger, 1990).

An affiliation network can also be represented as a bipartite graph (Wilson, 1982). In a bipartite graph nodes can be partitioned into two subsets and all lines are between nodes from different subsets. In the bipartite graph for an affiliation network the two sets of nodes are the set of actors, $\mathcal{N}$, and the set of events, $\mathcal{M}$, so there are $g + h$ nodes. Since the lines indicate ties of affiliation they are always between actors and events. Fig. 1 shows the hypothetical affiliation network as a bipartite graph.

The bipartite graph can also be presented in a sociomatrix, denoted by $\boldsymbol{X}^{\mathcal{N}\mathcal{M}}$. This matrix has $g + h$ rows and columns indexing actors and events, and has the form:

$$X^{\mathcal{N}\mathcal{M}} = \begin{bmatrix} 0 & A \\ A' & 0 \end{bmatrix} \tag{3}$$

where actors occupy the first g rows and columns and events occupy the last h rows and columns.

In studying an affiliation network one can look at properties of actors in the one-mode relation of actor co-memberships, properties of events in one-mode relation of event overlaps, or properties of both actors and events in the two-mode affiliation relation.

Table 3
Event overlap matrix

	m_1	m_2	m_3
m_1	3	2	2
m_2	2	4	2
m_3	2	2	4

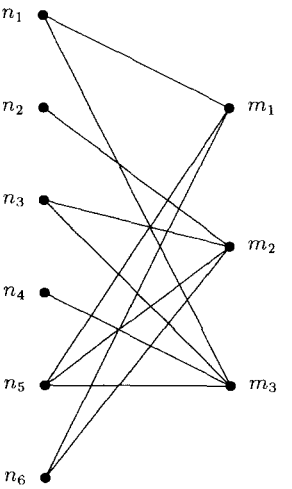


Fig. 1. Bipartite graph of affiliation network.

Centrality is one network property that frequently has been used to study actors or events in affiliation networks (Bonacich, 1972, 1991; Mariolis, 1975; Stokman, 1977; Bonacich and Domhoff, 1981; Mintz and Schwartz, 1981a,b; Mizruchi and Bunting, 1981; Mariolis and Jones, 1982; Mizruchi, 1982, 1992; Rosenthal et al., 1985; Mizruchi et al., 1986; Fernandez and McAdam, 1988). However, virtually all analyses of centrality in affiliation networks neglect the inherent properties of these networks: the presence of two modes, the duality of actors and events, and the non-dyadic affiliation relation. In the following sections I address the issue of conceptualization and measurement of centrality in affiliation networks, with special attention to the interpretation of existing centrality measures as applied to affiliation networks and to possible future directions for developing centrality approaches that are especially appropriate for affiliation networks.

2. Centrality

The general notion of centrality encompasses a number of different aspects of the 'importance' or 'visibility' of actors within a network. Discussions of centrality can be found in Freeman (1979), Knoke and Burt (1983), Friedkin (1991), Faust and Wasserman (1992) and Wasserman and Faust (1994). In general, there are four common motivations for centrality in one-mode dyadic networks:

- actors are central if they are active in the network (motivating degree centrality);
- actors are central if they can contact others through efficient (short) paths (motivating closeness centrality);
- actors are central if they have the potential to mediate flows of resources or information between other actors (motivating betweenness centrality); and
- actors are central if they have ties to other actors that are themselves central (motivating eigenvector centrality).

Freeman (1979) presents three sets of indices that formalize the ideas of degree (activity), closeness (efficiency), and betweenness (control) actor centrality and graph centralization. From a slightly different perspective, Friedkin (1991) reviews the theoretical basis for centrality measures within the framework of models of interpersonal influence. Within this framework he distinguishes between 'total', 'immediate', and 'mediative' effects centrality, roughly analogous to the total influence of an actor, an actor's closeness to others, and the betweenness of the actor. At the conclusion of his paper Friedkin cites Coombs's position on the theoretical basis for measurement and scaling models in general, commenting:

> Coombs reminds us that 'a measurement or scaling model is actually a theory about behavior, admittedly on a miniature level, but nevertheless theory' [Coombs, 1964, p. 5]. By this criterion, every new proposal of a centrality measure presents new theoretical material. (Friedkin, 1991, p. 1498.)

It is in this spirit that I intend to review centrality measures for affiliation networks. The theoretical foundations for affiliation networks as well as their formal properties differ in important ways from the more usual one-mode dyadic networks. Thus, centrality indices that have been developed for one-mode dyadic social networks may or may not be appropriate for studying affiliation networks. In addition, there may be theoretical insights gained from affiliation networks that will suggest new centrality approaches.

In the next section I discuss four motivations for centrality in affiliation networks, with special attention to those features of affiliation networks that are distinctive and lead to centrality motivations that are different from centrality motivations for one-mode networks.

3. Centrality motivations for affiliation networks

This section presents four ideas that can motivate centralities in affiliation networks. Two of these ideas arise from the formal properties of affiliation networks (two-mode, non-dyadic networks), and two arise from empirical and theoretical insights suggested by researchers who have studied affiliation networks. The four ideas that I present below are: centralities for both actors and events, centralities for subsets of actors and/or subsets of events, the importance of linkages created by actors and events, and the importance of subset–superset relationships in actors' affiliations and events' memberships.

3.1. Centralities for actors and events

Since affiliation networks are two-mode networks, a complete analysis should give centrality indices for both actors and events. As we shall see below, calculating indices for both actors and events is straightforward. The important question is: what, if anything, is the relationship between the centralities of events and the centralities of the actors that belong to the events? Different authors have taken different perspectives on this question. In their discussion of the relationship between the centrality of corporate

boards and the centrality of individuals who sit on the boards, Perrucci and Lewis (1989, p. 216) argue that

> These two networks can reveal considerable overlap in that the organizations that occupy central positions in their network are also those organizations that are the primary affiliations of the leaders that hold central positions in their network. On the other hand, the two networks can be independent in that the central organizations and the central leaders are drawn from two different subgroups.

Perrucci and Lewis view the relationship between the centrality of an actor and the centralities of the events to which the actor belongs as an empirical question. In their view, a given network may or may not show any relationship between actor and event centralities. Bonacich (1991, p. 256) takes exactly the opposite position. He argues that

> Centrality involves ... the 'duality' of groups and individuals. A central firm gets its central position from the board membership patterns of its members.... Dually a central individual should be one who belongs to a variety of important firms. One kind of centrality cannot be defined without reference to the other.

Bonacich's argument builds on the theoretically important duality in the relationship between actors and events. In keeping with this perspective, Bonacich (1991) presents a centrality measure for which the centrality of an event is proportional to the sum of the centralities of the actors that are members of the event, and the centrality of an actor is proportional to the sum of the centralities of the events to which it belongs. I discuss this measure, eigenvector centrality, in detail below.

Thus, the first important feature of centralities for affiliation networks is that there should be centrality scores for both actors and events. Furthermore, there should be a clearly specifiable relationship between these quantities.

3.2. Centralities for subsets

Another important feature of an affiliation network is that the affiliation relation is *non-dyadic* (Wilson, 1982; Seidman, 1985). Unlike a dyadic relation that links pairs of actors (and thus is binary), a non-dyadic relation links members of subsets of arbitrary size (and is thus n-ary). The affiliation relation relates each event to a *subset* of actors and relates each actor to a *subset* of events.

What does this imply for centralities in these networks? Existing centrality measures apply to one of two levels: centrality indices for actors or centralization indices for whole networks (Freeman, 1979). But if we take seriously the idea that actors in affiliation networks are defined by the collection of the events to which they belong and events are defined by their collection of members (Simmel, 1955; Breiger, 1974), then the centrality of an actor should be a function of the collection of events to which it belongs and the centrality of an event should be a function of the centrality of its collection of members. In contrast to a centrality index for a node in a graph, a centrality index for a collection of nodes should quantify the importance of that collection within the entire graph. For example, a centrality index for an event in an affiliation network

should quantify the importance of the collection of actors belonging to that event. The idea is to extend centrality indices for individual nodes to centrality indices for collections of nodes in a graph (or actors or events in an affiliation network).

These proposals of centralities for two modes and for subsets of entities build on formal properties of affiliation networks. Next I discuss theoretical ideas and empirical insights that have been used to describe the 'importance' of actors and events in affiliation networks, and which thus suggest theoretically and substantively grounded motivations for centralities in affiliation networks.

3.3. Linkages between actors and events

A theoretically important property of an affiliation network is that actors create linkages between events and events create linkages between actors. Multiple memberships of actors provide conduits for the flow of information between events or for the coordination of activities between events. Events as collections of actors provide the opportunity for contacts between actors, for sharing of information or other resources among actors, and facilitate formation of pair-wise ties between actors (McPherson, 1982; McPherson and Smith-Lovin, 1982). Bonacich and Domhoff (1981) observe that "no groups are directly related to each other; they are only indirectly related to each other through individuals" (p. 178), and similarly "central individuals link groups which tend to be disjoint" (p. 179).

The linking function of actors and events means that an actor is central in an affiliation network if it creates ties between events and an event is central if it creates ties between actors. An actor that belongs only to one event creates no ties between events. Thus, an actor's multiple memberships contribute to its centrality because that actor creates linkages between events. It is not simply the number of memberships that an actor has that is important. An actor that belongs to few events may nevertheless create a critical tie between two or more events that otherwise would not be linked. Similarly, an event is important because it creates linkages between its members. Although events with large membership lists create more ties between pairs of actors than do events with small membership lists, it is not simply the size of an event that is important. Even a small event may bring together actors that would not meet otherwise.

Thus, the linkages between events created by actors' multiple affiliations, and the linkages between actors created by events' collections of members are important considerations in determining the centrality of both actors and events in an affiliation network. Actors are always between events and events are always between actors. This suggests that some form of betweenness centrality will be appropriate for studying affiliation networks.

3.4. Subset–superset inclusions of actors and events

A fourth distinctive motivation for centrality in affiliation networks arises from observations about the structure of memberships in these networks. Memberships in affiliation networks can be viewed as patterns of inclusions among events' membership

lists, as patterns of inclusions among actors' affiliations, or both (Bonacich, 1978). Here, we shall want to look not at the number of overlaps for events or co-memberships for actors, but at the patterns of inclusions among memberships. Specifically, a number of authors have observed that in terms of their participation in events, some actors' affiliations are included within the affiliations of other actors. This leads to the distinction between 'primary' and 'secondary' actors (Davis et al., 1941; Doreian, 1979a,b; Freeman and White, 1993). Many of these insights have come from analyses of the data of Davis et al. (1941) on social activities of a community of Southern women. In their original analysis of these data, Davis et al. arrange the affiliation network matrix not only to highlight two 'cliques' of women, but also to indicate which women were primary and which were secondary within each 'clique'. In his reanalysis of these data using q-connectivity to locate cohesive subgroups, Doreian (1979a) finds essentially the same two subgroups that Davis et al. had found, and, moreover, he describes the internal structure of these subgroups. Doreian (1979a, pp. 224–225) comments that

> each of the subgroups can be viewed as having a core and a periphery, with higher connectivity within the core, lower connectivity with the periphery.

Furthermore, Doreian identifies one of the women as a 'central core member' of one of the cliques based on her high level of connectivity to others. In their reanalysis of these data using Galois lattices (which I discuss below), Freeman and White (1993) specifically describe the nature of the internal structure of the subgroups. They observe that for secondary actors

> the events they attended are all subsets of the events attended by one or more primary actors. (p. 138.)

Freeman and White also note that events' memberships can be characterized by similar subset–superset relationships.

These observations suggest that centrality indices for affiliation networks should capture the subset–superset relationships in the affiliations of actors and events. The distinction between primary and secondary actors means that secondary actors are more likely to participate in events when primary actors are also present, and are unlikely to participate on their own. Primary actors, on the other hand, participate in events even when secondary actors are not present. The participation of secondary actors is conditional on the participation of primary actors, implying that the events to which secondary actors belong are a subset of the events to which primary actors belong (Freeman and White, 1993). In terms of centrality in affiliation networks, less central actors participate in events only in the presence of more central actors. Importantly, it is not simply the greater number of events to which more or less central actors belong, but the patterning of these events such that the events to which less central actors belong are a subset of the events to which more central actors belong (Bonacich, 1978). Similarly, one can consider the relative centrality of events using these ideas. Less central events have membership lists that are contained within the memberships of more central events. Again, it is not simply the size of the events that determines their centrality, but rather it is the patterning of memberships in these events that is important.

3.5. Summary

In summary, the properties of affiliation networks suggest that centralities for these networks should do four things:

1. give centrality indices for both actors and events;
2. be extendable to subsets of actors and events;
3. focus on the linkages between actors and events through overlapping memberships;
4. capture subset–superset inclusion relations between actors and events.

Analyses of affiliation networks have seldom used these ideas to study the centrality of actors and events in affiliation networks. Rather, more conventional centrality motivations for one-mode networks have been widely employed to study affiliation networks.

In addition, virtually all analyses of affiliation networks study one-mode networks of actors or events derived from the original affiliation network. For example, many researchers have studied the affiliation of corporate executives with the corporate boards on which they sit. With the exception of Bonacich (1991) and Bonacich and Domhoff (1981), centrality analyses of these data always study either the board member ties of co-membership or the corporation ties of interlocking directorates, but do not study both simultaneously.

Studying only a single mode from the two-mode network ignores the fundamental duality inherent in the affiliation relation. The relationship between the centrality of actors and the centrality of the events to which they belong, or the relationship between the centrality of events and the centrality of their members, cannot be studied directly by looking at the one-mode networks separately.

In the next sections I consider five centrality indices: degree, eigenvector, closeness, betweenness, and flow betweenness. For each I describe the application of the index to affiliation network data and discuss the results with respect to the centrality motivations introduced above.

4. Centralities for affiliation networks

A review of analyses of centrality in affiliation networks reveals that virtually all applications use one of two approaches to centrality: (1) some form of degree centrality or (2) some form of eigenvector centrality. I begin by discussing degree and eigenvector centrality, since they have been widely used to study centrality in affiliation networks. I then discuss closeness, betweenness, and flow betweenness centralities. As we shall see, calculating basic centrality measures for both actors and events in an affiliation network is straightforward. To study both actors and events one can analyze the bipartite graph using standard centrality approaches. An important issue for each measure is to specify, if possible, the relationship between actor and event centralities.

Since centrality indices will be defined for several different kinds of graphs and networks (a one-mode dyadic network, a bipartite graph, a one-mode actor co-membership relation, or a one-mode event overlap relation) and indices will be presented for nodes in a graph, and for actors and events in affiliation networks, we shall need

notation that reflects these distinctions. $C_A(\bullet)$ will refer to centrality definition 'A' for entity '●'. For example, $C_C(\bullet)$ will refer to closeness centrality. The entity for which the centrality is defined can be an actor, denoted by n_i, an event, denoted by m_k, or a node in a generic graph, denoted p_i. In addition, when the centrality definition applies to a specific kind of relation or matrix (for example the bipartite graph, or the actor co-membership relation) I will use the following superscripts: $\mathcal{NM}$ for the bipartite graph, $\mathcal{N}$ for the one-mode actor co-membership relation, and $\mathcal{M}$ for the one-mode event overlap relation. Thus, $C_C\mathcal{NM}(n_i)$ will refer to the closeness centrality of an actor calculated on the bipartite graph. In the absence of a superscript, the definition refers to a one-mode dyadic graph or network.

4.1. Degree centralities

Degree centrality, denoted by $C_D(p_i)$, is one of the most straightforward centrality indices. In an affiliation network motivations for degree centrality are that actors are important because of their level of activity or the number of contacts that they have, and events are important because of the size of their memberships (Stokman, 1977; Mizruchi and Bunting, 1981; Mariolis and Jones, 1982; Fernandez and McAdam, 1988).

In a graph for a single dichotomous relation, the degree of a node is the number of nodes adjacent to it. For a nondirectional relation the degree of a node is equal to the sum of the values in the row (or column) of the sociomatrix; $C_D(p_i) = \sum_{j=1}^{g} x_{ij}$. In an affiliation network there are several ways to calculate degree centrality depending on whether one focuses on the bipartite graph (including both actors and events) or on the one-mode networks of actor co-memberships or event overlaps. Interpretations of the results will differ.

First, focus on the bipartite graph in which both actors and events are represented as nodes and lines represent affiliation ties between actors and events. In the bipartite graph the degree centrality of an actor is the number of events with which it is affiliated and the degree centrality of an event is the number of actors affiliated with it. These quantities are also equal to the entries on the main diagonal of the actor co-membership matrix $\boldsymbol{X}^{\mathcal{N}}$ (for actors) or the event overlap matrix $\boldsymbol{X}^{\mathcal{M}}$ (for events), or to the row total (for actors) or the column total (for events) of the affiliation matrix $\boldsymbol{A}$. Thus, actor and event degree centralities in the bipartite graph are:

$$C_D^{\mathcal{NM}}(n_i) = \sum_{k=1}^{g+h} x_{ik}^{\mathcal{NM}} = x_{ii}^{\mathcal{N}} = a_{i+} \tag{4}$$

$$C_D^{\mathcal{NM}}(m_k) = \sum_{i=1}^{g+h} x_{ik}^{\mathcal{NM}} = x_{kk}^{\mathcal{M}} = a_{+k}. \tag{5}$$

Although degree centrality is usually defined for dichotomous relations, there are also a number of ways to define degree-like centralities for the one-mode valued relations of actor co-memberships and event overlaps. Both of these are valued, nondirectional relations where the values indicate the number of co-memberships for pairs of actors or the number of overlaps for pairs of events.

Consider the entries in a given row (or column) of the actor co-membership matrix, $\boldsymbol{X}^{\mathcal{N}}$. The number of non-zero off-diagonal entries in the row gives the number of distinct actors with which the actor shares any event in common. However, we can also define other possible degree-like measures for this relation. Suppose we take the sum of the entries in a row. For actors in the co-membership relation this degree centrality is:

$$C_D^{\mathcal{N}}(n_i) = \sum_{j=1}^{g} x_{ij}^{\mathcal{N}} = x_{i+}^{\mathcal{N}}. \tag{6}$$

Interestingly, this quantity is equal to the sum of the sizes of the events to which the actor belongs:

$$C_D^{\mathcal{N}}(n_i) = \sum_{n_i \in m_k} x_{kk}^{\mathcal{M}} = \sum_{n_i \in m_k} a_{+k}. \tag{7}$$

We can think of $C_D^{\mathcal{N}}(n_i)$ as the actor's total 'volume' of activity. It is the number of contacts that an actor has with other actors, counting other actors each time they are encountered.

For events, the similar degree centrality is equal to

$$C_D^{\mathcal{M}}(m_k) = \sum_{l=1}^{h} x_{kl}^{\mathcal{M}} = x_{k+}^{\mathcal{M}}. \tag{8}$$

This quantity can be thought of as the total participation of the actors that belong to the event, and is equal to the sum of the number of memberships for all actors in the event:

$$C_D^{\mathcal{M}}(m_k) = \sum_{n_i \in m_k} x_{ii}^{\mathcal{N}} = \sum_{n_i \in m_k} a_{i+}. \tag{9}$$

If these centralities are calculated including the diagonal elements of $\boldsymbol{X}^{\mathcal{N}}$ or $\boldsymbol{X}^{\mathcal{M}}$, then they are also equal to the number of two-step walks from the actor or event node in the bipartite graph.

Taken together, Eqs. (7) and (9) show that for the one-mode networks of actor co-memberships and event overlaps there is a specifiable relationship between an actor's centrality and the centrality of the events to which it belongs, and between an event's centrality and the centrality of its members:

$$C_D^{\mathcal{N}}(n_i) = \sum_{n_i \in m_k} C_D^{\mathcal{NM}}(m_k) \tag{10}$$

$$C_D^{\mathcal{M}}(m_k) = \sum_{n_i \in m_k} C_D^{\mathcal{NM}}(n_i). \tag{11}$$

The degree centrality of an actor is equal to the sum of the sizes of its events (the degrees of these events in the bipartite graph) and the degree centrality of an event is equal to the sum of the number of memberships of its actors (the degrees of these actors in the bipartite graph).

Degree centrality indices for actors and events in the bipartite graph for the hypothetical example are presented in Table 4. Table 5 presents the degree centralities

Table 4
Centrality of actors and events in the bipartite graph

	Measure of centrality					
	$C_D^{\mathcal{NM}}$	$\sum d(i,k)$	$C_C^{\mathcal{NM}}$	$C_B^{\mathcal{NM}}$	$C_F^{\mathcal{NM}}$	$C_E^{\mathcal{NM}}$
n_1	2	17	0.4706	1.75	5	0.286
n_2	1	21	0.3810	0.00	0	0.154
n_3	2	15	0.5333	3.00	5	0.309
n_4	1	21	0.3810	0.00	0	0.154
n_5	3	13	0.6154	6.50	3	0.440
n_6	2	17	0.4706	1.75	5	0.286
m_1	3	16	0.5000	4.00	12	0.365
m_2	4	14	0.5714	10.50	19	0.428
m_3	4	14	0.5714	10.50	19	0.428

Table 5
Centrality measures for actors and events in one-mode networks

	Measure of centrality									
	$C_D^{\mathcal{N}a}$	$C_D^{\mathcal{N}b}$	$C_B^{\mathcal{N}01}$	$C_F^{\mathcal{N}}$	$C_E^{\mathcal{N}}$	$C_D^{\mathcal{M}a}$	$C_D^{\mathcal{M}b}$	$C_B^{\mathcal{M}01}$	$C_F^{\mathcal{M}}$	$C_E^{\mathcal{M}}$
n_1	5	7	0.33	7	0.404					
n_2	3	4	0.00	4	0.218					
n_3	6	8	1.17	10	0.437					
n_4	3	4	0.00	4	0.218					
n_5	8	11	1.17	13	0.622					
n_6	5	7	0.33	7	0.404					
m_1						4	7	0.00	2	0.515
m_2						4	8	0.00	2	0.606
m_3						4	8	0.00	2	0.606

[a]Excludes diagonal. [b]Includes diagonal.

for the one-mode networks of actor co-memberships and event overlaps, calculated both with diagonal entries included and with diagonal entries excluded. [1]

A number of authors have used versions of degree centrality to study centrality in affiliation networks. Romney and Weller (1984) use this index to measure the overall level of people's observed behavioral interaction in data from three groups studied by Bernard et al. (1980, Bernard et al., 1982). In their study of corporate interlock networks, Mariolis and Jones (1982) use this index as one measure of a firm's centrality, referring to it as 'the number of interlocks'. Similarly, in his study of delegations' co-sponsorship of bills within the United Nations, Stokman (1977) uses the number of co-sponsorship ties that a delegation has as an index of the delegation's centrality in the co-sponsorship network.

Some authors have criticized degree centrality because it does not consider the centrality of the actors (or events) to which an actor (or event) is adjacent. According to

[1] Centrality indices were calculated using *UCINET IV* (Borgatti et al., 1992).

this argument, two actors may be adjacent to the same number of others, but an actor is more central if it has ties to actors that themselves are quite central. One way to deal with this issue is to incorporate the centrality of the actors to which a given actor is adjacent into the centrality index. This is what eigenvector centrality does.

4.2. Eigenvector centrality

The rationale for eigenvector centrality in a one-mode network is that the centrality of an actor should be proportional to the strength of the actor's ties to other network members and the centrality of these other actors (Bonacich, 1972; Mizruchi, 1982; Mizruchi et al., 1986). Originally, Bonacich (1972) motivated this centrality index as a measure of popularity, related to the measures of relative standing or status proposed by Katz (1953) and Hubbell (1965). Recently, people have defended the use of this index as a measure of the extent to which actors are in a position to influence others in the network (Mizruchi and Bunting, 1981; Fernandez and McAdam, 1988; Friedkin, 1991). Friedkin (1991) describes this index as 'total effects centrality' in the context of network influence models.

Eigenvector centrality is widely used in studies of interlocking corporate boards of directors (Mariolis, 1975; Mintz and Schwartz, 1981a,b; Mizruchi and Bunting, 1981; Mariolis and Jones, 1982; Mizruchi, 1982; Roy, 1983; Rosenthal et al., 1985). For many of these researchers centrality *is* eigenvector centrality (Mariolis, 1975; Mintz and Schwartz, 1981a,b; Roy, 1983; Rosenthal et al., 1985).

In this section I review eigenvector centrality in general and then show how the centrality indices for actors are related to the centrality indices for events in an affiliation network.

Denoting the eigenvector centrality of node p_i in a one-mode network by $C_E(p_i)$, eigenvector centrality is expressed as

$$C_E(p_i) \approx C_E(p_j)\, x_{ij}. \tag{12}$$

The centrality of a node is proportional to the centrality of the nodes to which it is adjacent, weighted by the value of the tie between the nodes.

Finding centrality values, $C_E(p_i)$, that satisfy this equation for all nodes in a graph involves solving a system of simultaneous linear equations. This standard eigenvector–eigenvalue problem is expressed by the equation

$$\boldsymbol{X}\boldsymbol{c} = \lambda \boldsymbol{c} \tag{13}$$

where $\boldsymbol{X}$ is a $g \times g$ sociomatrix, λ is its largest eigenvalue, and $\boldsymbol{c}$ is a vector of centrality scores (the eigenvector corresponding to the largest eigenvalue).

I will denote the vector of eigenvector centrality scores for actors by $\boldsymbol{c}^{\mathcal{N}}$ and the vector of eigenvector centrality scores for events by $\boldsymbol{c}^{\mathcal{M}}$. One can find these scores by analyzing the one-mode actor co-membership matrix and the one-mode event overlap matrix:

$$\boldsymbol{X}^{\mathcal{N}}\boldsymbol{c}^{\mathcal{N}} = \boldsymbol{A}\boldsymbol{A}'\boldsymbol{c}^{\mathcal{N}} = \lambda^2 \boldsymbol{c}^{\mathcal{N}} \tag{14}$$

$$\boldsymbol{X}^{\mathcal{M}}\boldsymbol{c}^{\mathcal{M}} = \boldsymbol{A}'\boldsymbol{A}\boldsymbol{c}^{\mathcal{M}} = \lambda^2 \boldsymbol{c}^{\mathcal{M}}. \tag{15}$$

Bonacich suggests that the scores be scaled so that $\sum C_E^{\mathcal{N}}(n_i)^2 = \sum C_E^{\mathcal{M}}(m_k)^2 = 1$, though other scalings are possible. The actor and event eigenvector centralities are related to each other through the set of equations:

$$\boldsymbol{A}\boldsymbol{c}^{\mathcal{N}} = \lambda \boldsymbol{c}^{\mathcal{M}} \tag{16}$$

$$\boldsymbol{A}'\boldsymbol{c}^{\mathcal{M}} = \lambda \boldsymbol{c}^{\mathcal{N}}. \tag{17}$$

Eqs. (16) and (17) show the duality of actor and event centralities for the affiliation network. The centrality of an actor is proportional to the centralities of the events with which it is affiliated, and the centrality of an event is proportional to the centralities of its members.

Bonacich (1991) shows that for an affiliation network one can find the actor and event centrality scores by analyzing a number of different matrices. Specifically, one can analyze the affiliation network matrix, $\boldsymbol{A}$, the pair of one-mode matrices, $\boldsymbol{X}^{\mathcal{N}}$ and $\boldsymbol{X}^{\mathcal{M}}$ (Eqs. (14) and (15)), or the matrix for the bipartite graph, $\boldsymbol{X}^{\mathcal{N}\mathcal{M}}$. Analysis of the bipartite graph is expressed in the equation:

$$\lambda \begin{bmatrix} \boldsymbol{c}^{\mathcal{N}} \\ \boldsymbol{c}^{\mathcal{M}} \end{bmatrix} = \begin{bmatrix} 0 & \boldsymbol{A} \\ \boldsymbol{A}' & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{c}^{\mathcal{N}} \\ \boldsymbol{c}^{\mathcal{M}} \end{bmatrix} \tag{18}$$

(Bonacich, 1991, p. 158). The generality of this approach is apparent in the fact that the eigenvector centrality scores from any of these analyses are equivalent, once they are identically scaled.

For eigenvector centralities there is a clear and specifiable relationship between the centralities for actors and the centralities for events. This relationship is more apparent if we rewrite Eqs. (16) and (17) in terms of individual actor and event centrality indices. An actor's centrality is a function of the centralities of the events to which it belongs:

$$C_E^{\mathcal{N}}(n_i) = \frac{1}{\lambda} \sum_{k=1}^{h} C_E^{\mathcal{M}}(m_k) a_{ik}. \tag{19}$$

Similarly, the centrality of an event is a function of the centralities of its members:

$$C_E^{\mathcal{M}}(m_k) = \frac{1}{\lambda} \sum_{i=1}^{g} C_E^{\mathcal{N}}(n_i) a_{ik}. \tag{20}$$

Eigenvector centrality explicitly incorporates the duality between actor and event centralities (Bonacich, 1991). This duality is expressed in the pairs of Eqs. (16) and (17) or Eqs. (19) and (20) in which actors' centralities are proportional to the centralities of the events to which they belong, and events' centralities are proportional to the centralities of their members.

A similar relationship between actor and event scores results from a correspondence analysis of the affiliation matrix, $\boldsymbol{A}$ (Bonacich, 1991; Wasserman and Faust, 1994). One of the goals of correspondence analysis is to assign scores to the rows and columns of a data array so that a row's score is proportional to the weighted average of the column scores and a column's score is proportional to the weighted average of the row scores; the weights are the relative frequencies in the columns or rows, respectively. For an affiliation network, an actor's score is proportional to the average of the scores for the events with which it is affiliated and an event's score is proportional to the average of

the scores for the actors that belong to the event. Correspondence analysis of an affiliation matrix, $\boldsymbol{A}$, results in three sets of information:

- a set of g actor scores on each of W dimensions $\{u_{iw}\}$ for $i = 1,2,\ldots,g$ and $w = 1,2,\ldots,W$;
- a set of h event scores on each of W dimensions $\{v_{kw}\}$ for $k = 1,2,\ldots,h$ and $w = 1,2,\ldots,W$, and
- a set of W principal inertias $\{\eta_w^2\}$ for $w = 1,2,\ldots,W$.

Correspondence analysis scores for rows (actors) and columns (events) are related to each other through the following equations:

$$\eta_w u_{iw} = \sum_{k=1}^{h} \frac{a_{ik}}{a_{i+}} v_{kw}$$
$$\eta_w v_{kw} = \sum_{i=1}^{g} \frac{a_{ik}}{a_{+k}} u_{iw}. \tag{21}$$

These row and column scores are found through a singular value decomposition of an appropriately scaled affiliation matrix: $\tilde{\boldsymbol{A}} = \{a_{ik}/\sqrt{a_{i+}a_{+k}}\}$. Although the relationship between actor and event correspondence analysis scores, Eq. (21), appears similar to the relationship between actor and event eigenvector centrality scores, Eqs. (19) and (20), the resulting scores can be quite different. This difference is due to the fact that eigenvector centralities result from a decomposition of $\boldsymbol{A}$ whereas correspondence analysis scores result from a decomposition of $\tilde{\boldsymbol{A}}$.

The eigenvector centralities for actors and events in the hypothetical example are presented in Table 4 (for the bipartite graph) and Table 5 (for the one-mode relations of actor co-memberships and event overlaps). To illustrate the duality of actor and event eigenvector centralities, consider the eigenvector centrality of n_1. For the one-mode actor co-membership relation, $\lambda = 2.775$ and $C_E^{\mathcal{N}}(n_1) = 0.404$. Actor n_1 is affiliated with events m_1 and m_3, with centralities in the event overlap relation of $C_E^{\mathcal{M}}(m_1) = 0.515$ and $C_E^{\mathcal{M}}(m_3) = 0.606$. Thus, n_1's centrality is equal to

$$0.404 = \frac{(0.515)(1) + (0.606)(0) + (0.606)(1)}{2.775}. \tag{22}$$

A number of people have argued that the magnitudes of eigenvector centralities for events in the event overlap relation are undesirably affected by differences in the sizes of events (Mariolis, 1975; Mizruchi and Bunting, 1981; Mariolis and Jones, 1982; Mizruchi, 1982; Bonacich, 1991). Thus, they argue, steps should be taken to remove this effect. Two approaches have been proposed: (1) standardize the event overlap measure $x_{kl}^{\mathcal{M}}$ prior to analysis, or (2) remove from the centrality index that component which is due to the degree (i.e. the size) of the event.

The first way to accommodate different event sizes is to standardize the entries in the event overlap matrix, $\boldsymbol{X}^{\mathcal{M}}$, prior to finding eigenvector centralities. Mariolis (1975) suggests that standardized values $\boldsymbol{X}^{*\mathcal{M}} = \{x_{kl}^{*\mathcal{M}}\}$ be defined as:

$$x_{kl}^{*\mathcal{M}} = \frac{x_{kl}^{\mathcal{M}}}{\sqrt{x_{kk}^{\mathcal{M}} x_{ll}^{\mathcal{M}}}}. \tag{23}$$

Since $x_{kk}^{\mathcal{MM}}$ and $x_{ll}^{\mathcal{MM}}$ are the sizes of events m_k and m_l respectively, this standardization takes event sizes into account by dividing the measure of overlap by the square root of the product of the event sizes. However, Bonacich (1991) demonstrates that this standardization fails to remove size differences. Moreover, the event eigenvector centralities from this standardized overlap measure will not be identical to event eigenvector centralities for the bipartite graph, unless entries in $\boldsymbol{X}^{\mathcal{NM}}$ are also appropriately standardized. The appropriate standardization defines a new affiliation network matrix $\boldsymbol{A}^* = \{a_{ik}^*\}$ where $a_{ik}^* = a_{ik}/\sqrt{a_{+k}}$. The new sociomatrix for the bipartite graph, $\boldsymbol{X}^{*\mathcal{NM}}$, then has the form

$$\begin{bmatrix} 0 & \boldsymbol{A}^{\star} \\ \boldsymbol{A}^{\star\prime} & 0 \end{bmatrix}. \tag{24}$$

The event eigenvector centralities for $\boldsymbol{X}^{*\mathcal{NM}}$ will be identical to the event eigenvector centralities for $\boldsymbol{X}^{*\mathcal{M}}$, once they are scaled as described above. However, the actor eigenvector centralities from $\boldsymbol{X}^{*\mathcal{NM}}$ will *not* be identical to actor eigenvector centralities from $\boldsymbol{X}^{\mathcal{N}}$. Thus, by standardizing $\boldsymbol{X}^{\mathcal{M}}$ to 'remove' differences in event sizes, one loses the important duality between actor and event centralities.

A second way to deal with differences in event sizes exploits information about the components of the eigenvector centrality index. Bonacich (1991) shows that this index is the sum of a weighted series of components, each of which quantifies the contribution to the overall centrality of the node due to paths of length $1,2,\ldots,\infty$ from the node. One can remove the effect of event sizes by 'removing' the component of centrality due to paths of length 1 (the size of the event). The advantage of this approach is that one can analyze the bipartite graph and obtain both actor and event centrality scores, thus maintaining the duality between actor and event centralities.

One potentially undesirable feature of eigenvector centralities for studying affiliation networks is that actors that belong to a single event (and thus create no ties between events) and events that have only a single member (and thus create no ties between actors) can have non-zero eigenvector centrality. For example, in the hypothetical affiliation network, n_2, which belongs to only one event, has eigenvector centrality of 0.154. This is due to the centrality that actors and events derive from their ties to others that are central. Non-zero centrality is inconsistent with the theoretical argument that actors in affiliation networks are central to the extent that they link events, and events are central to the extent that they link actors. However, it is consistent with the 'status' or 'popularity' motivation for this index that central actors derive their centrality through their ties to central actors.

Both degree centrality and eigenvector centrality have been widely used to study affiliation networks. However, only Bonacich and Domhoff (1981) and Bonacich (1991) consider both actor and event centralities and the relationship between them. In the next sections I discuss centralities that were not designed to study affiliation networks and have not been used to analyze these networks.

4.3. Closeness centrality

Closeness centrality is based on the geodesic distances between nodes in a graph, and is not applicable to valued relations. Thus, I will restrict my attention to closeness

centralities for the bipartite graph. The basis for the closeness centrality index is the average geodesic distance that a node is from all other nodes in the graph. Thus it begins with the 'farness' of a node from other nodes in the graph. Distances between nodes are summarized in the matrix $\boldsymbol{D} = \{d(i,j)\}$, where the entry in cell (i,j) of this matrix is the geodesic distance from node i to node j. In other words, it is the length of any shortest path between the nodes. Since the bipartite graph is nondirectional, the distance from i to j is the same as the distance from j to i, and $\boldsymbol{D}$ is symmetric.

In general, closeness centrality, $C_C(p_i)$, is the inverse of the average geodesic distance between the node and all other nodes in the graph, and is calculated as

$$C_c(p_i) = \left[\frac{\sum_{j=1}^{g} d(i,j)}{g-1}\right]^{-1}. \tag{25}$$

If the graph is disconnected, then closeness centrality is undefined, since some pairs of nodes are unreachable and the distance between them is infinite.

The numerator of Eq. (25) is the sum of the distances from node i to all other nodes in the graph. For an actor in an affiliation network this is the distance from the actor to other actors plus the distance from the actor to all events. However, since the affiliation network is a bipartite graph, actors are only adjacent to events, and all paths emanating from an actor must first pass through the events to which the actor belongs. Similarly, since events are only adjacent to actors, all paths from events must pass through the actors that are their members.

Consider the relationship between the closeness centrality of an actor and the closeness centralities of the events to which the actor belongs, and the relationship between the closeness centrality of an event and the closeness centralities of its actors. First focus on the distances from an actor in the bipartite graph as a function of the distances from the events to which it belongs. The distance from node i representing an actor to any node j (either actor or event) is $d(i,j) = 1 + \min_k d(k,j)$, for event nodes k adjacent to i. Given this property, the numerator of Eq. (25) for the closeness centrality of an actor in the bipartite graph can be expressed as a function of the distances from the actor's events, k:

$$\sum_{j=1}^{g+h} d(i,j) = \sum_{j=1}^{g+h}\left[1 + \min_k d(k,j)\right] = g + h - 1 + \sum_{j=1}^{g+h} \min_k d(k,j) \tag{26}$$

for k adjacent to i, and j not equal to i. Combining Eqs. (25) and (26) gives the following expression for the closeness centrality of an actor as a function of the geodesic distances from its events to other actors and events:

$$C_C^{\mathcal{AM}}(n_i) = \left[1 + \frac{\sum_{j=1}^{g+h} \min_k d(k,j)}{g+h-1}\right]^{-1} \tag{27}$$

for events k adjacent to actor i. Thus, the closeness centrality of an actor is a function of the minimum distances from any of its events to other actors and events in the bipartite graph.

Similarly, the closeness centrality of an event is a function of the minimum distances from its actors to other actors and to other events. Eq. (27) can be written for events as:

$$C_C^{\mathcal{NM}}(m_k) = \left[1 + \frac{\sum_{i=1}^{g+h} \min_j d(i,j)}{g+h-1}\right]^{-1} \tag{28}$$

for actors i adjacent to event k.

The combination of Eqs. (27) and (28) shows that there is a clear relationship between the closeness centralities of actors and the closeness centralities of events in an affiliation network. The closeness centrality of an actor is a function of the minimum distances to its events, and the closeness centrality of an event is a function of the minimum distances to its actors.

Table 4 gives the closeness centralities for actors and events in the bipartite graph for the hypothetical example.

4.4. Betweenness centrality

Betweenness centrality for a one-mode dyadic network focuses on the extent to which actors sit on geodesic paths between other pairs of actors (Freeman, 1979). Since calculating betweenness centrality requires considering *all* geodesics in a graph, it seems unlikely that it will be possible to express the betweenness of a node in a graph (an actor or event in an affiliation network) as a function simply of the betweenness of the nodes to which it is adjacent. Nevertheless, this section presents some informal observations about the relationship between actor and event betweenness centralities in affiliation networks. Betweenness centrality is defined for dichotomous, nondirectional relations, so I will concentrate betweenness of actors and events in the bipartite graph for the affiliation network. However, at the end of this section I consider what would happen if one were to analyze the one-mode relations of actor co-memberships or event overlaps.

An intermediate step in calculating betweenness centrality is to find the 'partial betweenness' of nodes in the network (Freeman, 1979). Node p_i's partial betweenness counts the number of pairs of other nodes whose geodesic(s) contain node p_i. If there is more than one geodesic between a given pair of nodes, then p_i receives fractional credit, where the fraction is reciprocal of the number of geodesics between the pair. Let g_{jk} be the number of geodesics between p_j and p_k, and let $g_{jk}(p_i)$ be the number of geodesics between p_j and p_k that contain p_i. If all geodesics are equally likely, then the probability that a geodesic between p_j and p_k contains node p_i is equal to $g_{jk}(p_i)/g_{jk}$. The betweenness centrality of node p_i, denoted by $C_B(p_i)$, is defined as the sum of these quantities across all pairs of nodes:

$$C_B(p_i) = \sum_{j<k} \frac{g_{jk}(p_i)}{g_{jk}}. \tag{29}$$

For a graph with g nodes, $C_B(p_i)$ reaches its maximum value of $(g-1)(g-2)/2$ when node p_i is on geodesics between all other pairs of nodes.

Returning to an affiliation network, recall that linkages between pairs of actors are always through actors' participation in events, thus events are always on geodesics

between actors. Similarly, linkages between pairs of events are always through the joint memberships of actors, thus actors are always on geodesics between events. Consider the relationship between the betweenness centrality of an actor and the betweenness centrality of the events with which it is affiliated, and the relationship between the betweenness centrality of an event and the betweenness of its members.

In calculating the betweenness centrality of an event, m_k, in an affiliation network, focus on the collection of actors that belong to that event. Event m_k is on a geodesic between all pairs of actors that are members of it. Since event m_k has a_{+k} members, there are a_{+k}^2 pairs of actors that have a geodesic that contains m_k. If a given pair of actors, (n_i,n_j), only shares event m_k in common (thus $x_{ij}^{\mathcal{N}} = 1$) then m_k is on the only geodesic between them, and m_k's betweenness is incremented by 1. Actually, n_i and n_j share $x_{ij}^{\mathcal{N}}$ memberships, thus m_k's betweenness is incremented by $1/x_{ij}^{\mathcal{N}}$ for each pair of actors (n_i,n_j) in m_k. Thus, a portion of the betweenness centrality of event m_k can be expressed in terms of the number of co-memberships of pairs of its members as:

$$\frac{1}{2}\sum_{n_i,n_j\in m_k}\frac{1}{x_{ij}^{\mathcal{N}}}. \tag{30}$$

From this we see that an event gains betweenness centrality to the extent that pairs of its members meet only in that event.

In addition, an event gains betweenness centrality if an individual actor belongs only to that event. In that case, all geodesics from such an actor must contain the event. Since there are $g+h$ nodes (actors and events) in the affiliation network, an event gains $g+h-2$ betweenness 'points' for each of its members that belongs to no other event. This quantity is not independent of the count in Eq. (30).

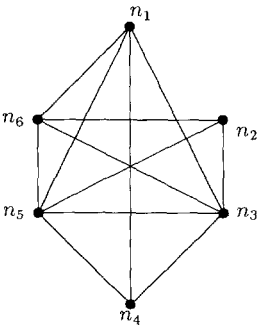
These observations suggest that the betweenness centrality of an event increases to the extent that its members belong to no other events. Since such single-membership actors are not between any pairs of events, an event gains betweenness centrality if it contains 'non-central' actors. Also, the betweenness centrality of an event increases to the extent that pairs of actors share only that event in common.

Similar properties of betweenness centrality hold for actors. An actor gains $g+h-2$ betweenness centrality 'points' if it is the only member of an event. An actor gains $1/x_{kl}^{\mathcal{N}}$ betweenness 'points' for all pairs of events to which it belongs.

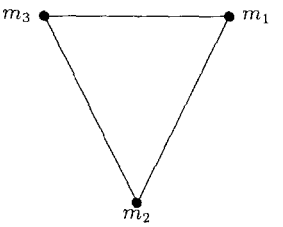
Betweenness centralities for the actors and events in the bipartite graph for the hypothetical example are presented in Table 4. Notice that $C_B(n_2) = C_B(n_4) = 0.00$ since actors n_2 and n_4 each belong to only one event, and thus create no linkages between either actors or events.

As mentioned above, betweenness centrality is defined for networks in which relations are dichotomous and nondirectional. However, sometimes it is tempting to analyze valued nondirectional relations by dichotomizing the original values and analyzing the new dichotomous relation. For example, one could define a dichotomous actor co-membership relation, $x_{ij}^{\mathcal{N}01}$, where

$$x_{ij}^{\mathcal{N}01} = \begin{cases} 0 & \text{if } x_{ij}^{\mathcal{N}} = 0 \\ 1 & \text{if } x_{ij}^{\mathcal{N}} > 0. \end{cases} \tag{31}$$



a. Dichotomous actor co-membership relation



b. Dichotomous event overlap relation

Fig. 2. Graphs for dichotomous one-mode relations.

In this dichotomous relation actors are adjacent if they ever belong to the same event. This relation has been defined by Seidman (1985, p. 368) as the 'q-overlap graph', where actors are adjacent if they share at least q events, and $q = 1$. Doreian (1969) presents a similar idea for valued graphs. One could also define a dichotomous event overlap relation $x_{kl}^{\mathcal{M}01}$ where

$$x_{kl}^{\mathcal{M}01} = \begin{cases} 0 & \text{if} \quad x_{kl}^{\mathcal{M}} = 0 \\ 1 & \text{if} \quad x_{kl}^{\mathcal{M}} > 0. \end{cases} \tag{32}$$

In this dichotomous relation events are adjacent if they share any actor in common. Stokman (1977) describes either of these relations as creating a new graph by 'induction' where nodes in the new graph are adjacent if they are adjacent to the same node in the original graph.

Graphs for these two dichotomous relations for the hypothetical example are presented in Fig. 2 Notice that the graph for the dichotomous event overlap relation is complete (all events are adjacent) since each pair of events shares at least one actor. Since the graph is complete there is no differentiation among events; all are equally central in this graph. In contrast, the actor co-membership graph is not complete (some pairs of actors never belong to the same event).

What happens if we now find actor and event betweenness centralities, $C_B^{\mathcal{M}01}(n_i)$ and $C_B^{\mathcal{M}01}(m_i)$, for these relations? These centralities are presented in Table 5. Several points are important to note about these results. First, as anticipated above, all events in this

example have equal betweenness centrality for the dichotomous one-mode network of event overlaps. In contrast, in the bipartite graph, events m_2 and m_3 are clearly more central than is event m_1. The dichotomous event overlap relation lacks information about the number of actors shared by pairs of events, and thus lacks information about the number of geodesics between events. Second, notice that for the dichotomous actor co-membership relation, actors n_2 and n_4 have betweenness centrality equal to 0. These actors are adjacent to other actors on this relation but they are not on geodesics between them. Actors n_2 and n_4 are each affiliated with only a single event. Thus, if one of these 'single-membership' actors is adjacent to two other actors (for example, n_2 is adjacent to n_5 and n_6) those two actors must be adjacent to each other since all three must belong to the single event to which the 'single-membership' actor belongs. Actors that belong to only one event cannot be on geodesics between other actors.

These observations about betweenness centrality in affiliation networks are informal. Clearly, it remains for further research to express the exact relationship between actor and event betweenness centrality.

4.5. *Flow betweenness centrality*

Recently, Freeman et al. (1991) have proposed an extension of betweenness centrality that is applicable to valued relations. For a pair of actors, the value of the relation might be "their amount of interaction, the time they spend in one another's company, the range of different social settings in which they interact..." (Freeman et al., 1991, p. 145). Thus, this approach is clearly applicable to one-mode relations of actor co-memberships or event overlaps.

Flow betweenness centrality extends betweenness centrality in two ways. First it considers all paths between nodes, rather than just geodesics. Second, it is appropriate for both graphs and valued graphs in which larger values indicate stronger ties between actors. Values in the graph are considered to represent the potential for 'flow' of information or resources between nodes, with the following constraints. First, the flow between a pair of adjacent nodes, f_{ij}, is constrained to be less than or equal to the value of the tie between the nodes: $f_{ij} \leq x_{ij}$. Second the flow 'into' a node is equal to the flow 'out of' the node. The fact that this approach focuses on the betweenness of nodes in a graph and can be used to study valued relations suggests that it should be quite appropriate for studying centrality in affiliation networks.

To define the flow betweenness centrality of node p_i, one considers the extent to which the maximum flow between other pairs of nodes in the graph depends on paths that include node p_i. Roughly, the flow centrality index for a node in a valued graph quantifies the extent to which flow between other pairs of nodes in the graph would be reduced if that node were removed from the graph. Defining $F_{kl}(p_i)$ as the amount of flow between nodes p_k and p_l that passes through p_i, the flow betweenness centrality of p_i is defined as the total flow between all pairs of nodes that depends on node p_i. The flow betweenness for node p_i, denoted by $C_F(p_i)$, is calculated as:

$$C_F(p_i) = \sum_{k<l} F_{kl}(p_i). \tag{33}$$

This index is described in detail in Freeman et al. (1991). Flow betweenness may be calculated for the one-mode networks of actor co-memberships and event overlaps, or for the bipartite graph.

Table 4 presents flow betweenness for actors and events in the bipartite graph. Table 5 presents results for the one-mode networks of actor co-memberships and event overlaps. Two results are worth noting in these tables. First, notice that actors n_2 and n_4 which each belong to only a single event have flow betweenness centralities equal to zero in the bipartite graph, but have flow centralities equal to four in the actor co-membership relation. In the one-mode network these actors cannot reside on geodesics between other actors, but they can reside on paths between other actors. Since flow betweenness considers 'flows' along all paths (not just geodesics), actors that belong to a single event can have non-zero flow betweenness in the one-mode actor co-membership relation.

A second puzzling result is seen in Table 4. Notice that for the bipartite graph actor n_5 (which is affiliated with all three events) has flow betweenness centrality equal to three, but actors n_1 and n_3 (which are each affiliated with two events) have centralities equal to five. However, looking at either the bipartite graph (Fig. 1) or the affiliation network matrix (Table 1), we see that the events to which actors n_1 and n_3 belong are subsets of the events to which n_5 belongs; n_1 and n_3 are never in an event unless n_5 is also a member. This anomalous result deserves further investigation.

5. Extensions

The centrality approaches discussed so far (degree, eigenvector, closeness, betweenness, and flow betweenness) were all designed for one-mode networks. As we have seen, all can be used to study affiliation networks by analyzing the bipartite graph. In this section I present two possible extensions of centrality using Galois lattices and graph covers. Both of these approaches are designed to study two-mode data, both focus on subsets and relations between subsets, and both allow insights into the difference between 'primary' and 'secondary' actors and events. Thus, these extensions incorporate the distinctive features of affiliation networks and might lead to centrality approaches that more naturally capture important features of these networks.

5.1. Galois lattices

Patterns of inclusion and overlap in events' memberships and in actors' affiliations are theoretically and empirically important components of actor and event centrality. As an early attempt to capture these relationships, Bonacich (1978) proposed using homomorphisms of Boolean algebras. The resulting homomorphisms were represented as lattices showing unions and intersections of events' members (with respect to a subset of actors), or of actors' affiliations (with respect to a subset of events). As noted by Bonacich, the height of an event or an actor in the lattice is related to its relative centrality. Bonacich's representation nicely shows the structure of inclusions among actors or among events, but it is limited in that it requires two separate representations

(one for actors and one for events). In addition, it captures only a portion of the inclusions among actors and events because the lattice for actors is constructed with respect to only a subset of events, and the lattice for events is constructed with respect to only a subset of actors. A Galois lattice provides a more appropriate approach since a single representation presents patterns of inclusions for all actors and events simultaneously.

Galois lattices have only recently been used to study affiliation networks (Schweizer, 1991; Freeman and White, 1993; Wasserman and Faust, 1994), though applications to other kinds of data are more widespread. General discussions of Galois lattices can be found in Birkhoff (1940) and Wille (1984).

A Galois lattice represents the relationship between two sets of objects in terms of inclusion mappings between subsets of objects from each set. For an affiliation network we have the two sets: $\mathcal{N} = \{n_1, n_2, \ldots, n_g\}$ and $\mathcal{M} = \{m_1, m_2, \ldots, m_h\}$. We also have the affiliation relation, denoted by α, from $\mathcal{N}$ to $\mathcal{M}$, where $n_i \alpha m_k$ if n_i 'is affiliated with' m_k. Thus $n_i \alpha m_k$ if actor n_i belongs to event m_k. The inverse relation, α^{-1}, is a mapping from $\mathcal{M}$ to $\mathcal{N}$, where $m_k \alpha^{-1} n_i$ if event m_k includes actor n_i.

Extending the relations α and α^{-1} from individual actors and events to subsets of actors and events, we get two complementary mappings: $\uparrow$ from subsets of actors to subsets of events and $\downarrow$ from subsets of events to subsets of actors. The $\uparrow$ mapping maps a subset of entities in $\mathcal{N}$ to a subset of entities in $\mathcal{M}$. For subsets $\mathcal{N}_S \subset \mathcal{N}$ and $\mathcal{M}_S \subset \mathcal{M}$, we define $\mathcal{N}_S \uparrow \mathcal{M}_S$ if $n_i \alpha m_k$ for all $n_i \in \mathcal{N}_S$, and $m_k \in \mathcal{M}_S$. For an affiliation network, the $\uparrow$ mapping maps a subset of actors to the subset of events attended by all of the actors in that subset.

The dual mapping, $\downarrow$, goes from subsets of $\mathcal{M}$ to subsets of $\mathcal{N}$. For subsets $\mathcal{N}_S \subset \mathcal{N}$ and $\mathcal{M}_S \subset \mathcal{M}$, we define $\mathcal{M}_S \downarrow \mathcal{N}_S$ if $m_k \alpha^{-1} n_i$ for all $n_i \in \mathcal{N}_S$ and $m_k \in \mathcal{M}_S$. In an affiliation network the $\downarrow$ mapping maps a subset of events to the subset of actors all of which attend all events in the subset.

A Galois lattice represents the relation between the sets $\mathcal{N}$ and $\mathcal{M}$ in terms of the $\uparrow$ and $\downarrow$ mappings between subsets of entities from each set. A Galois lattice is presented in a diagram in which points represent entities or subsets of entities from the two sets, and lines represent subset–superset relations in the following way. There is a line or sequence of lines in the diagram ascending from object $n_i \in \mathcal{N}$ to object $m_k \in \mathcal{M}$ if $n_i \alpha m_k$, and there is a line or sequence of lines descending from object $m_k \in \mathcal{M}$ to object $n_i \in \mathcal{N}$ if $m_k \alpha^{-1} n_i$.

Fig. 3 presents the Galois lattice for the hypothetical example of six actors and three events (this figure is adapted from Wasserman and Faust, 1994, p. 333). In this figure an ascending line or sequence of lines connects each actor to the subset of events with which it is affiliated and a descending line or sequence of lines connects each event to the actors that are affiliated with it. One can also look at lines or sequences of lines between actors or between events. A line or sequence of lines ascending from actor n_i to actor n_j means that the events to which actor n_j belongs are a subset of the events to which actor n_i belongs. Similarly, a line or sequence of lines descending from event m_k to event m_l means that the actors that are affiliated with event m_l are a subset of the actors that are affiliated with event m_k.

Viewing centrality in terms of subset–superset relations, actors that are relatively low

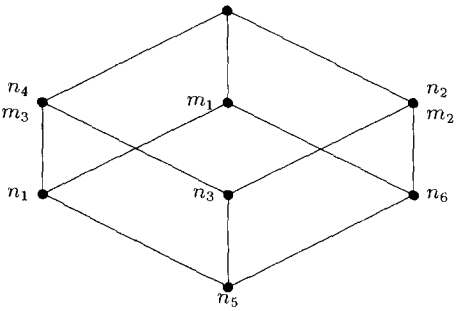


Fig. 3. Galois lattice for hypothetical example.

in the diagram are relatively more central than are actors that are relatively high in the diagram. Similarly, events that are relatively high in the diagram are relatively more central than are events that are relatively low in the diagram (Freeman and White, 1993). For the hypothetical example in Fig. 3, actor n_5 is the most central actor, followed by n_1, n_3, and n_6; n_4 and n_2 are least central. In this example the events are indistinguishable in terms of their centrality.

A Galois lattice is especially appropriate for studying affiliation networks because it explicitly represents the non-dyadic relation of inclusion between subsets of entities from the two modes. Furthermore, it represents an aspect of centrality in these networks, the subset–superset inclusions between actors and events, that is not represented by other centrality approaches. Freeman and White (1993) use Galois lattices in their reanalysis of the data of Davis et al. (1941) on Southern club women. Their results clearly showed which actors and events were primary and which were secondary in terms of the upward containment structure for actors and the downward containment structure for events.

5.2. *Graph covers*

The idea of *graph covers* was introduced by Seidman (1985) to formalize the extent to which actors potentially have access to information about different kinds of structural features of an affiliation network. An actor that is located in such a way that it can 'see' important features of other actors' membership patterns (such as pair-wise attendances or cohesive subgroups) might be well located to initiate or coordinate actions within the network. Seidman uses the idea of a graph cover to formalize exactly what features of an affiliation network an actor or a set of actors 'sees'.

We begin with a hypergraph $\mathscr{H}(\mathscr{N},\mathscr{M})$ consisting of a set of points $\mathscr{N}$ and a set of edges $\mathscr{M}$, where each edge is a subset of points from $\mathscr{N}$. For an affiliation network it seems natural to start with the set of points being the set of actors, $\mathscr{N}$, and the set of edges being the set of events, $\mathscr{M}$. (We could also reverse the roles of $\mathscr{N}$ and $\mathscr{M}$ and have the dual hypergraph $\mathscr{H}^*(\mathscr{M},\mathscr{N})$.) We next define a duality function, $\delta(n_i)$ for points. This duality function is a mapping from point n_i to the set of all edges with which it is incident. In an affiliation network, this function for actors maps the actor to

the set of all events with which it is affiliated. Formally we can express this duality function as

$$\delta(n_i) = \{m_k \in \mathscr{M} | n_i \in m_k\}. \tag{34}$$

Using the notation from Galois lattices (above), $\delta(n_i)$ is the subset of events $\mathscr{M}_s$ for which $n_i \alpha m_k$.

Seidman (1985) then uses this duality function to define two different kinds of graph covers: a *strict cover* and a *wide cover*. A set is a wide (or strict) cover (as defined below) of some set of objects $\mathscr{A}$, where the objects are a specified set of entities. For example, the set $\mathscr{A}$ might be a subset of actors $\{n_1, n_2, n_3\}$ or it might be a subset of pairs of actors $\{(n_1,n_2),(n_1,n_3),(n_2,n_3)\}$.

In general, for some set of objects $\mathscr{A}$, a set $\mathscr{N}_s$ is a *wide cover* of $\mathscr{A}$ if the union of the duals of elements of $\mathscr{A}$ is included in the union of the duals of elements of $\mathscr{N}_s$. Formally, for object set $\mathscr{A}$ (a set of actors), $\mathscr{N}_s$ is a wide cover of $\mathscr{A}$ if

$$\bigcup_{a \in A} \delta(a) \subset \bigcup_{n_i \in \mathscr{N}_s} \delta(n_i). \tag{35}$$

Since the duality function $\delta(a)$ maps an actor to the set of events with which it is affiliated, for a subset of actors the union of these mappings is the collection of all events with which any actor in the subset is affiliated. Consider the case in which $\mathscr{A} = \mathscr{N}_t$ (a subset of actors). Another subset of actors, $\mathscr{N}_s$, is a wide cover of $\mathscr{N}_t$ if some actor in $\mathscr{N}_s$ was present at each and every event at which any actor from $\mathscr{N}_t$ was present. The events to which members of $\mathscr{N}_t$ belong are a subset of the events to which members of $\mathscr{N}_s$ belong. If $\mathscr{N}_s$ consists of a single actor, $\mathscr{N}_s = \{n_i\}$, then n_i is a wide cover of the set of all actors that were never present unless n_i was present.

To illustrate, in the hypothetical affiliation network of six actors and three events, n_5 is a wide cover of the set consisting of the remaining five actors, $\{n_1,n_2,n_3,n_4,n_6\}$; n_3 is a wide cover of the set $\{n_2,n_4\}$; n_6 is a wide cover of $\{n_2\}$; and n_1 is a wide cover of $\{n_4\}$. The set $\{n_1,n_6\}$ is a wide cover of the set $\{n_2,n_4\}$.

The second kind of graph cover is a *strict cover*. A strict cover is more general than a wide cover. For object set $\mathscr{A}$ (a set of actors), $\mathscr{N}_s$ is a strict cover of $\mathscr{A}$ if there exists some $n_i \in \mathscr{N}_s$ such that

$$\delta(n_i) \bigcap_{a \in \mathscr{A}} \delta(a) \neq \varnothing. \tag{36}$$

This means that for each object $a \in \mathscr{A}$ there exists some $n_i \in \mathscr{N}_s$ such that the dual of a intersects the dual of n_i. In an affiliation network, $\mathscr{N}_s$ is a strict cover of $\mathscr{A}$ if, for every actor in $\mathscr{A}$, some actor in $\mathscr{N}_s$ belongs to some event with that actor. To illustrate, for the hypothetical example n_6 is a strict cover of $\{n_1,n_2,n_3,n_5\}$.

Every wide cover is a strict cover, but the reverse is not true. In an affiliation network, if $\mathscr{N}_s$ is a wide cover of $\mathscr{A}$, some actor from $\mathscr{N}_s$ is present each and every time any member of $\mathscr{A}$ is present. In other words, a member of $\mathscr{A}$ is never in an event unless a member of $\mathscr{N}_s$ is also present. In contrast, if $\mathscr{N}_s$ is a strict cover of $\mathscr{A}$, then each member of $\mathscr{A}$ must belong to at least one event with some member of $\mathscr{N}_s$, though members of $\mathscr{A}$ might belong to events where no member of $\mathscr{N}_s$ is present.

Returning to the Galois lattice diagram for an affiliation network, we see that a single

actor, n_i, is a wide cover of the set of all actors that reside on lines ascending from it. The events to which any of these other actors belong are a subset of the events to which n_i belongs. A set of actors is a wide cover of that set of actors that reside on lines ascending from any of the actors in the set. A single actor is a strict cover of the set of all actors with which it is ever a co-member. In the Galois lattice this is the set of all actors from which lines intersect either ascending or descending lines from n_i at any point before the topmost point in the diagram. Any subset of actors is a strict cover of the set of all actors that ever belonged to any event with any actor in the set. In the Galois lattice this is the set of all actors whose lines (ascending or descending) intersect a line from any member of the cover subset of actors, before the topmost point in the diagram.

Seidman (1985) elaborates in detail how graph covers can be used to study structural features of hypergraphs by varying the definition of the elements of $\mathscr{A}$. For example, if $\mathscr{A}$ consists of a set of pairs of actors, then a wide cover of $\mathscr{A}$ is a set of actors that are always present at any event at which any pair of actors in $\mathscr{A}$ is present.

5.3. Summary

Both the Galois lattice and the concept of graph covers nicely present subset–superset relationships among actors and events. This subset–superset relation is exactly the centrality idea of primary and secondary actors, as suggested by Davis et al. (1941), Doreian (1979a) and Freeman and White (1993). The Galois lattice incorporates the duality in the relation between actors and events in a single representation and makes the relative centrality of actors and events quite apparent in their vertical positions in the diagram. The distinction between primary events and actors and secondary events and actors is easily seen in the relative vertical positions of actors and events and in the patterns of inclusions among them.

The concept of graph cover formalizes the idea of the 'importance' of an actor (or a subset of actors) in terms of the 'view' that the actor has of the network. One might think of a graph cover in terms of the kind and extent of information about an affiliation network that is available to an actor or to a subset of actors. An actor that 'covers' a large number of other actors, or 'covers' a large portion of the co-memberships between pairs of actors potentially has access to information about those features of the network.

Both Galois lattices and graph covers provide a set of concepts for thinking about centrality in affiliation networks that is not provided by any of the other centrality indices. Both are naturally appropriate for two-mode non-dyadic networks and focus on subset–superset relations between actors and/or events in a way that captures theoretical and empirical insights about the relative importance of actors in these networks.

Neither the Galois lattice nor the formalism of a graph cover provides an index or quantification of actors' or events' centrality. One possibility would be to define a new relation between actors based on subset–superset relationships between their memberships. For example, one could define $x_{ij} = 1$ if $\delta(n_j) \subset \delta(n_i)$, and $x_{ij} = 0$ otherwise. The quantity $\sum_j x_{ij}$ would give the number of actors for which n_i is a wide cover, while $\sum_j x_{ji}$ would give the number of actors that are wide covers of n_i. Exploring other possible indices is a fruitful arena for further research.

Now let us turn to another illustration.

6. Another example

As another example, consider the affiliation network of club and board memberships of corporate executive officers studied by Galaskiewicz (1985) as part of his extensive research on the urban grants economy in Minneapolis-St.Paul. I will focus on a subset of 26 CEOs and 15 clubs/boards from Galaskiewicz's data. The affiliation network matrix for this example is presented in Table 6. Centralities for CEOs and for the boards/clubs are in Table 7 (for the bipartite graph) and Table 8 (for the one-mode relations).

Overall there is considerable agreement among the centrality measures in their identification of which clubs and CEOs are relatively central. Tables 9–11 present correlation matrices for the five centrality indices, calculated on the bipartite graph (including all nodes, and for clubs and CEOs separately) and on the one-mode relations of club overlaps and actor co-memberships. The correlations between the measures of centrality for the clubs in the bipartite graph range from a low of 0.890 to a high of 0.992 with a median of 0.968; for CEOs in the bipartite graph the range is from 0.041 to 0.867 with a median of 0.643. For the one-mode relations, the correlation between the centrality measures for clubs ranges from 0.918 to 0.993 with a median of 0.954, and for CEOs the range is from 0.348 to 0.989 with a median of 0.711. It appears that there is

Table 6
Affiliation network matrix for CEOs and clubs

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	0	0	1	1	0	0	0	0	1	0	0	0	0	0	0
2	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0
3	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0
4	0	1	1	0	0	0	0	0	0	0	0	0	0	0	1
5	0	0	1	0	0	0	0	0	0	0	0	0	1	1	0
6	0	1	1	0	0	0	0	0	0	0	0	0	0	1	0
7	0	0	1	1	0	0	0	0	0	1	1	0	0	0	0
8	0	0	0	1	0	0	1	0	0	1	0	0	0	0	0
9	1	0	0	1	0	0	0	1	0	1	0	0	0	0	0
10	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0
11	0	1	1	0	0	0	0	0	1	0	0	0	0	0	0
12	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0
13	0	0	1	1	1	0	0	0	1	0	0	0	0	0	0
14	0	1	1	1	0	0	0	0	0	0	1	1	1	0	1
15	0	1	1	0	0	1	0	0	0	0	0	0	1	0	1
16	0	1	1	0	0	1	0	1	0	0	0	0	0	1	0
17	0	1	1	0	1	0	0	0	0	0	1	1	0	0	1
18	0	0	0	1	0	0	0	0	1	0	0	1	1	0	1
19	1	0	1	1	0	0	1	0	1	0	0	0	0	0	0
20	0	1	1	1	0	0	0	0	0	0	1	0	0	0	1
21	0	0	1	1	0	0	0	1	0	0	0	0	0	0	0
22	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1
23	0	1	1	0	0	1	0	0	0	0	0	0	0	0	1
24	1	0	1	1	0	1	0	0	0	0	0	0	0	0	1
25	0	1	1	0	0	0	0	0	0	0	0	0	1	0	0
26	0	1	1	0	0	0	0	0	0	0	0	1	0	0	0

Table 7
Centrality of CEOs and clubs in the bipartite graph

	Measure of centrality					
	$C_D^{\mathcal{NM}}$	$\Sigma d(i,k)$	$C_C^{\mathcal{NM}}$	$C_B^{\mathcal{NM}}$	$C_E^{\mathcal{NM}}$	$C_F^{\mathcal{NM}}$
m_1	3	112	35.71	5.53	0.057	49
m_2	11	90	44.44	52.57	0.276	235
m_3	22	62	64.52	374.10	0.476	527
m_4	12	84	47.62	139.41	0.236	271
m_5	3	112	35.71	3.91	0.064	64
m_6	4	110	36.36	5.76	0.102	60
m_7	4	114	35.09	12.61	0.049	125
m_8	4	110	36.36	13.16	0.068	106
m_9	6	104	38.46	17.13	0.114	156
m_{10}	3	116	34.48	5.32	0.037	79
m_{11}	4	110	36.36	5.52	0.118	45
m_{12}	5	108	37.04	11.01	0.120	98
m_{13}	5	106	37.74	12.56	0.119	100
m_{14}	3	122	32.79	2.63	0.057	85
m_{15}	9	94	42.55	37.77	0.237	146
n_1	3	89	44.94	9.36	0.126	7
n_2	3	93	43.01	20.88	0.090	58
n_3	2	97	41.24	3.46	0.091	14
n_4	3	95	42.11	3.24	0.151	5
n_5	3	95	42.11	14.37	0.099	50
n_6	3	97	41.24	11.75	0.123	38
n_7	4	87	45.98	36.34	0.132	58
n_8	3	117	34.19	9.09	0.049	59
n_9	4	111	36.04	17.55	0.061	93
n_{10}	2	97	41.24	3.12	0.090	6
n_{11}	3	95	42.11	7.21	0.132	8
n_{12}	2	119	33.61	3.02	0.043	23
n_{13}	4	87	45.98	23.26	0.136	43
n_{14}	7	81	49.38	49.62	0.241	52
n_{15}	5	91	43.96	18.19	0.184	39
n_{16}	5	91	43.96	35.86	0.149	79
n_{17}	6	89	44.94	34.20	0.197	75
n_{18}	5	91	43.96	24.93	0.126	35
n_{19}	5	85	47.06	44.54	0.142	64
n_{20}	5	85	47.06	23.79	0.205	27
n_{21}	3	89	44.94	14.25	0.119	23
n_{22}	3	93	43.01	11.12	0.119	25
n_{23}	4	93	43.01	10.11	0.166	26
n_{24}	5	85	47.06	37.87	0.169	61
n_{25}	3	95	42.11	5.75	0.133	16
n_{26}	3	95	42.11	6.08	0.133	16

greater consistency across centrality measures in the identification of central clubs than in the identification of central CEOs.

We can also look at the relationship between centrality measures for the bipartite graph and for the one-mode networks. Table 12 presents the correlation between each

Table 8
Centrality measures for CEOs and clubs in one-mode networks

	Measure of centrality				
	$C_D^{\mathcal{N}}$ [a]	$C_D^{\mathcal{N}}$ [b]	$C_B^{\mathcal{N}01}$	$C_E^{\mathcal{N}}$	$C_F^{\mathcal{N}}$
m_1	11	14	1.85	0.080	55
m_2	36	47	4.07	0.391	122
m_3	62	84	10.70	0.673	280
m_4	38	50	6.97	0.333	200
m_5	10	13	1.09	0.091	35
m_6	15	19	1.27	0.144	52
m_7	9	13	0.70	0.070	45
m_8	11	15	2.10	0.096	49
m_9	16	22	1.99	0.162	51
m_{10}	8	11	0.95	0.052	41
m_{11}	18	22	1.51	0.167	55
m_{12}	18	22	0.29	0.170	51
m_{13}	18	23	2.11	0.169	57
m_{14}	8	11	0.17	0.080	28
m_{15}	34	43	3.24	0.336	106
n_1	37	40	3.85	0.178	407
n_2	26	29	2.60	0.127	278
n_3	25	27	0.15	0.128	248
n_4	39	42	0.15	0.213	337
n_5	27	30	0.15	0.140	274
n_6	33	36	0.00	0.174	312
n_7	37	41	3.85	0.187	392
n_8	16	19	0.13	0.069	207
n_9	18	22	0.43	0.086	206
n_{10}	26	28	0.15	0.127	269
n_{11}	36	39	0.15	0.187	345
n_{12}	14	16	0.13	0.061	172
n_{13}	39	43	3.85	0.192	429
n_{14}	61	68	3.85	0.341	586
n_{15}	46	51	0.15	0.261	406
n_{16}	39	44	1.09	0.211	387
n_{17}	48	54	0.15	0.278	431
n_{18}	32	37	3.09	0.178	294
n_{19}	42	47	3.85	0.201	504
n_{20}	53	58	3.85	0.289	483
n_{21}	35	38	3.85	0.168	388
n_{22}	32	35	1.24	0.168	301
n_{23}	42	46	0.15	0.235	354
n_{24}	45	50	3.85	0.239	437
n_{25}	35	38	0.15	0.188	323
n_{26}	35	38	0.15	0.188	324

[a]Excludes diagonal. [b]Includes diagonal.

kind of centrality index calculated on the bipartite graph and on the one-mode relation, for CEOs and clubs separately. For example, the first line in the body of the table gives the correlation between the degree centralities for CEOs and for clubs in the bipartite

Table 9
Correlations between centrality measures: clubs and CEOs in the bipartite graph

	$C_D^{\mathcal{NA}}$	$C_C^{\mathcal{NA}}$	$C_B^{\mathcal{NA}}$	$C_E^{\mathcal{NA}}$
$C_C^{\mathcal{NA}}$				
All nodes	0.694			
Clubs	0.992			
CEOs	0.627			
$C_B^{\mathcal{NA}}$				
All nodes	0.928	0.752		
Clubs	0.948	0.968		
CEOs	0.867	0.659		
$C_E^{\mathcal{NA}}$				
All nodes	0.889	0.872	0.816	
Clubs	0.980	0.967	0.890	
CEOs	0.780	0.823	0.576	
$C_F^{\mathcal{NA}}$				
All nodes	0.938	0.485	0.892	0.696
Clubs	0.977	0.969	0.960	0.921
CEOs	0.572	0.041	0.696	0.045

Table 10
Correlations between centrality measures: one-mode matrix of club overlaps

	$C_D^{\mathcal{A}}$	$C_C^{\mathcal{A}}$	$C_B^{\mathcal{A}}$	$C_E^{\mathcal{A}}$
$C_C^{\mathcal{A}}$	0.958			
$C_B^{\mathcal{A}}$	0.937	0.954		
$C_E^{\mathcal{A}}$	0.993	0.930	0.918	
$C_F^{\mathcal{A}}$	0.959	0.954	0.985	0.936

Table 11
Correlations between centrality measures: one-mode matrix of CEO co-memberships

	$C_D^{\mathcal{N}}$	$C_C^{\mathcal{N}}$	$C_B^{\mathcal{N}}$	$C_E^{\mathcal{N}}$
$C_C^{\mathcal{N}}$	0.734			
$C_B^{\mathcal{N}}$	0.449	0.687		
$C_E^{\mathcal{N}}$	0.989	0.664	0.348	
$C_F^{\mathcal{N}}$	0.942	0.788	0.641	0.888

Table 12
Correlations between centralities for the bipartite graph and the one-mode relations, for CEOs and clubs

Centrality measure	CEOs	Clubs
Degree	0.767	0.979
Betweenness[a]	0.634	0.946
Eigenvector	1.000	1.000
Flow	0.137	0.944

[a]Betweenness centralities for dichotomized one-mode relation.

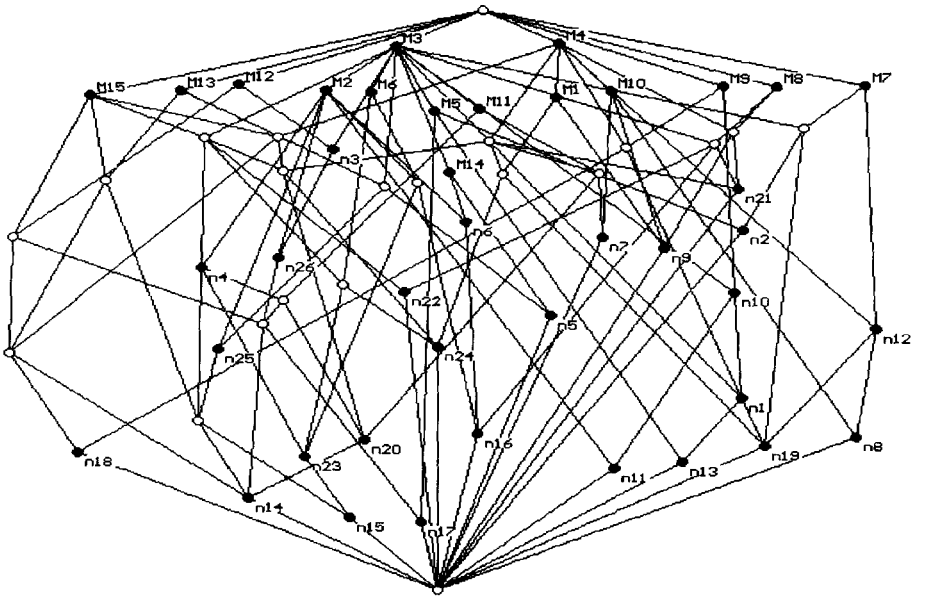


Fig. 4. Galois lattice for CEOs and clubs.

graph and in the one-mode actor co-membership relation (the $C_D^{\mathcal{N}\mathcal{A}}(n_i)$ and the $C_D^{\mathcal{N}}(n_i)$ and the $C_D^{\mathcal{N}\mathcal{A}}(m_k)$ and the $C_D^{\mathcal{A}}(m_k)$). For these data, there is greater discrepancy for actors (the CEOs) than for events (the clubs) between the centralities from the bipartite graph and the centralities from the one-mode relation.

Fig. 4 presents the Galois lattice for the CEOs and clubs. [2] First, consider the relative centrality of the clubs. Clubs 3 and 4 are the only clubs whose memberships contain the memberships of other clubs (club 4 contains clubs 1 and 10; club 3 contains clubs 2, 5, 6, 11 and 14). Clubs 3 and 4 are located toward the top of the diagram in Fig. 4 and the containments are indicated by the lines descending from these clubs. More complicated patterns of containment are present among the CEOs. CEO 14 contains 25, 26 and 20 (who in turn contains 4); 15 contains 23 and 25; 17 contains 26 (who in turn contains 3); 8 contains 12; 11 contains 10; 13 contains 1 (who in turn contains 10); 19 contains both 1 and 12; and 16 contains 6. This suggests at least three levels of centrality for CEOs. The most central CEOs are 8, 11, 13, 14, 15, 16, 17 and 19; second are 1, 6, 12, 20, 23, 25 and 26; and at the third level are 3, 4 and 23. The memberships of the remaining CEOs are not included within the memberships of others, nor do they include others, thus, their centrality relative to others cannot be determined using this approach. Further analyses show that other indices of centrality for the CEOs are consistent with these three levels in the Galois lattice. CEOs in the first level have higher centrality on other indices than do CEOs in the second level. CEOs in the second level have higher centrality values than do the CEOs in the third level. CEOs whose centrality is

[2] This figure was produced using the program *DIAGRAM* (Vogt and Bliegener, 1990).

indeterminate in the Galois lattice have moderate to high levels of centrality on other centrality indices.

In conclusion, let us consider some of the strengths and weaknesses of centrality indices for studying affiliation networks.

7. Discussion

The introduction to this paper presented four ideas that might motivate centrality approaches that would be especially appropriate for affiliation networks. Let us now consider existing centrality indices and possible extensions in light of these ideas.

First, it is important to have centrality scores for both actors and events in an affiliation network. This is not a problem for any of the approaches, since all can be used to analyze the bipartite graph. However, because of the duality in the relationship between actors and events, we might hope that the centralities of actors should be related in specifiable ways to the centralities of the events with which they are affiliated, and similarly the centralities of events should be related in specifiable ways to the centralities of the actors they include. These relationships hold for degree centralities calculated on the one-mode networks of actor co-memberships and event overlaps (Eqs. (10) and (11)), for closeness centrality in the bipartite graph (Eqs. (27) and (28)), and for eigenvector centralities in the bipartite graph or in the pair of one-mode matrices (Eqs. (16) and (17) or Eqs. (19) and (20)). Relationships between actor and event centralities are not as straightforward for betweenness centrality and flow betweenness centrality.

A second feature of affiliation networks is that the affiliation relation is non-dyadic, and thus focuses on subsets. To what extent do centralities in affiliation networks allow us to quantify the centrality of a subset of actors or a subset of events? In one sense, any index for which an actor's centrality is a function of the centralities of its events, or an event's centrality is a function of the centralities of its members, does capture the centralities of subsets. An actor's centrality is a function of the centrality of the subset of events to which it belongs. An event's centrality is a function of the subset of actors belonging to it. As noted above, these properties hold for degree, closeness, and eigenvector centralities, but not for betweenness and flow betweenness.

A third important theoretical property of an affiliation network is that actors create linkages between events and events create linkages between actors. Central actors should link events and central events should link actors. Theoretically, actors that do not link events, and events that do not link actors, should not be considered central. The property of linkages among actors and/or events seems most naturally embodied in betweenness, flow betweenness, and eigenvector centrality, but is also most problematic for these approaches. Part of the problem is that misleading results arise when one analyzes the one-mode relations of actor co-memberships or event overlaps. These relations lack important information about the pattern of affiliation ties. When used to analyze the bipartite graph, betweenness centrality and flow betweenness centrality give centralities equal to zero for actors that belong only to one event and to events with only a single member, as desired. However, eigenvector centrality gives non-zero values to such actors and events, even in the bipartite graph. This property is consistent with the

intuitive motivation for eigenvector centrality that a node's centrality is a function of the centralities of the nodes to which it is adjacent, but is inconsistent with the idea that centrality in affiliation networks is a result of the linkages created by actors and events.

The fourth empirically and theoretically important aspect of centrality in an affiliation network is the subset–superset (i.e. inclusion) relationships between actors' affiliations and events' memberships. These relationships capture the distinction between 'primary' and 'secondary' actors and events observed by numerous researchers. However, this distinction is not quantified by any of the existing centrality indices reviewed here. Subset–superset relationships are nicely displayed in the Galois lattice and are formalized by the idea of graph covers. Developing a centrality index based on these inclusion relations seems like a fruitful area for future research.

A final practical note is in order. Seidman (1981) discusses the misleading conclusions that can result when one studies one-mode networks of actor co-memberships or event overlaps for an affiliation network. Specifically, he demonstrates that cliques of actors in the co-membership relation (or events in the overlap relation) are not necessarily complete subgraphs in the affiliation relation. Thus, one is not necessarily justified in drawing conclusions about collections larger than pairs from the one-mode relations. As we have seen, similar cautions are in order for centrality analyses of the one-mode networks derived from the affiliation network. In going from the affiliation relation to either the actor co-membership relation or the event overlap relation, one loses information about the patterns of affiliation between actors and events. Thus, one needs to be cautious when interpreting centralities for these one-mode relations.

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